

November 8, 2017 Updated Instructional Segments Tables: These corrected instructional segment tables will be in the published version of the 2016 CA Science Framework.

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Kindergarten

Kindergarten Instructional Segment 1: Plant and Animal Needs
<i>Guiding Questions</i> • <i>How do we know that something is alive?</i> • <i>What do animals and plants need to survive?</i> • <i>Does what they need affect where they live?</i>
Students who demonstrate understanding can: K-LS1-1. Use observations to describe patterns of what plants and animals (including humans) need to survive. [Clarification Statement: Examples of patterns could include that animals need to take in food but plants do not; the different kinds of food needed by different types of animals; the requirement of plants to have light; and, that all living things need water.] K-ESS3-1. Use a model to represent the relationship between the needs of different plants or animals (including humans) and the places they live. [Clarification Statement: Examples of relationships could include that deer eat buds and leaves, therefore, they usually live in forested areas; and, grasses need sunlight so they often grow in meadows. Plants, animals, and their surroundings make up a system.] K-ESS3-3. Communicate solutions that will reduce the impact of humans on the land, water, air, and/or other living things in the local environment.* [Clarification Statement: Examples of human impact on the land could include cutting trees to produce paper and

using resources to produce bottles. Examples of solutions could include reusing paper and recycling cans and bottles.]		
The bundle of performance expectations above focuses on the following elements from the NRC document <i>A Framework for K–12 Science Education</i> :		
Highlighted Science and Engineering Practices [SEP-2] Developing and Using Models [SEP-4] Analyzing and Interpreting Data [SEP-8] Obtaining, Evaluating, and Communicating Information	Highlighted Disciplinary Core Ideas LS1.C: Organization for Matter and Energy Flow in Organisms ESS3.A: Natural Resources	Highlighted Crosscutting Concepts [CCC-1] Patterns [CCC-4] System and System Models
Highlighted California Environmental Principles & Concepts: Principle I The continuation and health of individual human lives and of human communities and societies depend on the health of the natural systems that provide essential goods and ecosystem services. Principle II The long-term functioning and health of terrestrial, freshwater, coastal and marine ecosystems are influenced by their relationships with human societies. Principle IV The exchange of matter between natural systems and human societies affects the long term functioning of both.		
CA CCSS Math Connections: MP. 2, 4, K.CC.1-3, K.MD.2-3		
CA CCSS ELA/Literacy Connections: : W.K.2, 8; SL.K.1, 4, 5; L.K.5c		
CA ELD Connections: ELD.PI.K.3		

Kindergarten Instructional Segment 2: Animals and Plants Can Change Their Environment
<i>Guiding Questions</i> How do animals and plants change their environment to survive? What do we (humans) do that changes our environment? What can we do to modify our impact on the environment?
Students who demonstrate understanding can: K-ESS2-2. Construct an argument supported by evidence for how plants and animals (including humans) can change the

environment to meet their needs. [Clarification Statement: Examples of plants and animals changing their environment could include a squirrel digs in the ground to hide its food and tree roots can break concrete.] K-ESS3-3. Communicate solutions that will reduce the impact of humans on the land, water, air, and/or other living things in the local environment.* [Clarification Statement: Examples of human impact on the land could include cutting trees to produce paper and using resources to produce bottles. Examples of solutions could include reusing paper and recycling cans and bottles.] K-2-ETS1-1. Ask questions, make observations, and gather information about a situation people want to change to define a simple problem that can be solved through the development of a new or improved object or tool. * The performance expectations marked with an asterisk integrate traditional science content with engineering through a practice or disciplinary core idea.		
The bundle of performance expectations above focuses on the following elements from the NRC document <i>A Framework for K–12 Science Education</i>:		
Highlighted Science and Engineering Practices [SEP-1] Asking Questions and Defining Problems [SEP-7] Engaging in argument from evidence	Highlighted Disciplinary Core Ideas ESS2.E: Biogeology ESS3.C: Human Impacts on Earth Systems ETS1.A: Defining and Delimiting Engineering Problems ETS1.B: Developing Possible Solutions	Highlighted Crosscutting Concepts [CCC-2] Cause and Effect [CCC-4] System and System Models
<i>Highlighted California Environmental Principles & Concepts:</i> Principle I The continuation and health of individual human lives and of human communities and societies depend on the health of the natural systems that provide essential goods and ecosystem services. Principle II The long-term functioning and health of terrestrial, freshwater, coastal and marine ecosystems are influenced by their relationships with human societies. Principle IV The exchange of matter between natural systems and human societies affects the long term functioning of both.		

CA CCSS Math Connections: K.MD.3

CA CCSS ELA/Literacy Connections: RI.K.1, 2, 10; SL.K. 2, 3, 5; W.K.2, 7, 8; L.K.1, 2

CA ELD Connections: ELD.PI.K.2, 5, 6; ELD.PII.K.3

Kindergarten Instructional Segment 3: Weather Patterns

Guiding Questions

What is the weather like today and how it is different from yesterday?

Can I predict tomorrow's weather?

What happens when the sun shines on different objects?

How can I protect myself from the sunlight?

How do we prepare for severe weather?

Students who demonstrate understanding can:

K-ESS2-1. Use and share observations of local weather conditions to describe patterns over time. [Clarification Statement: Examples of qualitative observations could include descriptions of the weather (such as sunny, cloudy, rainy, and warm); examples of quantitative observations could include numbers of sunny, windy, and rainy days in a month. Examples of patterns could include that it is usually cooler in the morning than in the afternoon and the number of sunny days versus cloudy days in different months.] [Assessment Boundary: Assessment of quantitative observations limited to whole numbers and relative measures such as warmer/cooler.]

K-ESS3-2. Ask questions to obtain information about the purpose of weather forecasting to prepare for, and respond to, severe weather.* [Clarification Statement: Emphasis is on local forms of severe weather.]

K-PS3-1. Make observations to determine the effect of sunlight on Earth's surface. [Clarification Statement: Examples of Earth's surface could include sand, soil, rocks, and water] [Assessment Boundary: Assessment of temperature is limited to relative measures such as warmer/cooler.]

K-PS3-2. Use tools and materials to design and build a structure that will reduce the warming effect of sunlight on an area* [Clarification Statement: Examples of structures could include umbrellas, canopies, and tents that minimize the warming effect of the sun.]

K-2-ETS1-1. Ask questions, make observations, and gather information about a situation people want to change to define a simple problem that can be solved through the development of a new or improved object or tool.

K–2-ETS1-2. Develop a simple sketch, drawing, or physical model to illustrate how the shape of an object helps it function as needed to solve a given problem.

K–2-ETS1-3. Analyze data from tests of two objects designed to solve the same problem to compare the strengths and weaknesses of how each performs.

* The performance expectations marked with an asterisk integrate traditional science content with engineering through a practice or disciplinary core idea.

The bundle of performance expectations above focuses on the following elements from the NRC document *A Framework for K–12 Science Education*:

Highlighted Science and Engineering Practices	Highlighted Disciplinary Core Ideas	Highlighted Crosscutting Concepts
[SEP-1] Asking Questions and Defining Problems [SEP-2] Developing and Using Models [SEP-3] Planning and Carrying Out Investigations [SEP-4] Analyzing and Interpreting Data [SEP-6] Constructing Explanations (for science) and Designing Solutions (for engineering) [SEP-8] Obtaining, Evaluating, and Communicating Information	PS3.B: Conservation of Energy and Energy Transfer ESS2.D: Weather and Climate ESS3.B: Natural Hazards ETS1.A: Defining and Delimiting Engineering Problems ETS1.B: Developing Possible Solutions ETS1.C: Optimizing the Design Solution	[CCC-1] Patterns [CCC-2] Cause and Effect [CCC-6] Structure and Function

Highlighted California Environmental Principles & Concepts:

Principle I The continuation and health of individual human lives and of human communities and societies depend on the health of the natural systems that provide essential goods and ecosystem services.

Principle III Natural systems proceed through cycles that humans depend upon, benefit from and can alter.

Principle IV The exchange of matter between natural systems and human societies affects the long term functioning of both.

CA CCSS Math Connections: K.CC.5-6; K.MD.2-3, 10; K.G.1

CA CCSS ELA/Literacy Connections: L.K.5c, 5d, 6; W.K.2, 3, 8; SL.K.1, 4, 5, 6

CA ELD Connections: PI.K.A.1, PI.K.A.3, PI.K.C.9

Kindergarten Instructional Segment 4: Pushes and Pulls

Guiding questions

What happens when you push or pull on an object?

How can you make an object move faster or in a different direction?

Students who demonstrate understanding can:

K-PS2-1. Plan and conduct an investigation to compare the effects of different strengths or different directions of pushes and pulls on the motion of an object. [Clarification Statement: Examples of pushes or pulls could include a string attached to an object being pulled, a person pushing an object, a person stopping a rolling ball, and two objects colliding and pushing on each other.] [Assessment Boundary: Assessment is limited to different relative strengths or different directions, but not both at the same time. Assessment does not include non-contact pushes or pulls such as those produced by magnets.]

K-PS2-2. Analyze data to determine if a design solution works as intended to change the speed or direction of an object with a push or a pull.* [Clarification Statement: Examples of problems requiring a solution could include having a marble or other object move a certain distance, follow a particular path, and knock down other objects. Examples of solutions could include tools such as a ramp to increase the speed of the object and a structure that would cause an object such as a marble or ball to turn.] [Assessment Boundary: Assessment does not include friction as a mechanism for change in speed.]

K-2-ETS1-1. Ask questions, make observations, and gather information about a situation people want to change to define a simple problem that can be solved through the development of a new or improved object or tool.

* The performance expectations marked with an asterisk integrate traditional science content with engineering through a practice or disciplinary core idea.

The bundle of performance expectations above focuses on the following elements from the NRC document *A Framework for K–12 Science Education*:

Highlighted Science and Engineering Practices	Highlighted Disciplinary Core Ideas	Highlighted Crosscutting Concepts
[SEP-1] Asking Questions and Defining Problems [SEP-3] Planning and Carrying Out Investigations [SEP-4] Analyzing and Interpreting Data	PS2.A: Forces and Motion PS2.B: Types of Interactions PS3.C: Relationship Between Energy and Forces ETS1.A: Defining Engineering Problems	[CCC-2] Cause and Effect [CCC-7] Stability and Change
CA CCSS Math Connections: MP.2; K.CC.4-6; K.MD.1-2; K.G.1, 4-6		
CA CCSS ELA/Literacy Connections: L.K.5b-c; L.1e-f; RL.K.10a-b; RI.K.1-10		
CA ELD Connections: ELD.PI.K.A1, PI.K.A3, PI.K.B5.		

Grade One

Grade One Instructional Segment 1: Plant shapes
<i>Guiding Questions</i> <i>How can we tell different types of plants apart?</i> <i>How do these differences help the plants?</i>
Students who demonstrate understanding can: 1-LS1-1. Use materials to design a solution to a human problem by mimicking how plants and/or animals use their external parts to help them survive, grow, and meet their needs.* [Clarification Statement: Examples of human problems that can be solved by mimicking plant or animal solutions could include designing clothing or equipment to protect bicyclists by mimicking turtle shells, acorn shells, and animal scales; stabilizing structures by mimicking animal tails and roots on plants; keeping out intruders by mimicking thorns on branches and animal quills; and, detecting intruders by mimicking eyes and ears.]

1-LS3-1. Make observations to construct an evidence-based account that young plants and animals are like, but not exactly like, their parents. [Clarification Statement: Examples of patterns could include features plants or animals share. Examples of observations could include leaves from the same kind of plant are the same shape but can differ in size; and, a particular breed of dog looks like its parents but is not exactly the same.] [Assessment Boundary: Assessment does not include inheritance or animals that undergo metamorphosis or hybrids.]

K-2-ETS1-2. Develop a simple sketch, drawing, or physical model to illustrate how the shape of an object helps it function as needed to solve a given problem.

PE Introduced, but not assessed until later segments:

1-ESS1-1. Use observations of the sun, moon, and stars to describe patterns that can be predicted. [Clarification Statement: Examples of patterns could include that the sun and moon appear to rise in one part of the sky, move across the sky, and set; and stars other than our sun are visible at night but not during the day.] [Assessment Boundary: Assessment of star patterns is limited to stars being seen at night and not during the day.] (IS4)

1-PS4-3. Plan and conduct an investigation to determine the effect of placing objects made with different materials in the path of a beam of light. [Clarification Statement: Examples of materials could include those that are transparent (such as clear plastic), translucent (such as wax paper), opaque (such as cardboard), and reflective (such as a mirror).] [Assessment Boundary: Assessment does not include the speed of light.] (IS3)

The bundle of performance expectations above focuses on the following elements from the NRC document *A Framework for K–12 Science Education*:

Highlighted Science and Engineering Practices	Highlighted Disciplinary Core Ideas	Highlighted Crosscutting Concepts
[SEP-1] Asking Questions and Defining Problems [SEP-2] Developing and Using Models [SEP-3] Planning and Carrying Out Investigations	LS1.A: Structure and Function LS3.A: Inheritance of Traits LS3.B: Variation of Traits ESS1.A: The Universe and its Stars	[CCC-1] Patterns [CCC-6] Structure and Function

[SEP-4] Analyzing and Interpreting Data [SEP-6] Constructing Explanations (for science) and Designing Solutions (for engineering)	PS4.B: Electromagnetic Radiation ETS1.B: Developing Possible Solutions	
<i>Highlighted California Environmental Principles & Concepts:</i> Principle I The continuation and health of individual human lives and of human communities and societies depend on the health of the natural systems that provide essential goods and ecosystem services. Principle III Natural systems proceed through cycles that humans depend upon, benefit from and can alter. Principle IV The exchange of matter between natural systems and human societies affects the long term functioning of both.		
CA CCSS Math Connections: 1.MD.2		
CA ELD Connections: ELD.PII.1.6		
CA CCSS ELA/Literacy Connections: W.1.1, 8; SL.1.4		

Grade One Instructional Segment 2: Animal Sounds
<i>Guiding questions</i> <i>How are parents and their children similar and different?</i> <i>How do animal parents and children interact to meet their needs?</i> <i>How do animals communicate and make sound?</i>
Students who demonstrate understanding can: 1-LS1-2. Read texts and use media to determine patterns in behavior of parents and offspring that help offspring survive. [Clarification Statement: Examples of patterns of behaviors could include the signals that offspring make (such as crying, cheeping, and other vocalizations) and the responses of the parents (such as feeding, comforting, and protecting the offspring).] 1-LS3-1. Make observations to construct an evidence-based account that young plants and animals are like, but not exactly like, their parents. [Clarification Statement: Examples of patterns could include features plants or animals share. Examples of observations could include leaves from the same kind of plant are the same shape but can differ in size; and, a particular breed of dog looks like its parents but is not exactly the same.] [Assessment Boundary: Assessment does not include inheritance or animals that undergo metamorphosis or hybrids.] (Revisited from IS1)

- 1-PS4-1.**Plan and conduct investigations to provide evidence that vibrating materials can make sound and that sound can make materials vibrate. [Clarification Statement: Examples of vibrating materials that make sound could include tuning forks and plucking a stretched string. Examples of how sound can make matter vibrate could include holding a piece of paper near a speaker making sound and holding an object near a vibrating tuning fork.]
- 1-PS4-4.**Use tools and materials to design and build a device that uses light or sound to solve the problem of communicating over a distance.* [Clarification Statement: Examples of devices could include a light source to send signals, paper cup and string “telephones,” and a pattern of drum beats.] [Assessment Boundary: Assessment does not include technological details for how communication devices work.]

* The performance expectations marked with an asterisk integrate traditional science content with engineering through a practice or disciplinary core idea.

The bundle of performance expectations above focuses on the following elements from the NRC document *A Framework for K–12 Science Education*:

Highlighted Science and Engineering Practices	Highlighted Disciplinary Core Ideas	Highlighted Crosscutting Concepts
[SEP-3] Planning and Carrying Out Investigations [SEP-6] Constructing Explanations (for science) and Designing Solutions (for engineering) [SEP-8] Obtaining, Evaluating, and Communicating Information	LS1.B: Growth and Development of Organisms LS3.A: Inheritance of Traits LS3.B: Variation of Traits PS4.A: Wave Properties PS4.C: Information Technologies and Instrumentation	[CCC-1] Patterns [CCC-2] Cause and Effect: Mechanism and Explanation

CA CCSS Math Connections: MP.5; 1.MD.1-2

CA CCSS ELA/Literacy Connections: RI.1.1, 3, 7, 9, 10; W.1.2, 8; SL.1.1, 2, 4, 5

CA ELD Connections: ELD.PI.1.1, 5, 6

Grade One Instructional Segment 3: Shadows and Light		
<i>Guiding questions</i> <i>What causes shadows?</i> <i>What happens when there is no light?</i>		
Students who demonstrate understanding can: 1-PS4-2. Make observations to construct an evidence-based account that objects can be seen only when illuminated. [Clarification Statement: Examples of observations could include those made in a completely dark room, a pinhole box, and a video of a cave explorer with a flashlight. Illumination could be from an external light source or by an object giving off its own light.] 1-PS4-3. Plan and conduct an investigation to determine the effect of placing objects made with different materials in the path of a beam of light. [Clarification Statement: Examples of materials could include those that are transparent (such as clear plastic), translucent (such as wax paper), opaque (such as cardboard), and reflective (such as a mirror).] [Assessment Boundary: Assessment does not include the speed of light.]		
The bundle of performance expectations above focuses on the following elements from the NRC document <i>A Framework for K–12 Science Education</i> :		
Highlighted Science and Engineering Practices [SEP-3] Planning and Carrying Out Investigations [SEP-6] Constructing Explanations (for science) and Designing Solutions (for engineering)	Highlighted Disciplinary Core Ideas PS4.B: Electromagnetic Radiation	Highlighted Crosscutting Concepts [CCC-2] Cause and Effect [CCC-4] Systems and system models
Highlighted California Environmental Principles & Concepts:		
Principle III Natural systems proceed through cycles that humans depend upon, benefit from and can alter.		
CA CCSS Math Connections:		
CA CCSS ELA/Literacy Connections: 1.W.3		

Grade One Instructional Segment 4: Patterns of Motion of Objects in the Sky		
<i>Guiding Questions</i> <i>What objects are in the sky and how do they seem to move?</i> <i>When will the sun set tomorrow?</i> <i>How does the moon's appearance change over each month?</i>		
Students who demonstrate understanding can: 1-ESS1-1. Use observations of the sun, moon, and stars to describe patterns that can be predicted. [Clarification Statement: Examples of patterns could include that the sun and moon appear to rise in one part of the sky, move across the sky, and set; and stars other than our sun are visible at night but not during the day.] [Assessment Boundary: Assessment of star patterns is limited to stars being seen at night and not during the day.] 1-ESS1-2. Make observations at different times of year to relate the amount of daylight to the time of year. [Clarification Statement: Emphasis is on relative comparisons of the amount of daylight in the winter to the amount in the spring or fall.] [Assessment Boundary: Assessment is limited to relative amounts of daylight, not quantifying the hours or time of daylight.]		
The bundle of performance expectations above focuses on the following elements from the NRC document <i>A Framework for K–12 Science Education</i>:		
Highlighted Science and Engineering Practices [SEP-3] Planning and Carrying Out Investigations [SEP-4] Analyzing and Interpreting Data	Highlighted Disciplinary Core Ideas ESS1.A: The Universe and its Stars ESS1.B: Earth and the Solar System	Highlighted Crosscutting Concepts [CCC-1] Patterns [CCC-2] Cause and Effect [CCC-4] System and System Models
<i>Highlighted California Environmental Principles & Concepts:</i> Principle III Natural systems proceed through cycles that humans depend upon, benefit from and can alter.		

CA CCSS Math Connections: 1.MD.3
CA CCSS ELA/Literacy Connections: RI.1.2, 4, 7, 10.a; L.1.4, 6; W.1,2
CA ELD Connections: ELD.PI.2.5, 6, 7

Grade Two

Grade Two Instructional Segment 1: Landscape Shapes		
Guiding Questions		
How can we describe the shape of land and water on Earth?		
Students who demonstrate understanding can:		
2-ESS2-2. Develop a model to represent the shapes and kinds of land and bodies of water in an area. [Assessment Boundary: Assessment does not include quantitative scaling in models.]		
2-ESS2-3. Obtain information to identify where water is found on Earth and that it can be solid or liquid.		
The bundle of performance expectations above focuses on the following elements from the NRC document <i>A Framework for K–12 Science Education</i> :		
Highlighted Science and Engineering Practices	Highlighted Disciplinary Core Ideas	Highlighted Crosscutting Concepts
[SEP-1] Asking questions (for science) and defining problems (for engineering) [SEP-2] Developing and Using Models [SEP-4] Analyzing and Interpreting Data [SEP-8] Obtaining, Evaluating, and Communicating Information	ESS2.B: Plate Tectonics and Large-Scale System Interactions ESS2.C: The Roles of Water in Earth's Surface Processes	[CCC-1] Patterns [CCC-3] Scale, Proportion, and Quantity [CCC-4] System and System Models
CA CCSS Math Connections: 2.MD.10, 2.G.1-2		
CA CCSS ELA/Literacy Connections: W.2.2, W.2.7-8, W.2.10, SL.2.2, SL.2.4e		
CA ELD Connections: ELD.PI.2.6, ELD.PI.2.10		

Grade Two Instructional Segment 2: Landscape Materials

Guiding Questions:

*How can we describe different materials?
How are materials similar and different from one another?
What sort of changes can happen to materials?
How do the properties of the materials relate to their use?*

Students who demonstrate understanding can:

- 2-PS1-1. Plan and conduct an investigation to describe and classify different kinds of materials by their observable properties.** [Clarification Statement: Observations could include color, texture, hardness, and flexibility. Patterns could include the similar properties that different materials share.]
- 2-PS1-2. Analyze data obtained from testing different materials to determine which materials have the properties that are best suited for an intended purpose.*** [Clarification Statement: Examples of properties could include, strength, flexibility, hardness, texture, and absorbency.] [Assessment Boundary: Assessment of quantitative measurements is limited to length.]
- 2-PS1-3. Make observations to construct an evidence-based account of how an object made of a small set of pieces can be disassembled and made into a new object.** [Clarification Statement: Examples of pieces could include blocks, building bricks, or other assorted small objects.]
- 2-PS1-4. Construct an argument with evidence that some changes in matter, caused by mixing, heating, or cooling can be reversed and some cannot.** [Clarification Statement: Examples of reversible changes could include materials such as water and butter at different temperatures. Examples of irreversible changes could include cooking an egg, freezing a plant leaf, and heating paper.]
- K–2-ETS1-3. Analyze data from tests of two objects designed to solve the same problem to compare the strengths and weaknesses of how each performs.**

*The performance expectations marked with an asterisk integrate traditional science content with engineering through a practice or disciplinary core idea.

The bundle of performance expectations above focuses on the following elements from the NRC document *A Framework for K–12 Science Education*:

Highlighted Science and Engineering Practices	Highlighted Disciplinary Core Ideas	Highlighted Crosscutting Concepts
[SEP-3] Planning and Carrying Out Investigations [SEP-4] Analyzing and Interpreting Data [SEP-6] Constructing Explanations (for science) and Designing Solutions (for engineering) [SEP-7] Engaging in argument from evidence	PS1.A: Structure and Properties of Matter PS1.B: Chemical Reactions	[CCC-1] Patterns [CCC-2 Cause and Effect: Mechanism and Explanation [CCC-5] Energy and Matter: Flows, Cycles, and Conservation
<i>Highlighted California Environmental Principles & Concepts:</i> Principle I The continuation and health of individual human lives and of human communities and societies depend on the health of the natural systems that provide essential goods and ecosystem services. Principle II The long-term functioning and health of terrestrial, freshwater, coastal and marine ecosystems are influenced by their relationships with human societies. Principle III Natural systems proceed through cycles that humans depend upon, benefit from and can alter. Principle IV The exchange of matter between natural systems and human societies affects the long term functioning of both. Principle V Decisions affecting resources and natural systems are complex and involve many factors.		
<i>CA CCSS ELA/Literacy Connections:</i> RI.2.3, 8; W.2.1, 8; L.2.6		
<i>CA ELD Connections:</i> ELD.PI.2.6, 10, 11, ELD.PII.2.5		

Grade Two Instructional Segment 3: Landscape Changes		
<i>Guiding Questions</i> <i>What evidence do natural processes leave behind as they shape the Earth?</i> <i>How do the material properties of rocks affect what happens to them in landscapes?</i>		
Students who demonstrate understanding can: 2-ESS1-1. Use information from several sources to provide evidence that Earth events can occur quickly or slowly. <i>[Clarification Statement: Examples of events and timescales could include volcanic explosions and earthquakes, which happen quickly, and erosion of rocks, which occurs slowly.] [Assessment Boundary: Assessment does not include quantitative measurements of timescales.]</i> 2-ESS2-1. Compare multiple solutions designed to slow or prevent wind or water from changing the shape of the land.* <i>[Clarification Statement: Examples of solutions could include different designs of dikes and windbreaks to hold back wind and water and different designs for using shrubs, grass, and trees to hold back the land.]</i> K–2-ETS1-2. Develop a simple sketch, drawing, or physical model to illustrate how the shape of an object helps it function as needed to solve a given problem.		
* The performance expectations marked with an asterisk integrate traditional science content with engineering through a practice or disciplinary core idea.		
The bundle of performance expectations above focuses on the following elements from the NRC document <i>A Framework for K–12 Science Education</i> :		
Highlighted Science and Engineering Practices	Highlighted Disciplinary Core Ideas	Highlighted Crosscutting Concepts
[SEP-2] Developing and Using Models [SEP-6] Constructing Explanations (for science) and Designing Solutions (for engineering) [SEP-8] Obtaining, Evaluating, and Communicating Information	ESS1.C: The History of Planet Earth ESS2.A: Earth Materials and Systems ETS1.B: Developing Possible Solutions ETS1.C: Optimizing the Design Solution	[CCC-7] Stability and Change

CA CCSS ELA/Literacy Connections: RI.2.1, 3

CA ELD Connections: ELD.PI.2.6, 11

Grade Two Instructional Segment 4: Biodiversity in Landscapes

Guiding Questions:

How can we determine what plants need to grow?

How do plants depend on animals?

How many types of living things live in a place? How can we tell?

Students who demonstrate understanding can:

2-LS2-1. Plan and conduct an investigation to determine if plants need sunlight and water to grow. [Assessment Boundary: Assessment is limited to testing one variable at a time.]

2-LS2-2. Develop a simple model that mimics the function of an animal in dispersing seeds or pollinating plants.*

2-LS4-1. Make observations of plants and animals to compare the diversity of life in different habitats. [Clarification Statement: Emphasis is on the diversity of living things in each of a variety of different habitats.] [Assessment Boundary: Assessment does not include specific animal and plant names in specific habitats.]

*The performance expectations marked with an asterisk integrate traditional science content with engineering through a practice or disciplinary core idea.

The bundle of performance expectations above focuses on the following elements from the NRC document *A Framework for K–12 Science Education*:

Highlighted Science and Engineering Practices	Highlighted Disciplinary Core Ideas	Highlighted Crosscutting Concepts
[SEP-2] Developing and Using Models	LS2.A: Interdependent Relationships in Ecosystems	[CCC-1] Patterns
[SEP-3] Planning and Carrying Out Investigations	LS4.D: Biodiversity and Humans	[CCC-2] Cause and Effect
	ETS1.B: Developing Possible Solutions	[CCC-6] Structure and Function

Highlighted California Environmental Principles & Concepts:

Principle II The long-term functioning and health of terrestrial, freshwater,

coastal and marine ecosystems are influenced by their relationships with human societies.
CA CCSS ELA/Literacy Connections: W.2.3, 4, 7, 8, 10
CA ELD Connections: ELD.PI.2.2, 6, ELD.PII.2.6

Grade Three

Grade Three Instructional Segment 1: Playground Forces
<i>Guiding Questions</i> <i>What happens when several different forces push or pull an object at once?</i> <i>How can an object be pushed or pulled but not move?</i> <i>What do we need to know to predict the motion of objects?</i> <i>How can some objects push or pull one another without even touching?</i>
Students who demonstrate understanding can: 3-PS2-1 Plan and conduct an investigation to provide evidence of the effects of balanced and unbalanced forces on the motion of an object. [Clarification Statement: Examples could include an unbalanced force on one side of a ball can make it start moving; and balanced forces pushing on a box from both sides will not produce any motion at all.] [Assessment Boundary: Assessment is limited to one variable at a time: number, size, or direction of forces. Assessment does not include quantitative force size, only qualitative and relative. Assessment is limited to gravity being addressed as a force that pulls objects down.] 3-PS2-2 Make observations and/or measurements of an object's motion to provide evidence that a pattern can be used to predict future motion. [Clarification Statement: Examples of motion with a predictable pattern could include a child swinging in a swing, a ball rolling back and forth in a bowl, and two children on a see-saw.] [Assessment Boundary: Assessment does not include technical terms such as period and frequency.] 3-PS2-3 Ask questions to determine cause and effect relationships of electric or magnetic interactions between two objects not in contact with each other. [Clarification Statement: Examples of an electric force could include the force on hair from an electrically charged balloon and the electrical forces between a charged rod and pieces of paper; examples of a magnetic force could include the force between two permanent magnets, the force between an electromagnet and steel paperclips, and the force exerted by one magnet versus the force exerted by two magnets. Examples of cause and effect relationships could include how the distance between objects affects strength of the force and how the orientation of

magnets affects the direction of the magnetic force.] [Assessment Boundary: Assessment is limited to forces produced by objects that can be manipulated by students, and electrical interactions are limited to static electricity.]

3-PS2-4 Define a simple design problem that can be solved by applying ideas about magnets.* [Clarification Statement: Examples of problems could include constructing a latch to keep a door shut and creating a device to keep two moving objects from touching each other.]

3-5-ETS1-1 Define a simple design problem reflecting a need or a want that includes specified criteria for success and constraints on materials, time, or cost. [This performance expectation does not have a clarification statement or an assessment boundary.]

3-5-ETS1-2. Generate and compare multiple possible solutions to a problem based on how well each is likely to meet the criteria and constraints of the problem. [This performance expectation does not have a clarification statement or an assessment boundary.]

The bundle of performance expectations above focuses on the following elements from the NRC document *A Framework for K–12 Science Education*:

Highlighted Science and Engineering Practices	Highlighted Disciplinary Core Ideas	Highlighted Crosscutting Concepts
[SEP-1] Asking Questions and Defining Problems [SEP-3] Planning and Carrying Out Investigations [SEP-6] Constructing Explanations (for science) and Designing Solutions (for engineering)	PS2.A: Forces and Motion PS2.B: Types of Interactions ETS1.A: Defining and Delimiting Engineering Problems ETS1.B: Developing Possible Solutions	[CCC-1] Patterns [CCC-2] Cause and effect
CA CCSS Math Connections: 3.OA.1-7, MP 5, 6		
CA CCSS ELA/Literacy Connections: RI.3.4; L.3.4, 5		
CA ELD Connections: ELD.PI.3.1, 5, 12		

Grade Three Instructional Segment 2: Life Cycles for Survival

Guiding Questions

What is the advantage of having a complicated lifecycle of growth and development?

How do animals' lifecycles help them survive?

How similar are animals and plants to their siblings and their parents?

How does being similar to parents help an animal survive?

Why do some animals live alone while others live in large groups?

Students who demonstrate understanding can:

3-LS1-1 Develop models to describe that organisms have unique and diverse life cycles but all have in common birth, growth, reproduction, and death. [Clarification Statement: Changes organisms go through during their life form a pattern.] [Assessment Boundary: Assessment of plant life cycles is limited to those of flowering plants. Assessment does not include details of human reproduction.]

3-LS2-1 Construct an argument that some animals form groups that help members survive.

3-LS3-1 Analyze and interpret data to provide evidence that plants and animals have traits inherited from parents and that variation of these traits exists in a group of similar organisms. [Clarification Statement: Patterns are the similarities and differences in traits shared between offspring and their parents or among siblings. Emphasis is on organisms other than humans.] [Assessment Boundary: Assessment does not include genetic mechanisms of inheritance and prediction of traits. Assessment is limited to non-human examples.]

3-LS4-2 Use evidence to construct an explanation for how the variations in characteristics among individuals of the same species may provide advantages in surviving, finding mates, and reproducing. [Clarification Statement: Examples of cause and effect relationships could be plants that have larger thorns than other plants may be less likely to be eaten by predators and animals that have better camouflage coloration than other animals may be more likely to survive and therefore more likely to leave offspring.]

The bundle of performance expectations above focuses on the following elements from the NRC document *A Framework for K–12 Science Education*:

Highlighted Science and Engineering Practices	Highlighted Disciplinary Core Ideas	Highlighted Crosscutting Concepts
[SEP-2] Developing and Using Models [SEP-4] Analyzing and Interpreting Data [SEP-6] Constructing Explanations (for science) and designing solutions (for engineering). [SEP-7] Engaging in Argument from Evidence	LS1.B: Growth and Development of Organisms LS2.D: Social Interactions and Group Behavior LS3.A: Inheritance of Traits LS3.B: Variation of Traits LS4.B: Natural Selection	[CCC-1] Patterns [CCC-2] Cause and effect [CCC-3] Scale, Proportion and quantity
<i>Highlighted California Environmental Principles & Concepts:</i> Principle I The continuation and health of individual human lives and of human communities and societies depend on the health of the natural systems that provide essential goods and ecosystem services. Principle II The long-term functioning and health of terrestrial, freshwater, coastal and marine ecosystems are influenced by their relationships with human societies. Principle III Natural systems proceed through cycles that humans depend upon, benefit from and can alter. Principle IV The exchange of matter between natural systems and human societies affects the long term functioning of both.		
<i>CA CCSS Math Connections:</i> 3.MD.4		
<i>CA ELD Connections:</i> ELD.3.PI.9		
<i>CA CCSS ELA/Literacy Connections:</i> RI.3.7; SL.3.1, 2, 3		

Grade Three Instructional Segment 3: Surviving in Different Environments
<i>Guiding questions</i> <i>How does the environment affect living organisms?</i> <i>How do organisms' traits help them survive in different environments?</i> <i>What happens to organisms when the environment changes?</i>
Students who demonstrate understanding can: 3-LS3-2 Use evidence to support the explanation that traits can be influenced by the environment. [Clarification Statement: Examples of the environment affecting a trait could include normally tall plants grown with insufficient water are stunted, and a pet dog that is given too much food and little exercise may become overweight.] 3-LS4-1 Analyze and interpret data from fossils to provide evidence of the organism and the environments in which they lived long ago. [Clarification Statement: Examples of data could include type, size, and distributions of fossil organisms. Examples of fossils and environments could include marine fossils found on dry land, tropical plant fossils found in Arctic areas, and fossils of extinct organisms.] [Assessment Boundary: Assessment does not include identification of specific fossils or present plants and animals. Assessment is limited to major fossil types and relative ages.] 3-LS4-3 Construct an argument with evidence that in a particular habitat some organisms can survive well, some survive less well, and some cannot survive at all. [Clarification Statement: Examples of evidence could include needs and characteristics of the organisms and habitats involved. The organisms and their habitat make up a system in which the parts depend on each other.] 3-LS4-4 Make a claim about the merit of a solution to a problem caused when the environment changes and the types of plants and animals that live there may change.* [Clarification Statement: Examples of environmental changes could include changes in land characteristics, water distribution, temperature, food, and other organisms.] [Assessment Boundary: Assessment is limited to a single environmental change. Assessment does not include the greenhouse effect or climate change.] 3-5-ETS1-1. Define a simple design problem reflecting a need or a want that includes specified criteria for success and constraints on materials, time, or cost. 3-5-ETS1-2. Generate and compare multiple possible solutions to a problem based on how well each is likely to meet the criteria and constraints of the problem.

*The performance expectations marked with an asterisk integrate traditional science content with engineering through a practice or disciplinary core idea.

The bundle of performance expectations above focuses on the following elements from the NRC document A Framework for K–12 Science Education:

Highlighted Science and Engineering Practices	Highlighted Disciplinary Core Ideas	Highlighted Crosscutting Concepts
[SEP-1] Asking Questions and Defining Problems [SEP-4] Analyzing and Interpreting Data [SEP-6] Constructing Explanations (for science) and Designing Solutions (for engineering) [SEP-7] Engaging in Argument from Evidence	LS3.A: Inheritance of Traits LS3.B: Variation of Traits LS2.C: Ecosystem Dynamics, Functioning, and Resilience LS4.A: Evidence of Common Ancestry and Diversity LS4.C: Adaptation LS4.D: Biodiversity and Humans ETS1.A: Defining and Delimiting Engineering Problems ETS1.B: Developing Possible Solutions	[CCC-1] Patterns [CCC-2] Cause and effect [CCC-3] Scale, Proportion, and Quantity [CCC-4] Systems and System Models

Highlighted California Environmental Principles & Concepts:

Principle II The long-term functioning and health of terrestrial, freshwater, coastal and marine ecosystems are influenced by their relationships with human societies.

Principle V Decisions affecting resources and natural systems are based on a wide range of considerations and decision-making processes.

CA CCSC Math Connections: MP.2, MP.5; 3.MD.3

CA CCSS ELA/Literacy Connections: W.3.1, 7; RI.3.1, 3, 5, 7; SL.3.1

CA ELD Connections: ELD.PI.3.1, 10, 11

Grade Three Instructional Segment 4: Weather Impacts

<i>Guiding Questions</i> <i>What is typical weather in my local region?</i> <i>How does it compare to other areas of California and the world?</i> <i>What weather patterns are common for different seasons?</i> <i>What weather-related hazards are in my region?</i> <i>How can we reduce weather-related hazards?</i>		
Students who demonstrate understanding can:		
3-ESS2-1. Represent data in tables and graphical displays to describe typical weather conditions expected during a particular season. [Clarification Statement: Examples of data at this grade level could include average temperature, precipitation, and wind direction.] [Assessment Boundary: Assessment of graphical displays is limited to pictographs and bar graphs. Assessment does not include climate change.] 3-ESS2-2. Obtain and combine information to describe climates in different regions of the world. 3-ESS3-1. Make a claim about the merit of a design solution that reduces the impacts of a weather-related hazard.* [Clarification Statement: Examples of design solutions to weather-related hazards could include barriers to prevent flooding, wind-resistant roofs, and lightning rods.] 3-5-ETS1-1. Define a simple design problem reflecting a need or a want that includes specified criteria for success and constraints on materials, time, or cost. 3-5-ETS1-2. Generate and compare multiple possible solutions to a problem based on how well each is likely to meet the criteria and constraints of the problem		
*The performance expectations marked with an asterisk integrate traditional science content with engineering through a practice or disciplinary core idea.		
The bundle of performance expectations above focuses on the following elements from the NRC document <i>A Framework for K–12 Science Education</i> :		
Highlighted Science and Engineering Practices [SEP-4] Analyzing and Interpreting Data [SEP-6] Constructing Explanations (for science) and	Highlighted Disciplinary Core Ideas ESS2.D Weather and Climate ESS3.B: Natural Hazards ETS1.A: Defining and Delimiting Engineering Problems ETS1.B: Developing	Highlighted Crosscutting Concepts [CCC-1] Patterns [CCC-2] Cause and effect [CCC-6] Structure and Function

Designing Solutions (for engineering) [SEP-7] Engaging in Argument from Evidence [SEP-8] Obtaining, Evaluating, and Communicating Information	possible solutions	
<i>Highlighted California Environmental Principles & Concepts:</i> Principle II The long-term functioning and health of terrestrial, freshwater, coastal and marine ecosystems are influenced by their relationships with human societies. Principle III Natural systems proceed through cycles that humans depend upon, benefit from and can alter. Principle IV The exchange of matter between natural systems and human societies affects the long term functioning of both. Principle V Decisions affecting resources and natural systems are complex and involve many factors.		
<i>CA CCSS Math Connections:</i> MP.5; 3.MD.3, 4		
<i>CA CCSS ELA/Literacy Connections:</i> W3.1B, W3.8, SL.3.1, SL.3.2, SL.3.3, SL.3.4, RI.3.1, RI.3.3, RI.3.4, RI.3.5, RI.3.7		
<i>CA ELD Connections:</i> 3.P1.A.1, 3.P1.A.2, 3.P1.B.5, 3.P1.C.9		

Grade Four

Grade Four Instructional Segment 1: Car Crashes		
<i>Guiding Questions</i> <i>Why do car crashes cause so much damage?</i> <i>What happens to energy when objects collide?</i>		
Students who demonstrate understanding can: 4-PS3-1. Use evidence to construct an explanation relating the speed of an object to the energy of that object. <i>**Clarification Statement: Examples of evidence relating speed and energy could include change of shape on impact or other results of collisions.] [Assessment Boundary: Assessment does not include quantitative measures of changes in the speed of an object or on any precise or quantitative definition of energy.]</i> 4-PS3-3. Ask questions and predict outcomes about the changes in energy that occur when objects collide. <i>[Clarification Statement: Emphasis is on the change in the energy due to the change in speed, not on the forces, as objects interact.] [Assessment Boundary: Assessment does not include quantitative measurements of energy.]</i> <i>**California clarification statements, marked with double asterisks, were incorporated by the California Science Expert Review Panel</i>		
The bundle of performance expectations above focuses on the following elements from the NRC document <i>A Framework for K–12 Science Education</i>:		
Highlighted Science and Engineering Practices [SEP-1] Asking Questions and Defining Problems [SEP-6] Constructing Explanations (for science) and Designing Solutions (for engineering)	Highlighted Disciplinary Core Ideas PS3.A: Definitions of Energy PS3.B: Conservation of Energy and Energy Transfer PS3.C: Relationship Between Energy and Forces	Highlighted Crosscutting Concepts [CCC-2] Cause and effect [CCC-5] Energy and matter: Flows, Cycles, and Conservation
<i>ELA/Literacy Connections: SL.3.3; W.3.7, 8; RI.3.3,; L.3.6</i>		
<i>ELD Connections: ELD.PI.3.1, 3, 12</i>		

Grade Four – Instructional Segment 2: Renewable Energy		
<i>Guiding Questions:</i> <i>How do we get electricity and fuel to run cars and power electronic devices?</i> <i>How does human use of natural resources affect the environment?</i>		
Students who demonstrate understanding can: 4-ESS3-1. Obtain and combine information to describe that energy and fuels are derived from natural resources and their uses affect the environment. [Clarification Statement: Examples of renewable energy resources could include wind energy, water behind dams, and sunlight; non-renewable energy resources are fossil fuels and fissile materials. Examples of environmental effects could include loss of habitat due to dams, loss of habitat due to surface mining, and air pollution from burning of fossil fuels.] 4-PS3-2. Make observations to provide evidence that energy can be transferred from place to place by sound, light, heat, and electric currents. [Assessment Boundary: Assessment does not include quantitative measurements of energy.] 4-PS3-4. Apply scientific ideas to design, test, and refine a device that converts energy from one form to another.* [Clarification Statement: Examples of devices could include electric circuits that convert electrical energy into motion energy of a vehicle, light, or sound; and, a passive solar heater that converts light into heat. Examples of constraints could include the materials, cost, or time to design the device.] [Assessment Boundary: Devices should be limited to those that convert motion energy to electric energy or use stored energy to cause motion or produce light or sound.] *The performance expectations marked with an asterisk integrate traditional science content with engineering through a practice or disciplinary core idea.		
The bundle of performance expectations above focuses on the following elements from the NRC document <i>A Framework for K–12 Science Education</i>:		
Highlighted Science and Engineering Practices [SEP-2] Developing and Using Models	Highlighted Disciplinary Core Ideas ESS3.A: Natural Resources PS3.A: Definitions of Energy	Highlighted Crosscutting Concepts [CCC-2] Cause and Effect: Mechanism and explanation [CCC-4] Systems and System Models

[SEP-3] Planning and Carrying Out Investigations [SEP-6] Constructing Explanations (for science) and Designing Solutions (for engineering) [SEP-8] Obtaining, Evaluating, and Communicating Information	PS3.B: Conservation of Energy and Energy Transfer PS3.D: Energy in Chemical Processes and Everyday Life	[CCC-5] Energy and matter: Flows, cycles, and conservation
<i>Highlighted California Environmental Principles & Concepts:</i> Principle I The continuation and health of individual human lives and of human communities and societies depend on the health of the natural systems that provide essential goods and ecosystem services. Principle II The long-term functioning and health of terrestrial, freshwater, coastal and marine ecosystems are influenced by their relationships with human societies. Principle IV The exchange of matter between natural systems and human societies affects the long term functioning of both. Principle V Decisions affecting resources and natural systems are based on a wide range of considerations and decision-making processes.		
CA CCSS ELA/Literacy Connections: RI.4.3, 5; W.4.1, 7		
CA ELD Connections: ELD.PI.4.2, 10a, 11		

Grade Four Instructional Segment 3: Sculpting Landscapes
<i>Guiding Questions:</i> <i>How do water, ice, wind, and vegetation sculpt landscapes?</i> <i>What factors affect how quickly landscapes change?</i> <i>How are landscape changes recorded by layers of rocks and fossils?</i> <i>How can people minimize the effects of changing landscape on property while still protecting the environment?</i>
Students who demonstrate understanding can:

- 4-ESS1-1. Identify evidence from patterns in rock formations and fossils in rock formations and fossils in rock layers for changes in a landscape over time to support an explanation for changes in a landscape over time.** [Clarification Statement: Examples of evidence from patterns could include rock layers with shell fossils above rock layers with plant fossils and no shells, indicating a change from land to water over time; and, a canyon with different rock layers in the walls and a river in the bottom, indicating that over time a river cut through the rock.] [Assessment Boundary: Assessment does not include specific knowledge of the mechanism of rock formation or memorization of specific rock formations and layers. Assessment is limited to relative time.]
- 4-ESS2-1. Make observations and/or measurements to provide evidence of the effects of weathering or the rate of erosion by water, ice, wind, or vegetation.** [Clarification Statement: Examples of variables to test could include angle of slope in the downhill movement of water, amount of vegetation, speed of wind, relative rate of deposition, cycles of freezing and thawing of water, cycles of heating and cooling, and volume of water flow.] [Assessment Boundary: Assessment is limited to a single form of weathering or erosion.]
- 4-ESS2-2. Analyze and interpret data from maps to describe patterns of Earth's features.** [Clarification Statement: Maps can include topographic maps of Earth's land and ocean floor, as well as maps of the locations of mountains, continental boundaries, volcanoes, and earthquakes.] (Introduced. Fully assessed in IS4)
- 4-ESS3-2. Generate and compare multiple solutions to reduce the impacts of natural Earth processes on humans.*** [Clarification Statement: Examples of solutions could include designing an earthquake resistant building and improving monitoring of volcanic activity.] [Assessment Boundary: Assessment is limited to earthquakes, floods, tsunamis, and volcanic eruptions.] (Introduced. Fully Assessed in IS4)
- 3-5-ETS1-2. Generate and compare multiple possible solutions to a problem based on how well each is likely to meet the criteria and constraints of the problem.**
- 3-5-ETS1-3. Plan and carry out fair tests in which variables are controlled and failure points are considered to identify aspects of a model or prototype that can be improved.**

*The performance expectations marked with an asterisk integrate traditional science content with engineering through a practice or disciplinary core idea.

The bundle of performance expectations above focuses on the following elements from the NRC document *A Framework for K–12 Science Education*:

Highlighted Science and Engineering Practices	Highlighted Disciplinary Core Ideas	Highlighted Crosscutting Concepts
[SEP-2] Developing and Using Models	ESS1.C: The History of Planet Earth	[CCC-1] Patterns
[SEP-3] Planning and Carrying Out Investigations	ESS2.A: Earth Materials and Systems ESS2.B: Plate Tectonics and Large-Scale System Interactions	[CCC-2] Cause and Effect: Mechanism and explanation
[SEP-4] Analyzing and Interpreting Data	ESS2.E: Biogeology	
[SEP-6] Constructing Explanations (for science) and Designing Solutions (for engineering)	ESS3.B: Natural Hazards ETS1.A: Defining Engineering Problems ETS1.B: Developing Possible Solutions ETS1.C: Optimizing the Design Solution	
<i>Highlighted California Environmental Principles & Concepts:</i> Principle III Natural systems proceed through cycles that humans depend upon, benefit from and can alter. Principle V Decisions affecting resources and natural systems are complex and involve many factors.		
CA CCSS ELA/Literacy Connections: W.4.3, 4, 7, 8, 10; L.4.1, 2, 5, 6		
CA ELD Connections: ELD.PI.4.6, 10.b		

Grade Four Instructional Segment 4: Earthquake Engineering
<i>Guiding Questions:</i> <i>How have earthquakes shaped California's history?</i> <i>How can we describe the amount of shaking in earthquakes?</i> <i>How can we minimize the damage from earthquakes?</i>
Students who demonstrate understanding can: 4-ESS2-2. Analyze and interpret data from maps to describe patterns of Earth's features. [Clarification Statement: Maps can include topographic maps of Earth's land and ocean floor, as well as maps of the locations of mountains, continental boundaries, volcanoes, and earthquakes.] 4-ESS3-2. Generate and compare multiple solutions to reduce the impacts of natural Earth processes on humans.* [Clarification Statement: Examples of solutions could include designing an earthquake resistant building and improving monitoring of volcanic activity.] [Assessment Boundary: Assessment is limited to earthquakes, floods, tsunamis, and volcanic eruptions.] 4-PS4-1. Develop a model of waves to describe patterns in terms of amplitude and wavelength and that waves can cause objects to move. [Clarification Statement: Examples of models could include diagrams, analogies, and physical models using wire to illustrate wavelength and amplitude of waves.] [Assessment Boundary: Assessment does not include interference effects, electromagnetic waves, non-periodic waves, or quantitative models of amplitude and wavelength.] 3–5-ETS1-1. Define a simple design problem reflecting a need or a want that includes specified criteria for success and constraints on materials, time, or cost. 3–5-ETS1-2. Generate and compare multiple possible solutions to a problem based on how well each is likely to meet the criteria and constraints of the problem. 3–5-ETS1-3. Plan and carry out fair tests in which variables are controlled and failure points are considered to identify aspects of a model or prototype that can be improved. *The performance expectations marked with an asterisk integrate traditional science content with engineering through a practice or disciplinary core idea.
The bundle of performance expectations above focuses on the following elements from the NRC document <i>A Framework for K–12 Science Education</i>:

Highlighted Science and Engineering Practices	Highlighted Disciplinary Core Ideas	Highlighted Crosscutting Concepts
[SEP-1] Asking Questions and Defining Problems [SEP-2] Developing and Using Models [SEP-3] Planning and Carrying Out Investigations [SEP-4] Analyzing and interpreting data [SEP-6] Constructing Explanations (for science) and Designing Solutions (for engineering)	ESS2.B: Plate Tectonics and Large-Scale System Interactions ESS3.B: Natural Hazards PS4.A: Wave Properties ETS1.A: Defining Engineering Problems ETS1.B: Developing Possible Solutions ETS1.C: Optimizing the Design Solution	[CCC-1] Patterns [CCC-2] Cause and Effect: Mechanism and explanation [CCC-6] Structure and Function
Highlighted California Environmental Principles & Concepts: Principle III Natural systems proceed through cycles that humans depend upon, benefit from and can alter. Principle IV The exchange of matter between natural systems and human societies affects the long term functioning of both. Principle V Decisions affecting resources and natural systems are based on a wide range of considerations and decision-making processes.		
<i>CA CCSS Math Connections:</i> 3.MD.7b; 4.NF.7, 5.G.1		
<i>CA CCSS ELA/Literacy Connections:</i> SL.4.2; W.4.8		
<i>CA ELD Connections:</i> ELD.PI.4.6, 11		

Grade Four Instructional Segment 5: Animal Senses

Guiding Questions:

How do the internal and external structures of animals help them sense and interpret their environment?

How do senses help animals survive, grow, and reproduce?

What role does light play in how we see?

How do humans encode information and transmit it across the world?

Students who demonstrate understanding can:

- 4-LS1-1. Construct an argument that plants and animals have internal and external structures that function to support survival, growth, behavior, and reproduction.** [Clarification Statement: Examples of structures could include thorns, stems, roots, colored petals, heart, stomach, lung, brain, and skin. **Each structure has specific functions within its associated system.] [Assessment Boundary: Assessment is limited to macroscopic structures within plant and animal systems.]
- 4-LS1-2. Use a model to describe that animals receive different types of information through their senses, process the information in their brain, and respond to the information in different ways.** [Clarification Statement: Emphasis is on systems of information transfer.] [Assessment Boundary: Assessment does not include the mechanisms by which the brain stores and recalls information or the mechanisms of how sensory receptors function.]
- 4-PS3-2. Make observations to provide evidence that energy can be transferred from place to place by sound, light, heat, and electric currents.** [Assessment Boundary: Assessment does not include quantitative measurements of energy.]
- 4-PS4-2. Develop a model to describe that light reflecting from objects and entering the eye allows objects to be seen.** [Assessment Boundary: Assessment does not include knowledge of specific colors reflected and seen, the cellular mechanisms of vision, or how the retina works.]
- 4-PS4-3. Generate and compare multiple solutions that use patterns to transfer information.*** [Clarification Statement: Examples of solutions could include drums sending coded information through sound waves, using a grid of 1s and 0s representing black and white to send information about a picture, and using Morse code to send text.]

*The performance expectations marked with an asterisk integrate traditional science content with engineering through a practice or disciplinary core idea.

The bundle of performance expectations above focuses on the following elements from the NRC document *A Framework for K–12 Science Education*:

Highlighted Science and Engineering Practices	Highlighted Disciplinary Core Ideas	Highlighted Crosscutting Concepts
[SEP-2] Developing and Using Models [SEP-3] Planning and Carrying Out Investigations [SEP-6] Constructing Explanations (for science) and Designing Solutions (for engineering) [SEP-7] Engaging in Argument from Evidence	LS1.A: Structure and Function LS1.D: Information Processing PS3.A: Definitions of Energy PS3.B: Conservation of Energy and Energy Transfer PS4.B: Electromagnetic Radiation PS4.C: Information Technologies and Instrumentation ETS1.C: Optimizing the Design Solution	[CCC-1] Patterns [CCC-2] Cause and effect [CCC-4] Systems and System Models [CCC-5] Energy and matter: Flows, Cycles, and Conservation [CCC-6] Structure and Function
<i>Highlighted California Environmental Principles & Concepts:</i> Principle I The continuation and health of individual human lives and of human communities and societies depend on the health of the natural systems that provide essential goods and ecosystem services. Principle II The long-term functioning and health of terrestrial, freshwater, coastal and marine ecosystems are influenced by their relationships with human societies.		
CA CCSS Math Connections: 4.OA.5; 4.MD.5, 6; 4.G.3; MP. 2, 4, 5, 6		
CA ELD Connections: ELD.PI.4.10		
CA CCSS ELA/Literacy Connections: W.4.1; RI.4.3, 7		

Grade Five

Grade Five Instructional Segment 1: What is Matter Made of?
<i>Guiding Questions:</i> <i>What causes different materials to have different properties?</i> <i>How do materials change when they dissolve, evaporate, melt, or mix together?</i> <i>What are the differences between solids, liquids, and gases?</i>
Students who demonstrate understanding can: 5-PS1-1. Develop a model to describe that matter is made of particles too small to be seen. [Clarification Statement: Examples of evidence supporting a model could include adding air to expand a basketball, compressing air in a syringe, dissolving sugar in water, and evaporating salt water.] [Assessment Boundary: Assessment does not include the atomic-scale mechanism of evaporation and condensation or defining the unseen particles.] 5-PS1-2. Measure and graph quantities to provide evidence that regardless of the type of change that occurs when heating, cooling, or mixing substances, the total weight of matter is conserved. [Clarification Statement: Examples of reactions or changes could include phase changes, dissolving, and mixing that forms new substances.] [Assessment Boundary: Assessment does not include distinguishing mass and weight.] 5-PS1-3. Make observations and measurements to identify materials based on their properties. [Clarification Statement: Examples of materials to be identified could include baking soda and other powders, metals, minerals, and liquids. Examples of properties could include color, hardness, reflectivity, electrical conductivity, thermal conductivity, response to magnetic forces, and solubility; density is not intended as an identifiable property.] [Assessment Boundary: Assessment does not include density or distinguishing mass and weight.] 5-PS1-4. Conduct an investigation to determine whether the mixing of two or more substances results in new substances. [**Clarification Statement: Examples of combinations that do not produce new substances could include sand and water. Examples of combinations that do produce new substances could include baking soda and vinegar or milk and vinegar.] 3–5-ETS1-3 Plan and carry out fair tests in which variables are controlled and failure points are considered to identify aspects of a model or prototype that can be improved. [Clarification Statement: Examples of models could include diagrams, and flow charts.]

****California clarification statements, marked with double asterisks, were incorporated by the California Science Expert Review Panel.**

The bundle of performance expectations above focuses on the following elements from the NRC document *A Framework for K–12 Science Education*:

Highlighted Science and Engineering Practices	Highlighted Disciplinary Core Ideas	Highlighted Crosscutting Concepts
[SEP-2] Developing and Using Models [SEP-3] Planning and Carrying Out Investigations [SEP-5] Using Mathematics and Computational Thinking [SEP-6] Constructing Explanations (for science) and Designing Solutions (for engineering) [SEP-7] Engaging in Argument from Evidence [SEP-8] Obtaining, Evaluating, and Communicating Information	PS1.A: Structure and Properties of Matter PS1.B: Chemical Reactions ETS1.B: Developing Possible Solutions ETS1.C: Optimizing the Design Solution	[CCC-2] Cause and effect [CCC-3] Scale, Proportion, and Quantity [CCC-4] Systems and System Models [CCC-5] Energy and matter: Flows, cycles and conservation

Highlighted California Environmental Principles & Concepts:

Principle IV The exchange of matter between natural systems and human societies affects the long term functioning of both.
CA CCSS Math Connections: 5.MD.3a,b; 5.MD.4
CA CCSS ELA/Literacy Connections: SL.5.1, 4, 5
CA ELD Connections: ELD.PI.5.1, 6

Grade Five Instructional Segment 2: From Matter to Organisms
Guiding Questions: <i>What matter do plants need to grow?</i> <i>How does matter move within an ecosystem?</i> <i>How does energy move within an ecosystem?</i>
Students who demonstrate understanding can: 5-LS1-1 Support an argument that plants get the materials they need for growth chiefly from air and water. [Clarification Statement: Emphasis is on the idea that plant matter comes mostly from air and water, not from the soil.] 5-LS2-1 Develop a model to describe the movement of matter among plants, animals decomposers, and the environment. [Clarification Statement: Emphasis is on the idea that matter that is not food (air, water, decomposed materials in soil) is changed by plants into matter that is food. Examples of systems could include organisms, ecosystems, and the Earth.] [Assessment Boundary: Assessment does not include molecular explanations.] 5-ESS2-1 Develop a model using an example to describe ways in which the geosphere, biosphere, hydrosphere, and/or atmosphere interact. [Clarification Statement: The geosphere, hydrosphere (including ice), atmosphere, and biosphere are each a system and each system is a part of the whole Earth System. Examples could include the influence of the ocean on ecosystems, landform shape, and climate; the influence of the atmosphere on landforms and ecosystems through weather and climate; and the influence of mountain ranges on winds and clouds in the atmosphere.] [Assessment Boundary: Assessment is limited to the interactions of two systems at a time.] (Introduced but not assessed until IS3) 5-PS3-1 Use models to describe that energy in animals' food (used for body repair, growth, motion, and to maintain body warmth) was once energy from the sun. [Clarification Statement: Examples of models could include diagrams, and flow charts.]
The bundle of performance expectations above focuses on the following elements from the NRC document <i>A Framework for K–12 Science Education</i>:

Highlighted Science and Engineering Practices	Highlighted Disciplinary Core Ideas	Highlighted Crosscutting Concepts
[SEP-2] Developing and Using Models [SEP-7] Engaging in Argument from Evidence	LS1.C: Organization for Matter and Energy Flow in Organisms LS2.A: Interdependent Relationships in Ecosystems LS2.B: Cycles of Matter and Energy Transfer in Ecosystems ESS2.A: Earth Materials and Systems PS3.D: Energy in Chemical Processes and Everyday Life	[CCC-4] System and system models [CCC-5] Energy and matter: Flows, Cycles, and Conservation
<i>Highlighted California Environmental Principles & Concepts:</i> Principle I The continuation and health of individual human lives and of human communities and societies depend on the health of the natural systems that provide essential goods and ecosystem services. Principle II The long-term functioning and health of terrestrial, freshwater, coastal and marine ecosystems are influenced by their relationships with human societies. Principle III Natural systems proceed through cycles that humans depend upon, benefit from and can alter. Principle IV The exchange of matter between natural systems and human societies affects the long term functioning of both. Principle V Decisions affecting resources and natural systems are complex and involve many factors.		
<i>CA CCSS for ELA/Literacy Connections:</i> W.5.1, SL.5.4, 6; L.5.6		
<i>CA ELD Connections:</i> ELD.PI.5.1, 3, 9, 11, 12		

Grade Five – Instructional Segment 3: Interacting Earth Systems
<i>Guiding Questions:</i> <i>How can we represent systems as complicated as the entire planet?</i> <i>Where does my tap water come from and where does it go?</i> <i>How much water do we need to live, to irrigate plants? How much water do we have?</i> <i>What can we do to protect Earth's resources?</i>
Students who demonstrate understanding can: 5-ESS2-1. Develop a model using an example to describe ways the geosphere, biosphere, hydrosphere, and/or atmosphere interact. [Clarification Statement: **The geosphere, hydrosphere (including ice), atmosphere, and biosphere are each a system and each system is a part of the whole Earth System. Examples could include the influence of the ocean on ecosystems, landform shape, and climate; the influence of the atmosphere on landforms and ecosystems through weather and climate; and the influence of mountain ranges on winds and clouds in the atmosphere. The geosphere, hydrosphere, atmosphere, and biosphere are each a system.] [Assessment Boundary: Assessment is limited to the interactions of two systems at a time.] 5-ESS2-2. Describe and graph the amounts and percentages of water and fresh water in various reservoirs to provide evidence about the distribution of water on Earth. [Assessment Boundary: Assessment is limited to oceans, lakes, rivers, glaciers, ground water, and polar ice caps, and does not include the atmosphere.] 5-ESS3-1. Obtain and combine information about ways individual communities use science ideas to protect the Earth's resources and environment. 3-5-ETS1-1 Define a simple design problem reflecting a need or a want that includes specified criteria for success and constraints on materials, time, or cost. 3-5-ETS1-2 Generate and compare multiple possible solutions to a problem based on how well each is likely to meet criteria and constraints of the problem. 3-5-ETS1-3. Plan and carry out fair tests in which variables are controlled and failure points are considered to identify aspects of a model or prototype that can be improved. **California clarification statements, marked with double asterisks, were incorporated by the California Science Expert Review Panel
The bundle of performance expectations above focuses on the following elements from the NRC document <i>A Framework for K–12 Science Education</i>:

Highlighted Science and Engineering Practices	Highlighted Disciplinary Core Ideas	Highlighted Crosscutting Concepts
[SEP-2] Developing and Using Models [SEP-5] Using Mathematics and Computational Thinking [SEP-6] Constructing Explanations (for science) and Designing Solutions (for engineering) [SEP-8] Obtaining, Evaluating, and Communicating Information	ESS2.A: Earth Materials and Systems ESS2.C: The Roles of Water in Earth's Surface Processes ESS3.C: Human Impacts on Earth Systems ETS1.A: Defining and Delimiting Engineering Problems ETS1.B: Developing Possible Solutions ETS1.C: Optimizing the Design Solution	[CCC-2] Cause and effect [CCC-3] Scale, proportion, and quantity [CCC-4] Systems and System Models
<i>Highlighted California Environmental Principles & Concepts:</i>		
Principle I The continuation and health of individual human lives and of human communities and societies depend on the health of the natural systems that provide essential goods and ecosystem services.		
Principle II The long-term functioning and health of terrestrial, freshwater, coastal and marine ecosystems are influenced by their relationships with human societies.		
Principle III Natural systems proceed through cycles that humans depend upon, benefit from and can alter.		
Principle IV The exchange of matter between natural systems and human societies affects the long term functioning of both.		
Principle V Decisions affecting resources and natural systems are based on a wide range of considerations and decision-making processes.		
CA CCSS Math Connections: 5.MD.1; 5.MD.5b; 6.RP.3; 5.NF.2; 5.G.2; MP. 2, 6		
CA CCSS ELA/Literacy Connections: SL.5.1, 4, 5		

Grade Five – Instructional Segment 4: Patterns in the Night Sky		
<i>Guiding Questions:</i> How far away are the stars? How can we tell? What trends and patterns are there in the movement of the Sun and stars?		
<i>Students who demonstrate understanding can:</i> 5-ESS1-1 Support an argument that differences in the apparent brightness of the sun compared to other stars is due to their relative distance from Earth. [Clarification Statement: Absolute brightness of stars is the result of a variety of factors. Relative distance from Earth is one factor that affects apparent brightness and is the one selected to be addressed by the performance expectation.] [Assessment Boundary: Assessment is limited to relative distances, not sizes, of stars. Assessment does not include other factors that affect apparent brightness (such as stellar masses, age, and stage).] 5-ESS1-2 Represent data in graphical displays to reveal patterns of daily changes in the length and direction of shadows, day and night, and the seasonal appearance of some stars in the night sky. [Clarification Statement: Examples of patterns in the sky could include the position and motion of Earth with respect to the sun and select stars that are visible only in particular months] [Assessment Boundary: Assessment does not include causes of seasons.] 5-PS2-1 Support an argument that the gravitational force exerted by Earth on objects is directed down. [Clarification Statement: “Down” is a local description of the direction that points toward the center of the spherical Earth.] [Assessment Boundary: Assessment does not include mathematical representation of gravitational force.]		
The bundle of performance expectations above focuses on the following elements from the NRC document <i>A Framework for K–12 Science Education</i>:		
Highlighted Science and Engineering Practices [SEP-4] Analyzing and Interpreting Data [SEP-7] Engaging in Argument from evidence	Highlighted Disciplinary Core Ideas ESS1.A: The Universe and its Stars ESS1.B: Earth and the Solar System PS2.B: Types of Interactions	Highlighted Crosscutting Concepts [CCC-1] Patterns [CCC-2] Cause and effect

		[CCC-3] Scale, proportion and quantity
CA CCSS Math Connections: 4.MD.6; 5.NF.6; 5.G.2		
CA CCSS ELA/Literacy Connections: RI.5.3; W.5.7, 10		
CA ELD Connections: ELD.PI.5.1, 5, 11		

Grades Six Through Eight Preferred Integrated Model: Chapter 5

Integrated Grade Six – Instructional Segment 1: Systems and Subsystems in Earth and Life Science
<i>Guiding Questions:</i> • How are living systems and Earth systems similar and different? • What is the value of creating a systems model?
Performance Expectations
<i>Students who demonstrate understanding can:</i> MS-LS1-1. Conduct an investigation to provide evidence that living things are made of cells; either one cell or many different numbers and types of cells. [Clarification Statement: Emphasis is on developing evidence that living things (including Bacteria, Archaea, and Eukarya) are made of cells, distinguishing between living and non-living things, and understanding that living things may be made of one cell or many and varied cells. Viruses, while not cells, have features that are both common with, and distinct from, cellular life.] MS-LS1-2. Develop and use a model to describe the function of a cell as a whole and ways parts of cells contribute to the function. [Clarification Statement: Emphasis is on the cell functioning as a whole system and the primary role of identified parts of the cell, specifically the nucleus, chloroplasts, mitochondria, cell

membrane, and cell wall.] [Assessment Boundary: Assessment of organelle structure/function relationships is limited to the cell wall and cell membrane. Assessment of the function of the other organelles is limited to their relationship to the whole cell. Assessment does not include the biochemical function of cells or cell parts.]

MS-LS1-3. Use argument supported by evidence for how the body is a system of interacting subsystems composed of groups of cells. [Clarification Statement: Emphasis is on the conceptual understanding that cells form tissues and tissues form organs specialized for particular body functions. Examples could include the interaction of subsystems within a system and the normal functioning of those systems.] [Assessment Boundary: Assessment does not include the mechanism of one body system independent of others. Assessment is limited to the circulatory, excretory, digestive, respiratory, muscular, and nervous systems.]

MS-LS1-8. Gather and synthesize information that sensory receptors respond to stimuli by sending messages to the brain for immediate behavior or storage as memories. [Assessment Boundary: Assessment does not include mechanisms for the transmission of this information.]

MS-ESS2-4. Develop a model to describe the cycling of water through Earth's systems driven by energy from the sun and the force of gravity. [Clarification Statement: Emphasis is on the ways water changes its state as it moves through the multiple pathways of the hydrologic cycle. Examples of models can be conceptual or physical.] [Assessment Boundary: A quantitative understanding of the latent heats of vaporization and fusion is not assessed.]

MS-ESS2-6. Develop and use a model to describe how unequal heating and rotation of the Earth cause patterns of atmospheric and

oceanic circulation that determine regional climates.

[Clarification Statement: Emphasis is on how patterns vary by latitude, altitude, and geographic land distribution. Emphasis of atmospheric circulation is on the sunlight-driven latitudinal banding, the Coriolis effect, and resulting prevailing winds; emphasis of ocean circulation is on the transfer of heat by the global ocean convection cycle, which is constrained by the Coriolis effect and the outlines of continents. Examples of models can be diagrams, maps and globes, or digital representations.] [Assessment Boundary: Assessment does not include the dynamics of the Coriolis effect.]

MS-ETS1-1. Define the criteria and constraints of a design problem with sufficient precision to ensure a successful solution, taking into account relevant scientific principles and potential impacts on people and the natural environment that may limit possible solutions.

MS-ETS1-2. Evaluate competing design solutions using a systematic process to determine how well they meet the criteria and constraints of the problem.

The bundle of performance expectations above focuses on the following elements from the NRC document *A Framework for K–12 Science Education*:

Highlighted Science and Engineering Practices	Highlighted Disciplinary Core Ideas	Highlighted Crosscutting Concepts
[SEP-1] Asking Questions and Defining Problems [SEP-2] Developing and Using Models	LS1.A: Structure and Function LS1.D: Information Processing	[CCC-2] Cause and Effect [CCC-3] Scale, proportion, and quantity

[SEP-6] Constructing Explanations (for science) and Designing Solutions (for engineering) [SEP-7] Engaging in Argument from Evidence [SEP-8] Obtaining, Evaluating, and Communicating Information	ESS2.C: The Role of Water in Earth's Surface Processes ESS2.D: Weather and Climate ETS1.A: Defining and Delimiting Engineering Problems ETS1.B: Developing Possible Solutions	[CCC-4] System and System Models [CCC-5] Energy and Matter: Flows, Cycles and Conservation [CCC-6] Structure and Function
<i>Highlighted California Environmental Principles & Concepts:</i> Principle II The long-term functioning and health of terrestrial, freshwater, coastal and marine ecosystems are influenced by their relationships with human societies. Principle III Natural systems proceed through cycles that humans depend upon, benefit from and can alter. Principle IV The exchange of matter between natural systems and human societies affects the long term functioning of both.		
<i>CA CCSS Math Connections:</i> 6.EE.9		
<i>CA ELD Connections:</i> ELD.PI.6.6a-b,10,9,11a		
<i>CA CCSS ELA/Literacy Connections:</i> RST.6-8.1, R.I.6.8, WHST.6-8.1,7,8,9, SL.6.5		

Integrated Grade Six – Instructional Segment 2: Earth System Interactions Cause Weather
<i>Guiding Questions:</i> • Why is the weather so different in different parts of California? • How is weather related to the transfer of energy? • How do models help us understand the different kinds of weather in California?
Performance Expectations
<i>Students who demonstrate understanding can:</i>

MS-ESS2-4. Develop a model to describe the cycling of water through Earth's systems driven by energy from the sun and the force of gravity. [Clarification Statement: Emphasis is on the ways water changes its state as it moves through the multiple pathways of the hydrologic cycle. Examples of models can be conceptual or physical.] [Assessment Boundary: A quantitative understanding of the latent heats of vaporization and fusion is not assessed.]

MS-ESS2-6. Develop and use a model to describe how unequal heating and rotation of the Earth cause patterns of atmospheric and oceanic circulation that determine regional climates. [Clarification Statement: Emphasis is on how patterns vary by latitude, altitude, and geographic land distribution. Emphasis of atmospheric circulation is on the sunlight-driven latitudinal banding, the Coriolis effect, and resulting prevailing winds; emphasis of ocean circulation is on the transfer of heat by the global ocean convection cycle, which is constrained by the Coriolis effect and the outlines of continents. Examples of models can be diagrams, maps and globes, or digital representations.] [Assessment Boundary: Assessment does not include the dynamics of the Coriolis effect].

MS-PS3-3. Apply scientific principles to design, construct, and test a device that either minimizes or maximizes thermal energy transfer. * [Clarification Statement: Examples of devices could include an insulated box, a solar cooker, and a Styrofoam cup.] [Assessment Boundary: Assessment does not include calculating the total amount of thermal energy transferred.]

MS-PS3-4. Plan an investigation to determine the relationships among the energy transferred, the type of matter, the mass, and the change in the average kinetic energy of the particles as measured by the temperature of the sample. [Clarification

Statement: Examples of experiments could include comparing final water temperatures after different masses of ice melted in the same volume of water with the same initial temperature, the temperature change of samples of different materials with the same mass as they cool or heat in the environment, or the same material with different masses when a specific amount of energy is added.] [Assessment Boundary: Assessment does not include calculating the total amount of thermal energy transferred.]

MS-PS3-5. Construct, use, and present arguments to support the claim that when the kinetic energy of an object changes, energy is transferred to or from the object. [Clarification Statement: Examples of empirical evidence used in arguments could include an inventory or other representation of the energy before and after the transfer in the form of temperature changes or motion of object.] [Assessment Boundary: Assessment does not include calculations of energy.]

MS-ETS1-1. Define the criteria and constraints of a design problem with sufficient precision to ensure a successful solution, taking into account relevant scientific principles and potential impacts on people and the natural environment that may limit possible solutions.

MS-ETS1-3. Analyze data from tests to determine similarities and differences among several design solutions to identify the best characteristics of each that can be combined into a new solution to better meet the criteria for success.

*The performance expectations marked with an asterisk integrate traditional science content with engineering through a practice or Disciplinary Core Idea.

The bundle of performance expectations above focuses on the following elements from the NRC document *A Framework for K–12 Science Education*:

Highlighted Science and Engineering Practices	Highlighted Disciplinary Core Ideas	Highlighted Crosscutting Concepts
[SEP-2] Developing and Using Models [SEP-3] Planning and Carrying Out Investigations [SEP-4] Analyzing and Interpreting Data [SEP-6] Constructing Explanations (for science) and Designing Solutions (for engineering) [SEP-7] Engaging in Argument from Evidence [SEP-8] Obtaining, Evaluating, and Communicating Information	ESS2.C: The Roles of Water in Earth's Surface Processes ESS2.D: Weather and Climate PS3.A: Definitions of Energy PS3.B: Conservation of Energy and Energy Transfer ETS1.A: Defining and Delimiting Engineering Problems ETS1.B: Developing Possible Solutions ETS1.C: Optimizing the Design Solution	[CCC-3] Scale, Proportion, and Quantity [CCC-4] System and System Models [CCC-5] Energy and Matter: Flows, Cycles, and Conservation
<i>Highlighted California Environmental Principles & Concepts:</i> Principle III Natural systems proceed through cycles that humans depend upon, benefit from and can alter. Principle IV The exchange of matter between natural systems and human societies affects the long term functioning of both. Principle V Decisions affecting resources and natural systems are based on a wide range of considerations and decision-making processes.		
<i>CA CCSS Math Connections:</i> MP.2, 6.RP.1, 6.SP.5		
<i>CA ELD Connections:</i> ELD.PI.6.6a-b,9,10,11a		
<i>CA CCSS ELA/Literacy Connections:</i> SL.6.5, RST.6-8.1,3,7,9 WHST.6-8.1,7,8		

Integrated Grade Six – Instructional Segment 3: Causes and Effects of Regional Climates

Guiding Questions:

- Why is the climate so different in different regions of the planet?
- Why are organisms so different in different regions of the planet?
- What makes organisms so similar to but also different from their parents?
- What makes animals behave the way they do, and how does their behavior affect their survival and reproduction?

Performance Expectations

Students who demonstrate understanding can:

MS-LS1-4. Use argument based on empirical evidence and scientific reasoning to support an explanation for how characteristic animal behaviors and specialized plant structures affect the probability of successful reproduction of animals and plants respectively. [Clarification Statement: Examples of behaviors that affect the probability of animal reproduction could include nest building to protect young from cold, herding of animals to protect young from predators, and vocalization of animals and colorful plumage to attract mates for breeding. Examples of animal behaviors that affect the probability of plant reproduction could include transferring pollen or seeds, and creating conditions for seed germination and growth. Examples of plant structures could include bright flowers attracting butterflies that transfer pollen, flower nectar and odors that attract insects that transfer pollen, and hard shells on nuts that squirrels bury.]

MS-LS1-5. Construct a scientific explanation based on evidence for how environmental and genetic factors influence the growth of organisms. [Clarification Statement: Examples of local environmental conditions could include availability of food, light,

space, and water. Examples of genetic factors could include large breed cattle and species of grass affecting growth of organisms. Examples of evidence could include drought decreasing plant growth, fertilizer increasing plant growth, different varieties of plant seeds growing at different rates in different conditions, and fish growing larger in large ponds than they do in small ponds.] *[Assessment Boundary: Assessment does not include genetic mechanisms, gene regulation, or biochemical processes.]* Introduced in IS3 but not assessed until IS4.

MS-LS1-8. Gather and synthesize information that sensory receptors respond to stimuli by sending messages to the brain for immediate behavior or storage as memories. *[Assessment Boundary: Assessment does not include mechanisms for the transmission of this information.]* Revisited from IS1.

MS-LS3-2. Develop and use a model to describe why asexual reproduction results in offspring with identical genetic information and sexual reproduction results in offspring with genetic variation. *[Clarification Statement: Emphasis is on using models such as Punnett squares, diagrams, and simulations to describe the cause and effect relationship of gene transmission from parent(s) to offspring and resulting genetic variation.]*

MS-ESS2-5. Collect data to provide evidence for how the motions and complex interactions of air masses results in changes in weather conditions. *[Clarification Statement: Emphasis is on how air masses flow from regions of high pressure to low pressure, causing weather (defined by temperature, pressure, humidity, precipitation, and wind) at a fixed location to change over time, and how sudden changes in weather can result when different air masses collide. Emphasis is on how weather can be predicted within probabilistic ranges. Examples of data can be*

provided to students (such as weather maps, diagrams, and visualizations) or obtained through laboratory experiments (such as with condensation).] [Assessment Boundary: Assessment does not include recalling the names of cloud types or weather symbols used on weather maps or the reported diagrams from weather stations.]

MS-ESS2-6. Develop and use a model to describe how unequal heating and rotation of the Earth cause patterns of atmospheric and oceanic circulation that determine regional climates.

[Clarification Statement: Emphasis is on how patterns vary by latitude, altitude, and geographic land distribution. Emphasis of atmospheric circulation is on the sunlight-driven latitudinal banding, the Coriolis effect, and resulting prevailing winds; emphasis of ocean circulation is on the transfer of heat by the global ocean convection cycle, which is constrained by the Coriolis effect and the outlines of continents. Examples of models can be diagrams, maps and globes, or digital representations.] [Assessment Boundary: Assessment does not include the dynamics of the Coriolis effect].

MS-PS3-4. Plan an investigation to determine the relationships among the energy transferred, the type of matter, the mass, and the change in the average kinetic energy of the particles as measured by the temperature of the sample. [Clarification Statement: Examples of experiments could include comparing final water temperatures after different masses of ice melted in the same volume of water with the same initial temperature, the temperature change of samples of different materials with the same mass as they cool or heat in the environment, or the same material with different masses when a specific amount of energy

is added.] [Assessment Boundary: Assessment does not include calculating the total amount of thermal energy transferred.]

The bundle of performance expectations above focuses on the following elements from the NRC document *A Framework for K–12 Science Education*:

Highlighted Science and Engineering Practices	Highlighted Disciplinary Core Ideas	Highlighted Crosscutting Concepts
[SEP-2] Developing and Using Models	LS1.B: Growth and Development of Organisms	[CCC-2] Cause and Effect
[SEP-3] Planning and Carrying Out Investigations	LS1.D: Information Processing	[CCC-3] Scale, Proportion, and Quantity
[SEP-6] Constructing Explanations (for science) and Designing Solutions (for engineering)	LS3.A: Inheritance of Traits	[CCC-4] System and System Models
[SEP-7] Engaging in Argument from Evidence	LS3.B: Variation of Traits	
[SEP-8] Obtaining, Evaluating, and Communicating Information	ESS2.C: The Roles of Water in Earth's Surface Processes	
	ESS2.D: Weather and Climate	
	PS3.A: Definitions of Energy	
	PS3.B: Conservation of Energy and Energy Transfer	

Highlighted California Environmental Principles & Concepts:

Principle II The long-term functioning and health of terrestrial, freshwater, coastal and marine ecosystems are influenced by their relationships with human societies.

Principle IV The exchange of matter between natural systems and human societies affects the long term functioning of both.
CA CCSS Math Connections: 6.SP.2, 4, 5, MP.2, MP.4
CA ELD Connections: ELD.PI.6.1,5,6 a-b, 9,10,11a
CA CCSSELA/Literacy Connections: RST.6-8.1,2,3,4,7,9 RI.6.8, WHST.6-8.1,2,7,8,9, SL.6.5
Integrated Grade Six – Instructional Segment 4: Effects of Global Warming on Living Systems
<i>Guiding Questions:</i> • How do human activities affect Earth’s systems? • How do we know our global climate is changing?
Performance Expectations
<i>Students who demonstrate understanding can:</i> MS-LS1-4. Use argument based on empirical evidence and scientific reasoning to support an explanation for how characteristic animal behaviors and specialized plant structures affect the probability of successful reproduction of animals and plants respectively. [Clarification Statement: Examples of behaviors that affect the probability of animal reproduction could include nest building to protect young from cold, herding of animals to protect young from predators, and vocalization of animals and colorful plumage to attract mates for breeding. Examples of animal behaviors that affect the probability of plant reproduction could include transferring pollen or seeds, and creating conditions for seed germination and growth. Examples of plant structures could include bright flowers attracting butterflies that transfer pollen, flower nectar and odors that attract insects that transfer pollen, and hard shells on nuts that squirrels bury.]

MS-LS1-5. Construct a scientific explanation based on evidence for how environmental and genetic factors influence the growth of organisms. [Clarification Statement: Examples of local environmental conditions could include availability of food, light, space, and water. Examples of genetic factors could include large breed cattle and species of grass affecting growth of organisms. Examples of evidence could include drought decreasing plant growth, fertilizer increasing plant growth, different varieties of plant seeds growing at different rates in different conditions, and fish growing larger in large ponds than they do in small ponds.]
[Assessment Boundary: Assessment does not include genetic mechanisms, gene regulation, or biochemical processes.]

MS-ESS3-3. Apply scientific principles to design a method for monitoring and minimizing a human impact on the environment.*
[Clarification Statement: Examples of the design process include examining human environmental impacts, assessing the kinds of solutions that are feasible, and designing and evaluating solutions that could reduce that impact. Examples of human impacts can include water usage (such as the withdrawal of water from streams and aquifers or the construction of dams and levees), land usage (such as urban development, agriculture, or the removal of wetlands), and pollution (such as of the air, water, or land).]

MS-ESS3-5. Ask questions to clarify evidence of the factors that have caused the rise in global temperatures over the past century.
[Clarification Statement: Examples of factors include human activities (such as fossil fuel combustion, cement production, and agricultural activity) and natural processes (such as changes in incoming solar radiation or volcanic activity). Examples of evidence can include tables, graphs, and maps of global and

regional temperatures, atmospheric levels of gases such as carbon dioxide and methane, and the rates of human activities. Emphasis is on the major role that human activities play in causing the rise in global temperatures.]

MS-ETS1-1. Define the criteria and constraints of a design problem with sufficient precision to ensure a successful solution, taking into account relevant scientific principles and potential impacts on people and the natural environment that may limit possible solutions.

MS-ETS1-2. Evaluate competing design solutions using a systematic process to determine how well they meet the criteria and constraints of the problem.

*The performance expectations marked with an asterisk integrate traditional science content with engineering through a practice or Disciplinary Core Idea.

The bundle of performance expectations above focuses on the following elements from the NRC document *A Framework for K–12 Science Education*:

Highlighted Science and Engineering Practices	Highlighted Disciplinary Core Ideas	Highlighted Crosscutting Concepts
[SEP-1] Asking Questions and Defining Problems	LS1.B: Growth and Development of Organisms	[CCC-2] Cause and Effect
[SEP-2] Developing and Using Models	ESS3.C: Human Impacts on Earth Systems	[CCC-4] System and System Models
[SEP-6] Constructing Explanations and Designing Solutions	ESS3.D: Global Climate Change	[CCC-5] Energy and Matter: Flows, Cycles, and Conservation
[SEP-7] Engaging in Argument from Evidence	ETS1.A: Defining and Delimiting Engineering Problems	[CCC-7] Stability and Change

[SEP-8] Obtaining, Evaluating, and Communicating Information]	ETS1.B: Developing Possible Solutions ETS1.B: Developing Possible Solutions	
<i>Highlighted California Environmental Principles & Concepts:</i> Principle I The continuation and health of individual human lives and of human communities and societies depend on the health of the natural systems that provide essential goods and ecosystem services. Principle II The long-term functioning and health of terrestrial, freshwater, coastal and marine ecosystems are influenced by their relationships with human societies. Principle IV The exchange of matter between natural systems and human societies affects the long term functioning of both.		
CA CCSS Math Connections: MP.2, 6RP.1,6.EE.6, 6.SP.2,4		
CA ELD Connections: ELD.PI.6.1,5,6 a-b, 9,10,11a		
CA CCSS ELA/Literacy Connections: RI.6.8,RST.6-8.1, 2 WHST.6-8.7, WHST.6-8.8		

Integrated Grade Seven – Instructional Segment 1: Organisms and Nonliving Things Are Made of Atoms
<i>Guiding Questions:</i> • How does the matter in living and nonliving things differ? • How does adding or removing thermal energy affect the physical states of matter? • How do interactions at the atomic level help us understand the observable properties of organisms and nonliving matter?
Performance Expectations
<i>Students who demonstrate understanding can:</i> MS-ESS3-1. Construct a scientific explanation based on evidence for how the uneven distributions of Earth’s mineral, energy, and groundwater resources are the result of past and current

geoscience processes. [Clarification Statement: Emphasis is on how these resources are limited and typically non-renewable, and how their distributions are significantly changing as a result of removal by humans. Examples of uneven distributions of resources as a result of past processes include but are not limited to petroleum (locations of the burial of organic marine sediments and subsequent geologic traps), metal ores (locations of past volcanic and hydrothermal activity associated with subduction zones), and soil (locations of active weathering and/or deposition of rock).] (Introduced, but not assessed until IS3)

MS-LS2-3. Develop a model to describe the cycling of matter and flow of energy among living and nonliving parts of an ecosystem.

[Clarification Statement: Emphasis is on describing the conservation of matter and flow of energy into and out of various ecosystems, and on defining the boundaries of the system.]

[Assessment Boundary: Assessment does not include the use of chemical reactions to describe the processes.] (Introduced, but not assessed until IS3)

MS-PS1-1. Develop models to describe the atomic composition of simple molecules and extended structures. [Clarification Statement:

Emphasis is on developing models of molecules that vary in complexity. Examples of simple molecules could include ammonia and methanol. Examples of extended structures could include sodium chloride or diamonds. Examples of molecular-level models could include drawings, 3D ball and stick structures, or computer representations showing different molecules with different types of atoms.] [Assessment Boundary: Assessment does not include valence electrons and bonding energy, discussing the ionic nature of subunits of complex structures, or a complete description of all individual atoms in a complex molecule or extended structure is not required.]

MS-PS1-3. Gather and make sense of information to describe that synthetic materials come from natural resources and impact society. [Clarification Statement: Emphasis is on natural resources that undergo a chemical process to form the synthetic material. Examples of new materials could include new medicine, foods, and alternative fuels.] [Assessment Boundary: Assessment is limited to qualitative information.] (Introduced, but assessed in IS4)

MS-PS1-4. Develop a model that predicts and describes changes in particle motion, temperature, and state of a pure substance when thermal energy is added or removed. [Clarification Statement: Emphasis is on qualitative molecular-level models of solids, liquids, and gases to show that adding or removing thermal energy increases or decreases kinetic energy of the particles until a change of state occurs. Examples of models could include drawings and diagrams. Examples of particles could include molecules or inert atoms. Examples of pure substances could include water, carbon dioxide, and helium.]

The bundle of performance expectations above focuses on the following elements from the NRC document *A Framework for K–12 Science Education*:

Highlighted Science and Engineering Practices	Highlighted Disciplinary Core Ideas	Highlighted Crosscutting Concepts
[SEP-2] Developing and Using Models [SEP-6] Constructing Explanations (for science) and Designing Solutions (for engineering)	ESS3.A: Natural Resources PS1.A: Structure and Properties of Matter PS1.B: Chemical Reactions	[CCC-2] Cause and Effect: Mechanism and Explanation [CCC-3] Scale, Proportion, and Quantity [CCC-6] Structure and Function

[SEP-8] Obtaining, Evaluating, and Communicating Information	PS3.A: Definitions of Energy	
<i>Highlighted California Environmental Principles & Concepts:</i> Principle I The continuation and health of individual human lives and of human communities and societies depend on the health of the natural systems that provide essential goods and ecosystem services. Principle II The long-term functioning and health of terrestrial, freshwater, coastal and marine ecosystems are influenced by their relationships with human societies.		
CA CCSS Math Connections: 6.EE.6, 6.RP.3, 6.NS.5, 7.EE.4, MP.2, MP.4		
CA ELD Connections: ELD.PI.6.1,5,6 a-b, 9,10,11a		
CA CCSSELA/Literacy Connections: RST.6-8.1,7, WHST.6-8.2,8		

Integrated Grade Seven – Instructional Segment 2: Matter Cycles and Energy Flows through Organisms and Rocks
<i>Guiding Questions:</i> • How do rocks and minerals record the flow of energy and cycling of matter in the Earth? • How do we get energy from our food? • How are hot objects different than cold objects? What changes when they heat up or cool down?
Performance Expectations
<i>Students who demonstrate understanding can:</i> MS-LS1-6. Construct a scientific explanation based on evidence for the role of photosynthesis in the cycling of matter and flow of energy into and out of organisms. [Clarification Statement: Emphasis is on tracing movement of matter and flow of energy.] [Assessment Boundary: Assessment does not include the biochemical mechanisms of photosynthesis.]

- MS-LS1-7. Develop a model to describe how food is rearranged through chemical reactions forming new molecules that support growth and/or release energy as this matter moves through an organism.** [Clarification Statement: Emphasis is on describing that molecules are broken apart and put back together and that in this process, energy is released.] [Assessment Boundary: Assessment does not include details of the chemical reactions for photosynthesis or respiration.]
- MS-ESS2-1. Develop a model to describe the cycling of Earth's materials and the flow of energy that drives this process.** [Clarification Statement: Emphasis is on the processes of melting, crystallization, weathering, deformation, and sedimentation, which act together to form minerals and rocks through the cycling of Earth's materials.] [Assessment Boundary: Assessment does not include the identification and naming of minerals.]
- MS-PS1-2. Analyze and interpret data on the properties of substances before and after the substances interact to determine if a chemical reaction has occurred.** [Clarification Statement: Examples of reactions could include burning sugar or steel wool, fat reacting with sodium hydroxide, and mixing zinc with hydrogen chloride.] [Assessment Boundary: Assessment is limited to analysis of the following properties: density, melting point, boiling point, solubility, flammability, and odor.]
- MS-PS1-5. Develop and use a model to describe how the total number of atoms does not change in a chemical reaction and thus mass is conserved.** [Clarification Statement: Emphasis is on law of conservation of matter and on physical models or drawings, including digital forms that represent atoms.] [Assessment Boundary: Assessment does not include the use of

atomic masses, balancing symbolic equations, or intermolecular forces.]

MS-PS1-6. Undertake a design project to construct, test, and modify a device that either releases or absorbs thermal energy by chemical processes.* [Clarification Statement: Emphasis is on the design, controlling the transfer of energy to the environment, and modification of a device using factors such as type and concentration of a substance. Examples of designs could involve chemical reactions such as dissolving ammonium chloride or calcium chloride.] [Assessment Boundary: Assessment is limited to the criteria of amount, time, and temperature of substance in testing the device.]

MS-ETS1-1. Define the criteria and constraints of a design problem with sufficient precision to ensure a successful solution, taking into account relevant scientific principles and potential impacts on people and the natural environment that may limit possible solutions.

MS-ETS1-2. Evaluate competing design solutions using a systematic process to determine how well they meet the criteria and constraints of the problem.

MS-ETS1-3. Analyze data from tests to determine similarities and differences among several design solutions to identify the best characteristics of each that can be combined into a new solution to better meet the criteria for success.

MS-ETS1-4. Develop a model to generate data for iterative testing and modification of a proposed object, tool, or process such that an optimal design can be achieved.

*The performance expectations marked with an asterisk integrate traditional science content with engineering through a practice or Disciplinary Core Idea.

The bundle of performance expectations above focuses on the following elements from the NRC document *A Framework for K–12 Science Education*:

Highlighted Science and Engineering Practices	Highlighted Disciplinary Core Ideas	Highlighted Crosscutting Concepts
[SEP-1] Asking Questions and Defining Problems [SEP-2] Developing and Using Models [SEP-4] Analyzing and Interpreting Data [SEP-6] Constructing Explanations (for science) and Designing Solutions (for engineering) [SEP-7] Engaging in Argument from Evidence	LS1.C: Organization for Matter and Energy Flow in Organisms PS1.A: Structure and Properties of Matter PS1.B: Chemical Reactions PS3.D: Energy in Chemical Processes and Everyday Life ESS2.A Earth's Materials and Systems ETS1.A: Defining and Delimiting Engineering Problems ETS1.B: Developing Possible Solutions ETS1.C: Optimizing the Design Solution	[CCC-1] Patterns [CCC-5] Energy and Matter: Flows, Cycles, and Conservation [CCC-7] Stability and Change

Highlighted California Environmental Principles & Concepts:

Principle III Natural systems proceed through cycles that humans depend upon, benefit from and can alter.

Principle IV The exchange of matter between natural systems and human societies affects the long term functioning of both.

Principle V: Decisions affecting resources and natural systems are based on a wide range of considerations and decision-making processes.

CA CCSS Math Connections: 6.EE.9, MP.2, MP.4, 6.SP.4,5, 7.EE.3, 7.SP.7

CA ELD Connections: ELD.PI.6.1,5,6 a-b, 9,10,11a

CA CCSSELA/Literacy Connections: RST.6-8.1,2,3,7, WHST.6-8.2,7,8,9, L.7.5

Integrated Grade Seven – Instructional Segment 3: Natural Processes and Human Activities Shape Earth’s Resources and Ecosystems

Guiding Questions:

- How can we use interactions between individual rocks or individual organisms to understand systems as big as the whole geosphere or whole ecosystem?
- How can we use patterns in geosphere interactions to predict the location of resources?
- How can we use patterns in ecosystem interactions to predict how organisms compete and share resources?

Performance Expectations

Students who demonstrate understanding can:

MS-LS2-1. Analyze and interpret data to provide evidence for the effects of resource availability on organisms and populations of organisms in an ecosystem. [Clarification Statement: Emphasis is on cause and effect relationships between resources and growth of individual organisms and the numbers of organisms in ecosystems during periods of abundant and scarce resources.]

MS-LS2-2. Construct an explanation that predicts patterns of interactions among organisms across multiple ecosystems. [Clarification Statement: Emphasis is on predicting consistent patterns of interactions in different ecosystems in terms of the relationships among and between organisms and abiotic components of ecosystems. Examples of types of interactions could include competitive, predatory, and mutually beneficial.]

MS-LS2-3. Develop a model to describe the cycling of matter and flow of energy among living and nonliving parts of an ecosystem.

[Clarification Statement: Emphasis is on describing the conservation of matter and flow of energy into and out of various ecosystems, and on defining the boundaries of the system.]

[Assessment Boundary: Assessment does not include the use of chemical reactions to describe the processes.]

MS-ESS2-3. Analyze and interpret data on the distribution of fossils and rocks, continental shapes, and seafloor structures to provide evidence of the past plate motions. [Clarification

Statement: Examples of data include similarities of rock and fossil types on different continents, the shapes of the continents (including continental shelves), and the locations of ocean structures (such as ridges, fracture zones, and trenches).]

[Assessment Boundary: Paleomagnetic anomalies in oceanic and continental crust are not assessed.]

MS-ESS3-1. Construct a scientific explanation based on evidence for how the uneven distributions of Earth’s mineral, energy, and groundwater resources are the result of past and current geoscience processes. [Clarification Statement: Emphasis is

on how these resources are limited and typically non-renewable, and how their distributions are significantly changing as a result of removal by humans. Examples of uneven distributions of resources as a result of past processes include but are not limited to petroleum (locations of the burial of organic marine sediments and subsequent geologic traps), metal ores (locations of past volcanic and hydrothermal activity associated with subduction zones), and soil (locations of active weathering and/or deposition of rock).]

MS-PS1-2. Analyze and interpret data on the properties of substances before and after the substances interact to determine if a chemical reaction has occurred. [Clarification Statement: Examples of reactions could include burning sugar or steel wool, fat reacting with sodium hydroxide, and mixing zinc with hydrogen chloride.] [Assessment Boundary: Assessment is limited to analysis of the following properties: density, melting point, boiling point, solubility, flammability, and odor.]

MS-PS1-3. Gather and make sense of information to describe that synthetic materials come from natural resources and impact society. [Clarification Statement: Emphasis is on natural resources that undergo a chemical process to form the synthetic material. Examples of new materials could include new medicine, foods, and alternative fuels.] [Assessment Boundary: Assessment is limited to qualitative information.] (Revisited from IS1, but not assessed until IS4)

MS-PS1-5. Develop and use a model to describe how the total number of atoms does not change in a chemical reaction and thus mass is conserved. [Clarification Statement: Emphasis is on law of conservation of matter and on physical models or drawings, including digital forms that represent atoms.] [Assessment Boundary: Assessment does not include the use of atomic masses, balancing symbolic equations, or intermolecular forces.]

The bundle of performance expectations above focuses on the following elements from the NRC document *A Framework for K–12 Science Education*:

Highlighted Science and Engineering Practices	Highlighted Disciplinary Core Ideas	Highlighted Crosscutting Concepts
[SEP-2] Developing and Using Models [SEP-4] Analyzing and Interpreting Data [SEP-6] Constructing Explanations (for science) and Designing Solutions (for engineering) [SEP-8] Obtaining, Evaluating, and Communicating Information	LS2.A: Interdependent Relationships in Ecosystems LS2.B: Cycles of Matter and Energy Transfer in Ecosystems ESS1.C: The History of Planet Earth ESS2.B: Plate Tectonics and Large-Scale System Interactions ESS3.A: Natural Resources PS1.A: Structure and Properties of Matter PS1.B: Chemical Reactions	[CCC-1] Patterns [CCC-2] Cause and Effect: Mechanism and Explanation [CCC-5] Energy and Matter: Flows, Cycles, and Conservation [CCC-6] Structure and Function

Highlighted California Environmental Principles & Concepts:

Principle I: The continuation and health of individual human lives and of human communities and societies depend on the health of the natural systems that provide essential goods and ecosystem services.

Principle II: The long-term functioning and health of terrestrial, freshwater, coastal and marine ecosystems are influenced by their relationships with human societies.

Principle III Natural systems proceed through cycles that humans depend upon, benefit from and can alter.

Principle IV The exchange of matter between natural systems and human societies affects the long term functioning of both.
Principle V Decisions affecting resources and natural systems are complex and involve many factors.
CA CCSS Math Connections: 6.EE.6,9, 6.SP.4,5, 6.RP.3,7.EE.4, MP.2
CA ELD Connections: ELD.PI.6.1,5,6 a-b, 9,10,11a
CA CCSSELA/Literacy Connections: RST.6-8.1,2,7,9, WHST.6-8.1,2,9, SL.7.1, SL.7.4, SL.7.5

Integrated Grade Seven – Instructional Segment 4: Sustaining Biodiversity and Ecosystems Service in a Changing World
<i>Guiding Questions:</i> • What natural processes and human activities threaten biodiversity and ecosystem services? • How can people help sustain biodiversity and ecosystem services in a changing world?
Performance Expectations
<i>Students who demonstrate understanding can:</i> MS-LS2-4. Construct an argument supported by empirical evidence that changes to physical or biological components of an ecosystem affect populations. [Clarification Statement: Emphasis is on recognizing patterns in data and making warranted inferences about changes in populations, and on evaluating empirical evidence supporting arguments about changes to ecosystems.] MS-LS2-5. Evaluate competing design solutions for maintaining biodiversity and ecosystem services.* [Clarification Statement: Examples of ecosystem services could include water purification, nutrient recycling, and prevention of soil erosion. Examples of

design solution constraints could include scientific, economic, and social considerations.]

MS-ESS2-2. Construct an explanation based on evidence for how geoscience processes have changed Earth's surface at varying time and spatial scales. [Clarification Statement: Emphasis is on how processes change Earth's surface at time and spatial scales that can be large (such as slow plate motions or the uplift of large mountain ranges) or small (such as rapid landslides or microscopic geochemical reactions), and how many geoscience processes (such as earthquakes, volcanoes, and meteor impacts) usually behave gradually but are punctuated by catastrophic events. Examples of geoscience processes include surface weathering and deposition by the movements of water, ice, and wind. Emphasis is on geoscience processes that shape local geographic features, where appropriate.]

MS-ESS3-2. Analyze and interpret data on natural hazards to forecast future catastrophic events and inform the development of technologies to mitigate their effects. [Clarification Statement: Emphasis is on how some natural hazards, such as volcanic eruptions and severe weather, are preceded by phenomena that allow for reliable predictions, but others, such as earthquakes, occur suddenly and with no notice, and thus are not yet predictable. Examples of natural hazards can be taken from interior processes (such as earthquakes and volcanic eruptions), surface processes (such as mass wasting and tsunamis), or severe weather events (such as hurricanes, tornadoes, and floods). Examples of data can include the locations, magnitudes, and frequencies of the natural hazards. Examples of technologies can be global (such as satellite systems to monitor hurricanes or

forest fires) or local (such as building basements in tornado-prone regions or reservoirs to mitigate droughts).]

MS-PS1-3. Gather and make sense of information to describe that synthetic materials come from natural resources and impact society. [Clarification Statement: Emphasis is on natural resources that undergo a chemical process to form the synthetic material. Examples of new materials could include new medicine, foods, and alternative fuels.] [Assessment Boundary: Assessment is limited to qualitative information.] (Assessed after being introduced in IS1 and IS3.)

MS-ETS1-1. Define the criteria and constraints of a design problem with sufficient precision to ensure a successful solution, taking into account relevant scientific principles and potential impacts on people and the natural environment that may limit possible solutions.

MS-ETS1-2. Evaluate competing design solutions using a systematic process to determine how well they meet the criteria and constraints of the problem.

*The performance expectations marked with an asterisk integrate traditional science content with engineering through a practice or Disciplinary Core Idea.

The bundle of performance expectations above focuses on the following elements from the NRC document *A Framework for K–12 Science Education*:

Highlighted Science and Engineering Practices	Highlighted Disciplinary Core Ideas	Highlighted Crosscutting Concepts
[SEP-1] Asking Questions and Defining Problems	LS2.C Ecosystem Dynamics, Functioning and Resilience	[CCC-1] Patterns

[SEP-4] Analyzing and Interpreting Data [SEP-6] Constructing Explanations (for science) and Designing Solutions (for engineering) [SEP-7] Engaging in Argument from Evidence [SEP-8] Obtaining, Evaluating, and Communicating Information	LS4.D Biodiversity and Humans	[CCC-2] Cause and Effect: Mechanism and Explanation
	ESS2.A Earth Materials and Systems	[CCC-3] Scale, Proportion and Quantity
	ESS2.C Roles of Water in Earth's Surface Processes	[CCC-6] Structure and Function
	ESS3.B Natural Hazards	[CCC-7] Stability and Change
	PS1.A: Structure and Properties of Matter	
	PS1.B: Chemical Reactions	
	ETS1.A: Defining and Delimiting Engineering Problems	
	ETS1.B: Developing Possible Solutions	
	ETS1.C Optimizing the Design Solution	

Highlighted California Environmental Principles & Concepts:

Principle I The continuation and health of individual human lives and of human communities and societies depend on the health of the natural systems that provide essential goods and ecosystem services.

Principle II The long-term functioning and health of terrestrial, freshwater, coastal and marine ecosystems are influenced by their relationships with human societies.

Principle III Natural systems proceed through cycles that humans depend upon, benefit from and can alter.

Principle IV The exchange of matter between natural systems and human societies affects the long term functioning of both.
Principle V Decisions affecting resources and natural systems are complex and involve many factors.
<i>CA CCSS Math Connections:</i> MP.2, MP.4, 6.EE.6, 6.RP.3, 7.EE.4
<i>CA ELD Connections:</i> ELD.PI.6.1,5,6 a-b, 9,10,11a
<i>CA CCSSELA/Literacy Connections:</i> RI.7.8, RST.6-8.1,7,8, WHST.6-8.1,2,9, SL.7.5

Integrated Grade Eight – Instructional Segment 1: Objects Move and Collide
<i>Guiding Questions:</i> • What are forces and how do they affect the motions of objects? • Do objects always need a force in order to keep moving? • What happens when a moving object collides with something? • How do fossils provide evidence of an ancient collision that wiped out the dinosaurs?
Performance Expectations
<i>Students who demonstrate understanding can:</i> MS-LS4-1. Analyze and interpret data for patterns in the fossil record that document the existence, diversity, extinction, and change of life forms throughout the history of life on Earth under the assumption that natural laws operate today as in the past. [Clarification Statement: Emphasis is on finding patterns of changes in the level of complexity of anatomical structures in organisms and the chronological order of fossil appearance in the rock layers.] [Assessment Boundary: Assessment does not include the names of individual species or geological eras in the fossil record.]

MS-PS2-1. Apply Newton's Third Law to design a solution to a problem involving the motion of two colliding objects.* [Clarification

Statement: Examples of practical problems could include the impact of collisions between two cars, between a car and stationary objects, and between a meteor and a space vehicle.] [Assessment Boundary: Assessment is limited to vertical or horizontal interactions in one dimension.]

MS-PS2-2. Plan an investigation to provide evidence that the change in an object's motion depends on the sum of the forces on the object and the mass of the object. [Clarification Statement: Emphasis is on balanced (Newton's First Law) and unbalanced forces in a system, qualitative comparisons of forces, mass and changes in motion (Newton's Second Law), frame of reference, and specification of units.] [Assessment Boundary: Assessment is limited to forces and changes in motion in one-dimension in an inertial reference frame and to change in one variable at a time. Assessment does not include the use of trigonometry.]

MS-PS3-1. Construct and interpret graphical displays of data to describe the relationships of kinetic energy to the mass of an object and to the speed of an object. [Clarification Statement: Emphasis is on descriptive relationships between kinetic energy and mass separately from kinetic energy and speed. Examples could include riding a bicycle at different speeds, rolling different sizes of rocks downhill, and getting hit by a wiffle ball versus a tennis ball.]

MS-ETS1-1. Define the criteria and constraints of a design problem with sufficient precision to ensure a successful solution, taking into account relevant scientific principles and potential impacts on people and the natural environment that may limit possible solutions.

MS-ETS1-2. Evaluate competing design solutions using a systematic process to determine how well they meet the criteria and constraints of the problem.

MS-ETS1-3. Analyze data from tests to determine similarities and differences among several design solutions to identify the best characteristics of each that can be combined into a new solution to better meet the criteria for success.

MS-ETS1-4. Develop a model to generate data for iterative testing and modification of a proposed object, tool, or process such that an optimal design can be achieved.

*The performance expectations marked with an asterisk integrate traditional science content with engineering through a practice or Disciplinary Core Idea.

The bundle of performance expectations above focuses on the following elements from the NRC document *A Framework for K–12 Science Education*:

Highlighted Science and Engineering Practices	Highlighted Disciplinary Core Ideas	Highlighted Crosscutting Concepts
[SEP-1] Asking Questions and Defining Problems [SEP-2] Developing and Using Models [SEP-3] Planning and Carrying Out Investigations [SEP-4] Analyzing and Interpreting Data [SEP-5] Using Mathematics and Computational	LS4.A Evidence of Common Ancestry and Diversity PS2.A Forces and Motion PS3.A Definitions of Energy ETS1.A Defining and Delimiting Engineering Problems ETS1.B Developing Possible Solutions	[CCC-1] Patterns [CCC-2] Cause and Effect: Mechanism and Explanation [CCC-3] Scale, Proportion, and Quantity [CCC-4] System and System Models [CCC-5] Energy and Matter: Flows, Cycles, and Conservation

Thinking [SEP-6] Constructing Explanations (for science) and Designing Solutions (for engineering) [SEP-7] Engaging in Argument from Evidence	ETS1.C Optimizing the Design Solution	[CCC-7] Stability and Change
<i>Highlighted California Environmental Principles & Concepts:</i> Principle II The long-term functioning and health of terrestrial, freshwater, coastal and marine ecosystems are influenced by their relationships with human societies. Principle V Decisions affecting resources and natural systems are complex and involve many factors.		
<i>CA CCSS Math Connections:</i> 6.EE.2,6, 6.NS.5, 6.RP.1,2, 7.EE.3,4, 7.RP.2, 7.SP.7, 8.EE.1,2, MP.2		
<i>CA ELD Connections:</i> ELD.PI.6.1,5,6 a-b, 9,10,11a		
<i>CA CCSSELA/Literacy Connections:</i> RST.6-8.1,3,7,9, WHST.6-8.7,8,9, SL.8.5		

Integrated Grade Eight – Instructional Segment 2: Noncontact Forces Influence Phenomena
<i>Guiding Questions:</i> • What causes the cyclical changes in the appearance of the Moon? • How can an object influence the motion of another object without touching it? • Does Earth’s force of gravity attract other objects equally?
Performance Expectations
<i>Students who demonstrate understanding can:</i> MS-ESS1-1. Develop and use a model of the Earth-Sun-Moon system to describe the cyclic patterns of lunar phases, eclipses of the Sun and Moon, and seasons. [Clarification Statement:

Examples of models can be physical, graphical, or conceptual.]

(Introduced, but seasons are not assessed until IS4)

MS-ESS1-2. Develop and use a model to describe the role of gravity in the motions within galaxies and the solar system. [Clarification

Statement: Emphasis for the model is on gravity as the force that holds together the solar system and Milky Way galaxy and controls orbital motions within them. Examples of models can be physical (such as the analogy of distance along a football field or computer visualizations of elliptical orbits) or conceptual (such as mathematical proportions relative to the size of familiar objects such as their school or state).] [Assessment Boundary: Assessment does not include Kepler's Laws of orbital motion or the apparent retrograde motion of the planets as viewed from Earth.]

MS-ESS1-3. Analyze and interpret data to determine scale properties of objects in the solar system. [Clarification Statement: Emphasis

is on the analysis of data from Earth-based instruments, space-based telescopes, and spacecraft to determine similarities and differences among solar system objects. Examples of scale properties include the sizes of an object's layers (such as crust and atmosphere), surface features (such as volcanoes), and orbital radius. Examples of data include statistical information, drawings and photographs, and models.] [Assessment Boundary: Assessment does not include recalling facts about properties of the planets and other solar system bodies.]

MS-PS2-3. Ask questions about data to determine the factors that affect the strength of electrical and magnetic forces. [Clarification

Statement: Examples of devices that use electrical and magnetic forces could include electromagnets, electric motors, or generators. Examples of data could include the effect of the

number of turns of wire on the electromagnet or the effect of increasing the number or strength of magnets on the speed of an electric motor.] [Assessment Boundary: Assessment about questions that require quantitative answers is limited to proportional reasoning and algebraic thinking.]

MS-PS2-4. Construct and present arguments using evidence to support the claim that gravitational interactions are attractive and depend on the masses of interacting objects. [Clarification Statement: Examples of evidence for arguments could include data generated from simulations or digital tools; and charts displaying mass, strength of interaction, distance from the Sun, and orbital periods of objects within the solar system.] [Assessment Boundary: Assessment does not include Newton's Law of Gravitation or Kepler's Laws.]

MS-PS2-5. Conduct an investigation and evaluate the experimental design to provide evidence that fields exist between objects exerting forces on each other even though the objects are not in contact. [Clarification Statement: Examples of this phenomenon could include the interactions of magnets, electrically-charged strips of tape, and electrically-charged pith balls. Examples of investigations could include first-hand experiences or simulations.]

MS-PS3-2. Develop a model to describe that when the arrangement of objects interacting at a distance changes, different amounts of potential energy are stored in the system. [Clarification Statement: Emphasis is on relative amounts of potential energy, not on calculations of potential energy. Examples of objects within systems interacting at varying distances could include: the Earth and either a roller coaster cart at varying positions on a hill or objects at varying heights on shelves, changing the

direction/orientation of a magnet, and a balloon with static electrical charge being brought closer to a classmate's hair. Examples of models could include representations, diagrams, pictures, and written descriptions of systems.]

[Assessment Boundary: Assessment is limited to two objects and electric, magnetic, and gravitational interactions.]

The bundle of performance expectations above focuses on the following elements from the NRC document *A Framework for K–12 Science Education*:

Highlighted Science and Engineering Practices	Highlighted Disciplinary Core Ideas	Highlighted Crosscutting Concepts
[SEP-1] Asking Questions and Defining Problems [SEP-2] Developing and Using Models [SEP-3] Planning and Carrying Out Investigations [SEP-4] Analyzing and Interpreting Data [SEP-7] Engaging in Argument from Evidence [SEP-8] Obtaining, Evaluating, and Communicating Information	ESS1.A: The Universe and Its Stars ESS1.B: Earth and the Solar System PS2.B: Types of Interactions PS3.A: Definitions of Energy PS3.C: Relationship Between Energy and Forces	CCC-1] Patterns [CCC-2] Cause and Effect: Mechanism and Explanation [CCC-3] Scale, Proportion and Quantity [CCC-4] System and System Models
CA CCSS Math Connections: 6.RP.1, 6.EE.2,6, 7.RP.2, 7.EE.3,4, MP.2, MP.4		
CA ELD Connections: ELD.PI.6.1,5,6 a-b, 9,10,11a		
CA CCSSELA/Literacy Connections: RST.6-8.1,3,7, WHST.6-8.1,7, SL.8.5		

Integrated Grade Eight – Instructional Segment 3: Evolution Explains Life's Unity and Diversity

Guiding Questions:

- What can we infer about the history of Earth and life on earth from the clues we can uncover in rock layers and the fossil record?
- What evidence supports Darwin's theory of biological evolution?
- How do evolution and natural selection explain life's unity and diversity?

Performance Expectations

Students who demonstrate understanding can:

MS-LS3-1. Develop and use a model to describe why structural changes to genes (mutations) located on chromosomes may affect proteins and may result in harmful, beneficial, or neutral effects to the structure and function of the organism.

[Clarification Statement: Emphasis is on conceptual understanding that changes in genetic material may result in making different proteins.] [Assessment Boundary: Assessment does not include specific changes at the molecular level, mechanisms for protein synthesis, or specific types of mutations.]

MS-LS4-1. Analyze and interpret data for patterns in the fossil record that document the existence, diversity, extinction, and change of life forms throughout the history of life on Earth under the assumption that natural laws operate today as in the past.

[Clarification Statement: Emphasis is on finding patterns of changes in the level of complexity of anatomical structures in organisms and the chronological order of fossil appearance in the rock layers.] [Assessment Boundary: Assessment does not include the names of individual species or geological eras in the fossil record.]

MS-LS4-2. Apply scientific ideas to construct an explanation for the anatomical similarities and differences among modern

organisms and between modern and fossil organisms to infer evolutionary relationships. [Clarification Statement: Emphasis is on explanations of the evolutionary relationships among organisms in terms of similarity or differences of the gross appearance of anatomical structures.]

MS-LS4-3. Analyze displays of pictorial data to compare patterns of similarities in the embryological development across multiple species to identify relationships not evident in the fully formed anatomy. [Clarification Statement: Emphasis is on inferring general patterns of relatedness among embryos of different organisms by comparing the macroscopic appearance of diagrams or pictures.] [Assessment Boundary: Assessment of comparisons is limited to gross appearance of anatomical structures in embryological development.]

MS-LS4-4. Construct an explanation based on evidence that describes how genetic variations of traits in a population increase some individuals' probability of surviving and reproducing in a specific environment. [Clarification Statement: Emphasis is on using simple probability statements and proportional reasoning to construct explanations.]

MS-LS4-5. Gather and synthesize information about the technologies that have changed the way humans influence the inheritance of desired traits in organisms. [Clarification Statement: Emphasis is on synthesizing information from reliable sources about the influence of humans on genetic outcomes in artificial selection (such as genetic modification, animal husbandry, gene therapy); and, on the impacts these technologies have on society as well as the technologies leading to these scientific discoveries.]

MS-LS4-6. Use mathematical representations to support explanations of how natural selection may lead to increases and decreases

of specific traits in populations over time. [Clarification Statement: Emphasis is on using mathematical models, probability statements, and proportional reasoning to support explanations of trends in changes to populations.]

MS-ESS1-4. Construct a scientific explanation based on evidence from rock strata for how the geologic time scale is used to organize Earth’s 4.6-billion-year-old history. [Clarification Statement: Emphasis is on how analyses of rock formations and the fossils they contain are used to establish relative ages of major events in Earth’s history. Examples of Earth’s major events could range from being very recent (such as the last Ice Age or the earliest fossils of homo sapiens) to very old (such as the formation of Earth or the earliest evidence of life). Examples can include the formation of mountain chains and ocean basins, the evolution or extinction of particular living organisms, or significant volcanic eruptions.] [Assessment Boundary: Assessment does not include recalling the names of specific periods or epochs and events within them.]

*The performance expectations marked with an asterisk integrate traditional science content with engineering through a practice or Disciplinary Core Idea.

The bundle of performance expectations above focuses on the following elements from the NRC document *A Framework for K–12 Science Education*:

Highlighted Science and Engineering Practices	Highlighted Disciplinary Core Ideas	Highlighted Crosscutting Concepts
[SEP-2] Developing and Using Models [SEP-4] Analyzing and Interpreting Data [SEP-5] Using	LS3.A Inheritance of Traits LS3.B Variation of Traits	[CCC-1] Patterns [CCC-2] Cause and Effect

Mathematics and Computational Thinking [SEP-6] Constructing Explanations (for science) and Designing Solutions (for engineering) [SEP-7] Engaging in Argument from Evidence [SEP-8] Obtaining, Evaluating, and Communicating Information	LS4.A Evidence of Common Ancestry and Diversity	[CCC-6] Structure and Function
	LS4.B Natural Selection	[CCC-7] Stability and Change
	LS4.C Adaptation	
	ESS1.C The History of Planet Earth	
<i>Highlighted California Environmental Principles & Concepts:</i> Principle II The long-term functioning and health of terrestrial, freshwater, coastal and marine ecosystems are influenced by their relationships with human societies.		
CA CCSS Math Connections: 6.RP.1, 6.SP.5, 6.EE.6, 7.RP.2, MP.4		
CA ELD Connections: ELD.PI.6.1,5,6 a-b, 9,10,11a		
CA CCSSELA/Literacy Connections: RST.6-8.1,4,7,9, WHST.6-8.2,8,9, SL.8.1,4,5		

Integrated Grade Eight – Instructional Segment 4: Sustaining Local and Global Biodiversity	
<i>Guiding Questions:</i> • What are the characteristic properties and behaviors of waves? • What human activities harm Earth’s biodiversity and what human activities help sustain local and global biodiversity? • How does communication technology encode information and how can digital technologies be used to help sustain biodiversity?	
Performance Expectations	

Students who demonstrate understanding can:

MS-LS4-4. Construct an explanation based on evidence that describes how genetic variations of traits in a population increase some individuals' probability of surviving and reproducing in a specific environment. [Clarification Statement: Emphasis is on using simple probability statements and proportional reasoning to construct explanations.]

MS-LS4-6. Use mathematical representations to support explanations of how natural selection may lead to increases and decreases of specific traits in populations over time. [Clarification Statement: Emphasis is on using mathematical models, probability statements, and proportional reasoning to support explanations of trends in changes to populations.]

MS-ESS1-1. Develop and use a model of the Earth-Sun-Moon system to describe the cyclic patterns of lunar phases, eclipses of the Sun and Moon, and seasons. [Clarification Statement: Examples of models can be physical, graphical, or conceptual.]

MS-ESS3-4. Construct an argument supported by evidence for how increases in human population and per-capita consumption of natural resources impact Earth's systems. [Clarification Statement: Examples of evidence include grade-appropriate databases on human populations and the rates of consumption of food and natural resources (such as freshwater, mineral, and energy). Examples of impacts can include changes to the appearance, composition, and structure of Earth's systems as well as the rates at which they change. The consequences of increases in human populations and consumption of natural resources are described by science, but science does not make the decisions for the actions society takes.]

MS-PS4-1. Use mathematical representations to describe a simple model for waves that includes how the amplitude of a wave is related to the energy in a wave. [Clarification Statement: Emphasis is on describing waves with both qualitative and quantitative thinking.] [Assessment Boundary: Assessment does not include electromagnetic waves and is limited to standard repeating waves.]

MS-PS4-2. Develop and use a model to describe that waves are reflected, absorbed, or transmitted through various materials. [Clarification Statement: Emphasis is on both light and mechanical waves. Examples of models could include drawings, simulations, and written descriptions.] [Assessment Boundary: Assessment is limited to qualitative applications pertaining to light and mechanical waves.]

MS-PS4-3. Integrate qualitative scientific and technical information to support the claim that digitized signals are a more reliable way to encode and transmit information than analog signals. [Clarification Statement: Emphasis is on a basic understanding that waves can be used for communication purposes. Examples could include using fiber optic cable to transmit light pulses, radio wave pulses in Wi-Fi devices, and conversion of stored binary patterns to make sound or text on a computer screen.] [Assessment Boundary: Assessment does not include binary counting. Assessment does not include the specific mechanism of any given device.]

MS-ETS1-1. Define the criteria and constraints of a design problem with sufficient precision to ensure a successful solution, taking into account relevant scientific principles and potential impacts on people and the natural environment that may limit possible solutions.

MS-ETS1-2. Evaluate competing design solutions using a systematic process to determine how well they meet the criteria and constraints of the problem.

*The performance expectations marked with an asterisk integrate traditional science content with engineering through a practice or Disciplinary Core Idea.

The bundle of performance expectations above focuses on the following elements from the NRC document *A Framework for K–12 Science Education*:

Highlighted Science and Engineering Practices	Highlighted Disciplinary Core Ideas	Highlighted Crosscutting Concepts
[SEP-1] Asking Questions and Defining Problems	LS4.B: Natural Selection	[CCC-1] Patterns
[SEP-2] Developing and Using Models	LS4.C: Adaptation	[CCC-2] Cause and Effect
[SEP-4] Analyzing and Interpreting Data	ESS1.A: The Universe and Its Stars	[CCC-6] Structure and Function
[SEP-5] Using Mathematics and Computational Thinking	ESS1.B: Earth and the Solar System	[CCC-7] Stability and Change
[SEP-6] Constructing Explanations (for science) and Designing Solutions (for engineering)	ESS3.C: Human Impacts on Earth Systems	
[SEP-7] Engaging in Argument from Evidence	PS4.A: Waves Properties	
[SEP-8] Obtaining, Evaluating, and Communicating Information	PS4.B: Electromagnetic Radiation	
	PS4.C: Information Technologies and Instrumentation	
	ETS1.A: Defining and Delimiting Engineering Problems	
	ETS1.B: Developing Possible Solutions	

<i>Highlighted California Environmental Principles & Concepts:</i> Principle I The continuation and health of individual human lives and of human communities and societies depend on the health of the natural systems that provide essential goods and ecosystem services. Principle II The long-term functioning and health of terrestrial, freshwater, coastal and marine ecosystems are influenced by their relationships with human societies.
<i>CA CCSS Math Connections:</i> 6.RP.1,3, 6.SP.5, 6.EE.6, 7.EE.3,4, 7.RP.2, 8.F.3, MP.2, MP.4
<i>CA ELD Connections:</i> ELD.PI.6.1,5,6 a-b, 9,10,11a
<i>CA CCSSELA/Literacy Connections:</i> RST.6-8.1,9, WHST.6-8.1,2,9, SL.8.1,4,5

<i>Highlighted California Environmental Principles & Concepts:</i> Principle II The long-term functioning and health of terrestrial, freshwater, coastal and marine ecosystems are influenced by their relationships with human societies.
<i>CA CCSC Math Connections:</i> 7.SP. 1-3, 8.SP.4
<i>CA CCSC ELA/Literacy Connections:</i> RI.8.2, RI.8.8, SL.8.1,4,6
<i>CA ELD Connections:</i> ELD.PI.8.1,2,4,6a-b,9

Grades Six Through Eight- Discipline Specific Course Model Chapter 6

Discipline Specific Grade Six – Instructional Segment 1: Earth’s Place in the Solar System
<i>Guiding Questions:</i> • What causes the cycles of stars, planets, and moons? • How can we represent the vastness of the solar system and compare objects as large as planets and moons?
Performance Expectations

Students who demonstrate understanding can:

MS-ESS1-1. Develop and use a model of the Earth-Sun-Moon system to describe the cyclic patterns of lunar phases, eclipses of the Sun and Moon, and seasons. [Clarification Statement: Examples of models can be physical, graphical, or conceptual.]

MS-ESS1-2. Develop and use a model to describe the role of gravity in the motions within galaxies and the solar system. [Clarification Statement: Emphasis for the model is on gravity as the force that holds together the solar system and Milky Way galaxy and controls orbital motions within them. Examples of models can be physical (such as the analogy of distance along a football field or computer visualizations of elliptical orbits) or conceptual (such as mathematical proportions relative to the size of familiar objects such as their school or state).] [Assessment Boundary: Assessment does not include Kepler's Laws of orbital motion or the apparent retrograde motion of the planets as viewed from Earth.]

MS-ESS1-3. Analyze and interpret data to determine scale properties of objects in the solar system. [Clarification Statement: Emphasis is on the analysis of data from Earth-based instruments, space-based telescopes, and spacecraft to determine similarities and differences among solar system objects. Examples of scale properties include the sizes of an object's layers (such as crust and atmosphere), surface features (such as volcanoes), and orbital radius. Examples of data include statistical information, drawings and photographs, and models.] [Assessment Boundary: Assessment does not include recalling facts about properties of the planets and other solar system bodies.]

The bundle of performance expectations above focuses on the following elements from the NRC document <i>A Framework for K–12 Science Education</i> :		
Highlighted Science and Engineering Practices [SEP-2] Developing and Using Models [SEP-4] Analyzing and Interpreting Data	Highlighted Disciplinary Core Ideas ESS1.A: The Universe and Its Stars ESS1.B: Earth and the Solar System <i>Other Necessary DCI:</i> PS2.B: Types of Interactions	Highlighted Crosscutting Concepts [CCC-1] Patterns [CCC-2] Cause and Effect: Mechanism and Explanation [CCC-3] Scale, Proportion, and Quantity [CCC-4] Systems and System Models
CA CCSS Math Connections: MP.2, MP.4, 6.RP.1, 7RP.2a-d, 6.EE.6, 7.EE.4a-d		
CA ELD Connections: ELD.PI.6.6a-b,10,9,11a		
CA CCSS ELA/Literacy Connections: RST.6-8.1,7; SL.6.5		

Discipline Specific Grade Six – Instructional Segment 2: Atmosphere: Flows of Energy

Guiding Questions:

- Why is it cold at the North Pole?
- What causes California's summers to be hot and dry? What causes the changes between summer and winter?
- Why is there more rain in Northern California than Southern California?
- What effect do humans have on Earth's climate?

Performance Expectations

Students who demonstrate understanding can:

MS-ESS1-1. Develop and use a model of the Earth-Sun-Moon system to describe the cyclic patterns of lunar phases, eclipses of the Sun and Moon, and seasons. [Clarification Statement: Examples of models can be physical, graphical, or conceptual.]
(Continued from IS1)

MS-ESS2-6. Develop and use a model to describe how unequal heating and rotation of the Earth cause patterns of atmospheric and oceanic circulation that determine regional climates.
[Clarification Statement: Emphasis is on how patterns vary by latitude, altitude, and geographic land distribution. Emphasis of atmospheric circulation is on the sunlight-driven latitudinal banding, the Coriolis effect, and resulting prevailing winds; emphasis of ocean circulation is on the transfer of heat by the global ocean convection cycle, which is constrained by the Coriolis effect and the outlines of continents. Examples of models can be diagrams, maps and globes, or digital representations.] [Assessment Boundary: Assessment does not include the dynamics of the Coriolis effect.]

MS-ESS3-4. Construct an argument supported by evidence for how increases in human population and per-capita consumption of natural resources impact Earth's systems. [Clarification Statement: Examples of evidence include grade-appropriate databases on human populations and the rates of consumption of food and natural resources (such as freshwater, mineral, and energy). Examples of impacts can include changes to the appearance, composition, and structure of Earth's systems as well as the rates at which they change. The consequences of increases in human populations and consumption of natural resources are described by science, but science does not make the decisions for the actions society takes.]

MS-ESS3-5. Ask questions to clarify evidence of the factors that have caused the rise in global temperatures over the past century. [Clarification Statement: Examples of factors include human activities (such as fossil fuel combustion, cement production, and agricultural activity) and natural processes (such as changes in incoming solar radiation or volcanic activity). Examples of evidence can include tables, graphs, and maps of global and regional temperatures, atmospheric levels of gases such as carbon dioxide and methane, and the rates of human activities. Emphasis is on the major role that human activities play in causing the rise in global temperatures.]

The bundle of performance expectations above focuses on the following elements from the NRC document *A Framework for K–12 Science Education*:

Highlighted Science and Engineering Practices	Highlighted Disciplinary Core Ideas	Highlighted Crosscutting Concepts
[SEP-1] Asking Questions and Defining Problems [SEP-2] Developing and Using Models [SEP-7] Engaging in Argument from Evidence	ESS1.A: The Universe and Its Stars ESS1.B: Earth and the Solar System ESS2.C: The Role of Water in Earth's Surface Processes ESS2.D: Weather and Climate ESS3.C: Human Impacts on Earth Systems ESS3.D: Global Climate Change <i>Other necessary DCI:</i> PS3.B: Conservation of Energy and Energy Transfer PS4.B: Electromagnetic Radiation	[CCC-1] Patterns [CCC-2] Cause and Effect: Mechanism and Explanation [CCC-4] System and System Models [CCC-7] Stability and Change

Highlighted California Environmental Principles & Concepts:

Principle I: The continuation and health of individual human lives and of human communities and societies depend on the health of the natural systems that provide essential goods and ecosystem services.

Principle II: The long-term functioning and health of terrestrial, freshwater, coastal and marine ecosystems are influenced by their relationships with human societies.

CA CCSS Math Connections: MP.2, MP.4, 6.RP.1, 6.EE.6, 7.EE.4a-b, 7.RP.2a-d

CA ELD Connections: ELD.PI.6.6a-b,10,9,11a

CA CCSS ELA/Literacy Connections: SL.6.5, RST.6-8.1, WHST.6-8.1a-f, 9

Discipline Specific Grade Six – Instructional Segment 3:

Atmosphere/Hydrosphere: Cycles of Matter

Guiding Questions:

- How do we predict tomorrow's weather?
- How do the atmosphere and hydrosphere interact to control our valuable water resources?

Performance Expectations

Students who demonstrate understanding can:

MS-ESS2-1. Develop a model to describe the cycling of Earth's materials and the flow of energy that drives this process. **[Clarification Statement: Emphasis is on the processes of melting, crystallization, weathering, deformation, and sedimentation, which act together to form minerals and rocks through the cycling of Earth's materials.]**
[Assessment Boundary: Assessment does not include the identification and naming of minerals.]

MS-ESS2-4. Develop a model to describe the cycling of water through Earth's systems driven by energy from the Sun and the force of gravity.

[Clarification Statement: Emphasis is on the ways water changes its state as it moves through the multiple pathways of the hydrologic cycle. Examples of models can be conceptual or physical.] [Assessment Boundary: A quantitative understanding of the latent heats of vaporization and fusion is not assessed.]

MS-ESS2-5. Collect data to provide evidence for how the motions and complex interactions of air masses results in changes in weather conditions. **[Clarification Statement: Emphasis is on how air masses flow from regions of high pressure to low pressure, causing weather (defined by temperature, pressure, humidity, precipitation, and wind) at a fixed location to change over time, and how sudden changes in weather can result when different air masses collide. Emphasis is on how weather can be predicted within probabilistic ranges. Examples of data can be provided to students (such as weather maps, diagrams, and visualizations) or obtained through laboratory experiments (such as with condensation).] [Assessment Boundary: Assessment does not include recalling the names of cloud types or weather symbols used on weather maps or the reported diagrams from weather stations.]**

MS-ESS2-6. Develop and use a model to describe how unequal heating and rotation of the Earth cause patterns of atmospheric and oceanic circulation that determine regional climates. **[Clarification Statement: Emphasis is on how patterns vary by latitude, altitude, and geographic land distribution. Emphasis of atmospheric circulation is on the sunlight-driven latitudinal**

banding, the Coriolis effect, and resulting prevailing winds; emphasis of ocean circulation is on the transfer of heat by the global ocean convection cycle, which is constrained by the Coriolis effect and the outlines of continents. Examples of models can be diagrams, maps and globes, or digital representations.] [Assessment Boundary: Assessment does not include the dynamics of the Coriolis effect.]

MS-ESS3-2. Analyze and interpret data on natural hazards to forecast future catastrophic events and inform the development of technologies to mitigate their effects. **[Clarification Statement: Emphasis is on how some natural hazards, such as volcanic eruptions and severe weather, are preceded by phenomena that allow for reliable predictions, but others, such as earthquakes, occur suddenly and with no notice, and thus are not yet predictable. Examples of natural hazards can be taken from interior processes (such as earthquakes and volcanic eruptions), surface processes (such as mass wasting and tsunamis), or severe weather events (such as hurricanes, tornadoes, and floods). Examples of data can include the locations, magnitudes, and frequencies of the natural hazards. Examples of technologies can be global (such as satellite systems to monitor hurricanes or forest fires) or local (such as building basements in tornado-prone regions or reservoirs to mitigate droughts).]**

MS-ESS3-4. Construct an argument supported by evidence for how increases in human population and per-capita consumption of natural resources impact Earth's systems. **[Clarification Statement: Examples of evidence include grade-appropriate databases on human populations and the rates of consumption of food and natural resources (such as freshwater, mineral, and**

energy). Examples of impacts can include changes to the appearance, composition, and structure of Earth's systems as well as the rates at which they change. The consequences of increases in human populations and consumption of natural resources are described by science, but science does not make the decisions for the actions society takes.]

The bundle of performance expectations above focuses on the following elements from the NRC document *A Framework for K–12 Science Education*:

Highlighted Science and Engineering Practices	Highlighted Disciplinary Core Ideas	Highlighted Crosscutting Concepts
[SEP-2] Developing and Using Models [SEP-3] Planning and Carrying Out Investigations [SEP-4] Analyzing and Interpreting Data [SEP-7] Engaging in Argument from Evidence	ESS2.A: Earth's Materials and Systems ESS2.C: The Role of Water in Earth's Surface Processes ESS2.D: Weather and Climate ESS3.B: Natural Hazards ESS3.C: Human Impacts on Earth Systems <i>Other necessary DCI:</i> PS3.B: Conservation of Energy and Energy Transfer	[CCC-1] Patterns [CCC-2] Cause and Effect [CCC-4] System and System Models [CCC-5] Energy and Matter: Flows, Cycles, and Conservation [CCC-7] Stability and Change

Highlighted California Environmental Principles & Concepts:

Principle I The continuation and health of individual human lives and of human communities and societies depend on the health of the natural systems that provide essential goods and ecosystem services.

Principle II: The long-term functioning and health of terrestrial, freshwater, coastal and marine ecosystems are influenced by their relationships with human societies.

CA CCSS Math Connections: MP.2, 6.NS.5, 6.EE.6, 6.RP.1, 7.RP.2a-d, 7.EE.4a-b

CA ELD Connections: ELD.PI.6.6a-b,10,9,11a

CA CCSS ELA/Literacy Connections: SL.6.5, RST.6-8.1,7,9, WHST.6-8.2a-f,8,9

**Discipline Specific Grade Six – Instructional Segment 4: Geosphere:
Surface Processes**

Guiding Questions:

- How can we read layers of rock like the pages of a history book to reconstruct what happened during Earth's past?
- What is the relationship between the way rocks are built up (deposition) and the way rocks are broken down (erosion)?
- How does our understanding of erosion and deposition help us find valuable energy and water resources and make ourselves safer from landslides?

Performance Expectations

Students who demonstrate understanding can:

MS-ESS1-4. Construct a scientific explanation based on evidence from rock strata for how the geologic time scale is used to organize Earth's 4.6-billion-year-old history. [Clarification Statement: Emphasis is on how analyses of rock formations and

the fossils they contain are used to establish relative ages of major events in Earth's history. Examples of Earth's major events could range from being very recent (such as the last Ice Age or the earliest fossils of homo sapiens) to very old (such as the formation of Earth or the earliest evidence of life). Examples can include the formation of mountain chains and ocean basins, the evolution or extinction of particular living organisms, or significant volcanic eruptions.] [Assessment Boundary: Assessment does not include recalling the names of specific periods or epochs and events within them.]

MS-ESS2-1. Develop a model to describe the cycling of Earth's materials and the flow of energy that drives this process. [Clarification Statement: Emphasis is on the processes of melting, crystallization, weathering, deformation, and sedimentation, which act together to form minerals and rocks through the cycling of Earth's materials.] [Assessment Boundary: Assessment does not include the identification and naming of minerals.]

MS-ESS2-2. Construct an explanation based on evidence for how geoscience processes have changed Earth's surface at varying time and spatial scales. [Clarification Statement: Emphasis is on how processes change Earth's surface at time and spatial scales that can be large (such as slow plate motions or the uplift of large mountain ranges) or small (such as rapid landslides or microscopic geochemical reactions), and how many geoscience processes (such as earthquakes, volcanoes, and meteor impacts) usually behave gradually but are punctuated by catastrophic events. Examples of geoscience processes include surface weathering and deposition by the movements of water, ice, and wind. Emphasis is on geoscience processes that shape local geographic features, where appropriate.]

MS-ESS3-1. Construct a scientific explanation based on evidence for how the uneven distributions of Earth's mineral, energy, and

groundwater resources are the result of past and current geoscience processes. [Clarification Statement: Emphasis is on how these resources are limited and typically non-renewable, and how their distributions are significantly changing as a result of removal by humans. Examples of uneven distributions of resources as a result of past processes include but are not limited to petroleum (locations of the burial of organic marine sediments and subsequent geologic traps), metal ores (locations of past volcanic and hydrothermal activity associated with subduction zones), and soil (locations of active weathering and/or deposition of rock).]

MS-ESS3-2. Analyze and interpret data on natural hazards to forecast future catastrophic events and inform the development of technologies to mitigate their effects. [Clarification Statement: Emphasis is on how some natural hazards, such as volcanic eruptions and severe weather, are preceded by phenomena that allow for reliable predictions, but others, such as earthquakes, occur suddenly and with no notice, and thus are not yet predictable. Examples of natural hazards can be taken from interior processes (such as earthquakes and volcanic eruptions), surface processes (such as mass wasting and tsunamis), or severe weather events (such as hurricanes, tornadoes, and floods). Examples of data can include the locations, magnitudes, and frequencies of the natural hazards. Examples of technologies can be global (such as satellite systems to monitor hurricanes or forest fires) or local (such as building basements in tornado-prone regions or reservoirs to mitigate droughts).]

The bundle of performance expectations above focuses on the following elements from the NRC document *A Framework for K–12 Science Education*:

Highlighted Science and Engineering Practices	Highlighted Disciplinary Core Ideas	Highlighted Crosscutting Concepts
[SEP-2] Developing and Using Models [SEP-4] Analyzing and Interpreting Data [SEP-6] Constructing Explanations (for science) and Designing Solutions (for engineering)	ESS1.C: The History of Planet Earth ESS2.A: Earth's Materials and Systems ESS2.C: The Roles of Water in Earth's Surface Processes ESS3.A: Natural Resources ESS3.B: Natural Hazards Other necessary DCI: LS4.A: Evidence of Common Ancestry and Diversity	[CCC-1] Patterns [CCC-2] Cause and Effect: Mechanism and Explanation [CCC-3] Scale, Proportion, and Quantity [CCC-7] Stability and Change
Highlighted California Environmental Principles & Concepts: Principle I The continuation and health of individual human lives and of human communities and societies depend on the health of the natural systems that provide essential goods and ecosystem services. Principle II: The long-term functioning and health of terrestrial, freshwater, coastal and marine ecosystems are influenced by their relationships with human societies.		

CA CCSS Math Connections: MP2,6.EE.6, 7.EE.4a-b
CA ELD Connections: ELD.PI.6.6a-b,10,9,11a
CA CCSS ELA/Literacy Connections: SL.6.5, RST.6-8.1,7 WHST.6-8.2a-f

Discipline Specific Grade Six – Instructional Segment 5: Geosphere: Internal Processes
<i>Guiding Questions:</i> • How can the shapes of landforms at the surface help us understand processes that are going on deep within the Earth? • How can understanding plate motions help us locate resources (energy, mineral, and water) and protect ourselves from natural hazards?
Performance Expectations
<i>Students who demonstrate understanding can:</i> MS-ESS2-1. Develop a model to describe the cycling of Earth’s materials and the flow of energy that drives this process. [Clarification Statement: Emphasis is on the processes of melting, crystallization, weathering, deformation, and sedimentation, which act together to form minerals and rocks through the cycling of Earth’s materials.] [Assessment Boundary: Assessment does not include the identification and naming of minerals.] MS-ESS2-2. Construct an explanation based on evidence for how geoscience processes have changed Earth’s surface at varying time and spatial scales. [Clarification Statement: Emphasis is on how processes change Earth’s surface at time and spatial scales that can be large (such as slow plate motions or the uplift of large mountain ranges) or small (such as rapid landslides or microscopic geochemical reactions), and how many geoscience processes (such as earthquakes, volcanoes, and meteor impacts) usually behave gradually but are punctuated by catastrophic events. Examples of geoscience processes include surface weathering and deposition by the movements of water, ice,

and wind. Emphasis is on geoscience processes that shape local geographic features, where appropriate.]

MS-ESS2-3. Analyze and interpret data on the distribution of fossils and rocks, continental shapes, and seafloor structures to provide evidence of the past plate motions. [Clarification Statement: Examples of data include similarities of rock and fossil types on different continents, the shapes of the continents (including continental shelves), and the locations of ocean structures (such as ridges, fracture zones, and trenches).] [Assessment Boundary: Paleomagnetic anomalies in oceanic and continental crust are not assessed.]

MS-ESS3-1. Construct a scientific explanation based on evidence for how the uneven distributions of Earth's mineral, energy, and groundwater resources are the result of past and current geoscience processes. [Clarification Statement: Emphasis is on how these resources are limited and typically non-renewable, and how their distributions are significantly changing as a result of removal by humans. Examples of uneven distributions of resources as a result of past processes include but are not limited to petroleum (locations of the burial of organic marine sediments and subsequent geologic traps), metal ores (locations of past volcanic and hydrothermal activity associated with subduction zones), and soil (locations of active weathering and/or deposition of rock).]

MS-ESS3-2. Analyze and interpret data on natural hazards to forecast future catastrophic events and inform the development of technologies to mitigate their effects. [Clarification Statement: Emphasis is on how some natural hazards, such as volcanic eruptions and severe weather, are preceded by phenomena that allow for reliable predictions, but others, such as earthquakes, occur suddenly and with no notice, and thus are not yet

predictable. Examples of natural hazards can be taken from interior processes (such as earthquakes and volcanic eruptions), surface processes (such as mass wasting and tsunamis), or severe weather events (such as hurricanes, tornadoes, and floods). Examples of data can include the locations, magnitudes, and frequencies of the natural hazards. Examples of technologies can be global (such as satellite systems to monitor hurricanes or forest fires) or local (such as building basements in tornado-prone regions or reservoirs to mitigate droughts).]

The bundle of performance expectations above focuses on the following elements from the NRC document *A Framework for K–12 Science Education*:

Highlighted Science and Engineering Practices	Highlighted Disciplinary Core Ideas	Highlighted Crosscutting Concepts
[SEP-2] Developing and Using Models	ESS1.C: The History of Planet Earth	[CCC-1] Patterns
[SEP-3] Planning and Carrying Out Investigations	ESS2.A: Earth’s Materials and Systems	[CCC-2] Cause and Effect: Mechanism and Explanation
[SEP-4] Analyzing and Interpreting Data	ESS2.B: Plate Tectonics and Large-Scale System Interactions	[CCC-3] Scale, Proportion, and Quantity.
[SEP-6] Constructing Explanations (for science) and Designing Solutions (for engineering)	ESS2.C: The Roles of Water in Earth’s Surface Processes	[CCC-5] Energy and Matter: Flows, Cycles, and Conservation

	ESS3.A: Natural Resources ESS3.B: Natural Hazards Other Necessary DCI: LS4.A: Evidence of Common Ancestry and Diversity	[CCC-7] Stability and Change
Highlighted California Environmental Principles & Concepts: Principle I The continuation and health of individual human lives and of human communities and societies depend on the health of the natural systems that provide essential goods and ecosystem services. Principle II: The long-term functioning and health of terrestrial, freshwater, coastal and marine ecosystems are influenced by their relationships with human societies.		
CA CCSS Math Connections: MP2,6.EE.6, 7.EE.4a-b		
CA ELD Connections: ELD.PI.6.6a-b,10,9,11a		
CA CCSS ELA/Literacy Connections: SL.6.5, RST.6-8.1,7 WHST.6-8.2a-f		

Discipline Specific Grade Seven – Instructional Segment 1: Interdependent Ecosystems
<i>Guiding Question:</i> How do parts of an ecosystem interact?
Performance Expectations
<i>Students who demonstrate understanding can:</i> MS-LS2-2. Construct an explanation that predicts patterns of interactions among organisms across multiple ecosystems. [Clarification Statement: Emphasis is on predicting consistent patterns of

interactions in different ecosystems in terms of the relationships among and between organisms and abiotic components of ecosystems. Examples of types of interactions could include competitive, predatory, and mutually beneficial.]

MS-LS2-3. Develop a model to describe the cycling of matter and flow of energy among living and nonliving parts of an ecosystem. [Clarification Statement: Emphasis is on describing the conservation of matter and flow of energy into and out of various ecosystems, and on defining the boundaries of the system.] [Assessment Boundary: Assessment does not include the use of chemical reactions to describe the processes.]

The bundle of performance expectations above focuses on the following elements from the NRC document *A Framework for K–12 Science Education*:

Highlighted Science and Engineering Practices	Highlighted Disciplinary Core Ideas	Highlighted Crosscutting Concepts
[SEP-2] Developing and Using Models [SEP-6] Constructing Explanations (for science) and Designing Solutions (for engineering)	LS2.A: Interdependent Relationships in Ecosystems LS2.B: Cycle of Matter and Energy Transfer in Ecosystems	[CCC-1] Patterns [CCC-4] System and System Models [CCC-5] Energy and Matter: Flows, Cycles, and Conservation

Highlighted California Environmental Principles & Concepts:

- Principle I The continuation and health of individual human lives and of human communities and societies depend on the health of the natural systems that provide essential goods and ecosystem services.
- Principle II The long-term functioning and health of terrestrial, freshwater, coastal and marine ecosystems are influenced by their relationships with human societies.
- Principle III Natural systems proceed through cycles that humans depend upon, benefit from and can alter.

Principle IV The exchange of matter between natural systems and human societies affects the long term functioning of both.
Principle V Decisions affecting resources and natural systems are based on a wide range of considerations and decision-making processes.
CA CCSS Math Connections: 6.EE.9, 6.SP.5a-d
CA ELD Connections: ELD.PI.7.6a-b,10,9,11a
CA CCSS ELA/Literacy Connections: SL.7.1a-d,4,5; RST.6-8.1, WHST.6-8.2a-f, 9

Discipline Specific Grade Seven – Instructional Segment 2: Photosynthesis and Respiration
<i>Guiding Questions:</i> • How do plants and animals get their energy? • What processes allow energy and matter to be exchanged in ecosystems?
Performance Expectations
<i>Students who demonstrate understanding can:</i> MS-LS1-6. Construct a scientific explanation based on evidence for the role of photosynthesis in the cycling of matter and flow of energy into and out of organisms. [Clarification Statement: Emphasis is on tracing movement of matter and flow of energy.] [Assessment Boundary: Assessment does not include the biochemical mechanisms of photosynthesis.] MS-LS1-7. Develop a model to describe how food is rearranged through chemical reactions forming new molecules that support growth and/or release energy as this matter moves through an organism. [Clarification Statement: Emphasis is on describing that molecules are broken apart and put back together and that in this process, energy is released.] [Assessment Boundary:

Assessment does not include details of the chemical reactions for photosynthesis or respiration.]

Other necessary PEs introduced, but not assessed until Grade Eight:

MS-PS1-1. Develop models to describe the atomic composition of simple molecules and extended structures. [Clarification Statement:

Emphasis is on developing models of molecules that vary in complexity. Examples of simple molecules could include ammonia and methanol. Examples of extended structures could include sodium chloride or diamonds. Examples of molecular-level models could include drawings, 3D ball and stick structures, or computer representations showing different molecules with different types of atoms.] [Assessment Boundary: Assessment does not include valence electrons and bonding energy, discussing the ionic nature of subunits of complex structures, or a complete description of all individual atoms in a complex molecule or extended structure is not required.]

MS-PS1-5. Develop and use a model to describe how the total number of atoms does not change in a chemical reaction and thus mass is conserved. [Clarification Statement: Emphasis is on

law of conservation of matter and on physical models or drawings, including digital forms that represent atoms.]

[Assessment Boundary: Assessment does not include the use of atomic masses, balancing symbolic equations, or intermolecular forces.]

The bundle of performance expectations above focuses on the following elements from the NRC document *A Framework for K–12 Science Education*:

Highlighted Science and Engineering Practices	Highlighted Disciplinary Core Ideas	Highlighted Crosscutting Concepts
[SEP-2] Developing and Using Models [SEP-5] Using Mathematics and Computational Thinking [SEP-6] Constructing Explanations (for science) and Designing Solutions (for engineering).	LS1.C: Organization for Matter and Energy Flow in Organisms PS3.D: Energy in Chemical Processes and Everyday Life Other Necessary DCIs: PS1.A: Structure and Properties of Matter PS1.B: Chemical Reactions	[CCC-5] Energy and Matter: Flows, Cycles, and Conservation [CCC-4] Systems and System Models [CCC-5] Energy and Matter: Flows, Cycles, and Conservation
Highlighted California Environmental Principles & Concepts:		
Principle II The long-term functioning and health of terrestrial, freshwater, coastal and marine ecosystems are influenced by their relationships with human societies.		
Principle V Decisions affecting resources and natural systems are based on a wide range of considerations and decision-making processes.		
CA CCSS Math Connections: 6.EE.9		
CA ELD Connections: ELD.PI.7.6a-b,10,9,11a		
CA CCSS ELA/Literacy Connections: SL.7.1a-d,4,5; RST.6-8.1,2, WHST.6-8.2a-f, 9		

Discipline Specific Grade Seven – Instructional Segment 3: Cells and Body Systems
<i>Guiding Questions:</i> • How do the parts of a cell sustain life? • How do cells work together to make a complex organism?
Performance Expectations
<i>Students who demonstrate understanding can:</i> MS-LS1-1. Conduct an investigation to provide evidence that living things are made of cells; either one cell or many different numbers and types of cells. [Clarification Statement: Emphasis is on developing evidence that living things (**including Bacteria, Archaea, and Eukarya) are made of cells, distinguishing between living and non-living things, and understanding that living things may be made of one cell or many and varied cells. **Viruses, while not cells, have features that are both common with, and distinct from, cellular life.] MS-LS1-2. Develop and use a model to describe the function of a cell as a whole and ways parts of cells contribute to the function. [Clarification Statement: Emphasis is on the cell functioning as a whole system and the primary role of identified parts of the cell, specifically the nucleus, chloroplasts, mitochondria, cell membrane, and cell wall.] [Assessment Boundary: Assessment of organelle structure/function relationships is limited to the cell wall and cell membrane. Assessment of the function of the other organelles is limited to their relationship to the whole cell. Assessment does not include the biochemical function of cells or cell parts.]

MS-LS1-3. Use argument supported by evidence for how the body is a system of interacting subsystems composed of groups of cells.

[Clarification Statement: Emphasis is on the conceptual understanding that cells form tissues and tissues form organs specialized for particular body functions. Examples could include the interaction of subsystems within a system and the normal functioning of those systems.] [Assessment Boundary: Assessment does not include the mechanism of one body system independent of others. Assessment is limited to the circulatory, excretory, digestive, respiratory, muscular, and nervous systems.]

MS-LS1-8. Gather and synthesize information that sensory receptors respond to stimuli by sending messages to the brain for immediate behavior or storage as memories.

[Assessment Boundary: Assessment does not include mechanisms for the transmission of this information.]

The bundle of performance expectations above focuses on the following elements from the NRC document *A Framework for K–12 Science Education*:

Highlighted Science and Engineering Practices	Highlighted Disciplinary Core Ideas	Highlighted Crosscutting Concepts
[SEP-2] Developing and Using Models	LS1.A: Structure and Function	[CCC-2] Cause and Effect
[SEP-3] Planning and Carrying Out Investigations	LS1.B: Growth and Development of Organisms	[CCC-3] Scale, Proportion, and Quantity
[SEP-7] Engaging in Argument from Evidence	LS1.D: Information Processing	[CCC-4] System and System Models

[SEP-8] Obtaining, Evaluating, and Communicating Information		[CCC-6] Structure and Function
CA CCSS Math Connections: 6.EE.9		
CA ELD Connections: ELD.PI.7.6a-b,10,9,11a		
CA CCSS ELA/Literacy Connections: RI.7.8,SL.7.1a-d,4,5; RST.6-8.1, WHST.6-8.1a-e,2a-f, 7,8,9		
Discipline Specific Grade Seven – Instructional Segment 4: Evidence of Evolution		
<i>Guiding Questions:</i> • In what ways are humans similar to dinosaurs? • How do rocks tell us about the history of life?		
Performance Expectations		
<i>Students who demonstrate understanding can:</i> MS-LS4-1. Analyze and interpret data for patterns in the fossil record that document the existence, diversity, extinction, and change of life forms throughout the history of life on Earth under the assumption that natural laws operate today as in the past. [Clarification Statement: Emphasis is on finding patterns of changes in the level of complexity of anatomical structures in organisms and the chronological order of fossil appearance in the rock layers.] [Assessment Boundary: Assessment does not include the names of individual species or geological eras in the fossil record.] MS-LS4-2. Apply scientific ideas to construct an explanation for the anatomical similarities and differences among modern		

organisms and between modern and fossil organisms to infer evolutionary relationships. [Clarification Statement: Emphasis is on explanations of the evolutionary relationships among organisms in terms of similarity or differences of the gross appearance of anatomical structures.]

MS-LS4-3. Analyze displays of pictorial data to compare patterns of similarities in the embryological development across multiple species to identify relationships not evident in the fully formed anatomy. [Clarification Statement: Emphasis is on inferring general patterns of relatedness among embryos of different organisms by comparing the macroscopic appearance of diagrams or pictures.] [Assessment Boundary: Assessment of comparisons is limited to gross appearance of anatomical structures in embryological development.]

MS-LS4-4. Construct an explanation based on evidence that describes how genetic variations of traits in a population increase some individuals' probability of surviving and reproducing in a specific environment. [Clarification Statement: Emphasis is on using simple probability statements and proportional reasoning to construct explanations.]

The bundle of performance expectations above focuses on the following elements from the NRC document *A Framework for K–12 Science Education*:

Highlighted Science and Engineering Practices [SEP-1] Asking Questions and Defining Problems	Highlighted Disciplinary Core Ideas LS4.A: Evidence of Common Ancestry and Diversity	Highlighted Crosscutting Concepts [CCC-1] Patterns
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[SEP-4] Analyzing and Interpreting Data [SEP-6] Constructing Explanations (for science) and Designing Solutions (for engineering)	LS4.B: Natural Selection	[CCC-2] Cause and Effect: Mechanism and Explanation
<i>Highlighted California Environmental Principles & Concepts:</i>		
Principle II The long-term functioning and health of terrestrial, freshwater, coastal and marine ecosystems are influenced by their relationships with human societies.		
<i>CA CCSS Math Connections:</i> 6.EE.6, SP.5a-d, 6.RP.1, 7.RP.2a-d, MP.4		
<i>CA ELD Connections:</i> ELD.PI.7.6a-b,10,9,11a		
<i>CA CCSS ELA/Literacy Connections:</i> SL.7.1a-d,4; RST.6-8.1,7,9; WHST.6-8.2a-f, 9		

Discipline Specific Grade Seven – Instructional Segment 5: Inheritance and Genetics
<i>Guiding Questions:</i> • How do cells know what to do and how to accomplish it? • Why do children look like their parents? • What causes differences between individuals?
Performance Expectations
<i>Students who demonstrate understanding can:</i> MS-LS3-1. Develop and use a model to describe why structural changes to genes (mutations) located on chromosomes may affect proteins and may result in harmful, beneficial, or neutral

effects to the structure and function of the organism.

[Clarification Statement: Emphasis is on conceptual understanding that changes in genetic material may result in making different proteins.] [Assessment Boundary: Assessment does not include specific changes at the molecular level, mechanisms for protein synthesis, or specific types of mutations.]

MS-LS3-2. Develop and use a model to describe why asexual reproduction results in offspring with identical genetic information and sexual reproduction results in offspring with genetic variation. [Clarification Statement: Emphasis is on using models such as Punnett squares, diagrams, and simulations to describe the cause and effect relationship of gene transmission from parent(s) to offspring and resulting genetic variation.]

MS-LS4-5. Gather and synthesize information about the technologies that have changed the way humans influence the inheritance of desired traits in organisms. [Clarification Statement: Emphasis is on synthesizing information from reliable sources about the influence of humans on genetic outcomes in artificial selection (such as genetic modification, animal husbandry, gene therapy); and, on the impacts these technologies have on society as well as the technologies leading to these scientific discoveries.]

MS-ETS1-1. Define the criteria and constraints of a design problem with sufficient precision to ensure a successful solution, taking into account relevant scientific principles and potential impacts on people and the natural environment that may limit possible solutions.

The bundle of performance expectations above focuses on the following elements from the NRC document *A Framework for K–12 Science Education*:

Highlighted Science and Engineering Practices	Highlighted Disciplinary Core Ideas	Highlighted Crosscutting Concepts
[SEP-1] Asking Questions and Defining Problems [SEP-2] Developing and Using Models	LS1.B: Growth and Development of Organisms LS3.A: Inheritance of Traits LS3.B: Variation of Traits ETS1.A: Defining and Delimiting Engineering Problems	[CCC-2] Cause and Effect: Mechanism and Explanation [CCC-6] Structure and Function
CA CCSS Math Connections: 6.SP.5a-d, MP.4		
CA ELD Connections: ELD.PI.7.6a-b,10,9,11a		
CA CCSS ELA/Literacy Connections: SL.7.5; RST.6-8.1,4,7; WHST.6-8.8		

Discipline Specific Grade Seven – Instructional Segment 6: Natural Selection
<i>Guiding Questions:</i> • How do specific traits help organisms access or utilize resources more efficiently? • How does a population benefit from having diversity within it? • What does it mean to have “survival of the fittest?”
Performance Expectations
<i>Students who demonstrate understanding can:</i>

MS-LS1-4. Use argument based on empirical evidence and scientific reasoning to support an explanation for how characteristic animal behaviors and specialized plant structures affect the probability of successful reproduction of animals and plants respectively. [Clarification Statement: Examples of behaviors that affect the probability of animal reproduction could include nest building to protect young from cold, herding of animals to protect young from predators, and vocalization of animals and colorful plumage to attract mates for breeding. Examples of animal behaviors that affect the probability of plant reproduction could include transferring pollen or seeds, and creating conditions for seed germination and growth. Examples of plant structures could include bright flowers attracting butterflies that transfer pollen, flower nectar and odors that attract insects that transfer pollen, and hard shells on nuts that squirrels bury.]

MS-LS2-1. Analyze and interpret data to provide evidence for the effects of resource availability on organisms and populations of organisms in an ecosystem. [Clarification Statement: Emphasis is on cause and effect relationships between resources and growth of individual organisms and the numbers of organisms in ecosystems during periods of abundant and scarce resources.]

MS-LS4-4. Construct an explanation based on evidence that describes how genetic variations of traits in a population increase some individuals' probability of surviving and reproducing in a specific environment. [Clarification Statement: Emphasis is on using simple probability statements and proportional reasoning to construct explanations.]

- MS-LS4-6. Use mathematical representations to support explanations of how natural selection may lead to increases and decreases of specific traits in populations over time. [Clarification Statement: Emphasis is on using mathematical models, probability statements, and proportional reasoning to support explanations of trends in changes to populations over time.] [Assessment Boundary: Assessment does not include Hardy Weinberg calculations.]**
- MS-ETS1-1. Define the criteria and constraints of a design problem with sufficient precision to ensure a successful solution, taking into account relevant scientific principles and potential impacts on people and the natural environment that may limit possible solutions.**
- MS-ETS1-2. Evaluate competing design solutions using a systematic process to determine how well they meet the criteria and constraints of the problem.**
- MS-ETS1-4. Develop a model to generate data for iterative testing and modification of a proposed object, tool, or process such that an optimal design can be achieved.**

The bundle of performance expectations above focuses on the following elements from the NRC document *A Framework for K–12 Science Education*:

Highlighted Science and Engineering Practices	Highlighted Disciplinary Core Ideas	Highlighted Crosscutting Concepts
[SEP-1] Asking Questions and Defining Problems	LS1.B: Growth and Development of Organisms	[CCC-2] Cause and Effect
[SEP-2] Developing and Using Models	LS2.A: Interdependent Relationships in Ecosystems	[CCC-6] Structure and Function

[SEP-4] Analyzing and Interpreting Data [SEP-5] Using mathematics and Computational Thinking [SEP-6] Constructing Explanations (for science) and Designing Solutions (for engineering) [SEP-7] Engaging in Argument from Evidence	LS4.B: Natural Selection LS4.C: Adaptation ETS1.A: Defining and Delimiting Engineering Problems ETS1.B: Developing Possible Solutions ETS1.C: Optimizing the Design Solution	
<i>Highlighted California Environmental Principles & Concepts:</i> Principle I The continuation and health of individual human lives and of human communities and societies depend on the health of the natural systems that provide essential goods and ecosystem services. Principle II The long-term functioning and health of terrestrial, freshwater, coastal and marine ecosystems are influenced by their relationships with human societies. Principle III Natural systems proceed through cycles that humans depend upon, benefit from and can alter. Principle IV The exchange of matter between natural systems and human societies affects the long term functioning of both. Principle V Decisions affecting resources and natural systems are based on a wide range of considerations and decision-making processes.		
<i>CA CCSC Math Connections: 6.SP.2, 6.RP.1, 7.RP.2a-d, MP.4</i>		

CA CCSC ELA/Literacy Connections: RST.6–8.1,7,9; RI.7.8,WHST.6–8.1a-e,2a-f; SL.7.1a-d, SL.7.4

CA ELD Connections: ELD.PI.1.1, 3, 6a, 6b, 10b, 11a

Discipline Specific Grade Seven – Instructional Segment 7:

Ecosystem Interactions, Revisited

Guiding Questions:

- How does natural selection relate to ecosystem changes?
- How do people affect ecosystems? Which activities have a positive impact and which negative?

Performance Expectations

Students who demonstrate understanding can:

MS-LS2-4. Construct an argument supported by empirical evidence that changes to physical or biological components of an ecosystem affect populations. [Clarification Statement: Emphasis is on recognizing patterns in data and making warranted inferences about changes in populations, and on evaluating empirical evidence supporting arguments about changes to ecosystems.]

MS-LS2-5. Evaluate competing design solutions for maintaining biodiversity and ecosystem services.* [Clarification Statement: Examples of ecosystem services could include water purification, nutrient recycling, and prevention of soil erosion. Examples of design solution constraints could include scientific, economic, and social considerations.]

*The performance expectations marked with an asterisk integrate traditional science content with engineering through a practice or disciplinary core idea.

The bundle of performance expectations above focuses on the following elements from the NRC document *A Framework for K–12 Science Education*:

Highlighted Science and Engineering Practices	Highlighted DisciplinaryCore Ideas	Highlighted Crosscutting Concepts
[SEP-7] Engaging in Argument from Evidence	LS2.C: Ecosystem Dynamics, Functioning, and Resilience LS4.C: Adaptation LS4.D: Biodiversity and Humans ETS1.B: Developing Possible Solutions	[CCC-4] System and System Models [CCC-7] Stability and Change

Highlighted California Environmental Principles & Concepts:

- Principle I The continuation and health of individual human lives and of human communities and societies depend on the health of the natural systems that provide essential goods and ecosystem services.
- Principle II The long-term functioning and health of terrestrial, freshwater, coastal and marine ecosystems are influenced by their relationships with human societies.
- Principle III Natural systems proceed through cycles that humans depend upon, benefit from and can alter.
- Principle IV The exchange of matter between natural systems and human societies affects the long term functioning of both.
- Principle V Decisions affecting resources and natural systems are based on a wide range of considerations and decision-making processes.

CA CCSC Math Connections: 6.RP.3a-d, MP.4

CA CCSC ELA/Literacy Connections: RST.6–8.1,8; RI.7.8,WHST.6–8.1a-e,9

Discipline Specific Grade Eight – Instructional Segment 1:

Energy of Motion

Guiding Questions:

- How can understanding energy and forces help make us safer in car crashes?
- What happens to energy when objects collide or otherwise interact?
- Why do objects sometimes appear to slow down on their own?

Performance Expectations

Students who demonstrate understanding can:

MS-PS2-1. Apply Newton’s Third Law to design a solution to a problem involving the motion of two colliding objects.* [Clarification

Statement: Examples of practical problems could include the impact of collisions between two cars, between a car and stationary objects, and between a meteor and a space vehicle.] [Assessment Boundary: Assessment is limited to vertical or horizontal interactions in one dimension.]

MS-PS2-2. Plan an investigation to provide evidence that the change in an object’s motion depends on the sum of the forces on the object and the mass of the object. [Clarification Statement:

Emphasis is on balanced (Newton’s First Law) and unbalanced forces in a system, qualitative comparisons of forces, mass and changes in motion (Newton’s Second Law), frame of reference, and specification of units.] [Assessment Boundary: Assessment is limited to forces and changes in motion in one-dimension in an inertial reference frame and to change in one variable at a time. Assessment does not include the use of trigonometry.]

- MS-PS3-1.** Construct and interpret graphical displays of data to describe the relationships of kinetic energy to the mass of an object and to the speed of an object. [Clarification Statement: Emphasis is on descriptive relationships between kinetic energy and mass separately from kinetic energy and speed. Examples could include riding a bicycle at different speeds, rolling different sizes of rocks downhill, and getting hit by a wiffle ball versus a tennis ball.]
- MS-PS3-5.** Construct, use, and present arguments to support the claim that when the kinetic energy of an object changes, energy is transferred to or from the object. [Clarification Statement: Examples of empirical evidence used in arguments could include an inventory or other representation of the energy before and after the transfer in the form of temperature changes or motion of object.] [Assessment Boundary: Assessment does not include calculations of energy.]
- MS-ETS1-1.** Define the criteria and constraints of a design problem with sufficient precision to ensure a successful solution, taking into account relevant scientific principles and potential impacts on people and the natural environment that may limit possible solutions.
- MS-ETS1-2.** Evaluate competing design solutions using a systematic process to determine how well they meet the criteria and constraints of the problem.
- MS-ETS1-3.** Analyze data from tests to determine similarities and differences among several design solutions to identify the best characteristics of each that can be combined into a new solution to better meet the criteria for success.
- MS-ETS1-4.** Develop a model to generate data for iterative testing and modification of a proposed object, tool, or process such that an optimal design can be achieved.

*The performance expectations marked with an asterisk integrate traditional science content with engineering through a practice or disciplinary core idea.

The bundle of performance expectations above focuses on the following elements from the NRC document <i>A Framework for K–12 Science Education</i>		
Highlighted Science and Engineering Practices	Highlighted Disciplinary Core Ideas	Highlighted Crosscutting Concepts
[SEP-1] Asking Questions and Defining Problems	PS2.A: Forces and Motion	[CCC-2] Cause and Effect: Mechanism and Explanation
[SEP-2] Developing and Using Models	PS3.A Definitions of Energy	[CCC-3] Scale, Proportion, and Quantity
[SEP-3] Planning and Carrying Out Investigations	PS3.B: Conservation of Energy and Energy Transfer	[CCC-4] Systems and System Models
[SEP-4] Analyzing and Interpreting Data	ETS1.A: Defining and Delimiting an Engineering Problem	[CCC-5] Energy and Matter: Flows, Cycles, and Conservation
[SEP-6] Constructing Explanations (for science) and Designing Solutions (for engineering)	ETS1.B: Developing Possible Solutions	[CCC-6] Structure and Function
[SEP-7] Engaging in Argument from Evidence	ETS1.C: Optimizing the Design Solution	[CCC-7] Stability and Change
<i>Highlighted California Environmental Principles & Concepts:</i>		

Principle V Decisions affecting resources and natural systems are based on a wide range of considerations and decision-making processes.
CA CCSC Math Connections: 6.NS.5, 6.RP.1,2, 6.EE.2a-c, 7.EE.3,4; 7.SP.7a-b; 7.RP.2a-d; 8.EE.1,2; 8.F.3, MP.2
CA CCSC ELA/Literacy Connections: RST.6–8.1,3,7,9; WHST.6–8.7,8,9
CA ELD Connections: ELD.PI.8.1, 3, 6a, 6b, 10b, 11a

Discipline Specific Grade Eight – Instructional Segment 2: Gravity, Energy Related to Position
<i>Guiding Questions:</i> • What affects the strength of the force of gravity? • How do roller coasters get the energy to go so fast? • Do heavy objects fall faster than lighter ones?
Performance Expectations
<i>Students who demonstrate understanding can:</i> MS-PS2-2. Plan an investigation to provide evidence that the change in an object's motion depends on the sum of the forces on the object and the mass of the object. [Clarification Statement: Emphasis is on balanced (Newton's First Law) and unbalanced forces in a system, qualitative comparisons of forces, mass and changes in motion (Newton's Second Law), frame of reference, and specification of units.] [Assessment Boundary: Assessment is limited to forces and changes in motion in one-dimension in an inertial reference frame and to change in one variable at a time. Assessment does not include the use of trigonometry.] MS-PS2-4. Construct and present arguments using evidence to support the claim that gravitational interactions are attractive and depend on the masses of interacting objects. [Clarification

Statement: Examples of evidence for arguments could include data generated from simulations or digital tools; and charts displaying mass, strength of interaction, distance from the Sun, and orbital periods of objects within the solar system.]

[Assessment Boundary: Assessment does not include Newton's Law of Gravitation or Kepler's Laws.]

MS-PS2-5. Conduct an investigation and evaluate the experimental design to provide evidence that fields exist between objects exerting forces on each other even though the objects are not in contact. [Clarification Statement: Examples of this phenomenon could include the interactions of magnets, electrically-charged strips of tape, and electrically-charged pith balls. Examples of investigations could include first-hand experiences or simulations.] [Assessment Boundary: Assessment is limited to electric and magnetic fields, and is limited to qualitative evidence for the existence of fields.]

MS-PS3-2. Develop a model to describe that when the arrangement of objects interacting at a distance changes, different amounts of potential energy are stored in the system. [Clarification Statement: Emphasis is on relative amounts of potential energy, not on calculations of potential energy. Examples of objects within systems interacting at varying distances could include: the Earth and either a roller coaster cart at varying positions on a hill or objects at varying heights on shelves, changing the direction/orientation of a magnet, and a balloon with static electrical charge being brought closer to a classmate's hair. Examples of models could include representations, diagrams, pictures, and written descriptions of systems.] [Assessment Boundary: Assessment is limited to two objects and electric, magnetic, and gravitational interactions.]

The bundle of performance expectations above focuses on the following elements from the NRC document <i>A Framework for K–12 Science Education</i> :		
Highlighted Science and Engineering Practices [SEP-2] Developing and Using Models [SEP-3] Planning and Carrying Out Investigations [SEP-7] Engaging in Argument from Evidence	Highlighted Disciplinary Core Ideas PS2.A: Forces and Motion PS2.B Types of Interactions PS3.A Definitions of Energy PS3.B Relationship Between Energy and Force	Highlighted Crosscutting Concepts [CCC-2] Cause and Effect: Mechanism and Explanation [CCC-4] Systems and System Models [CCC-7] Stability and Change Influence of Science, Engineering, and Technology on Society and the Natural World
CA CCSC Math Connections: 6.EE.2a-c; 7.EE.3,4; MP.2		
CA CCSC ELA/Literacy Connections: RST.6–8.3; RI.7.8,WHST.6–8.1a-e,7; SL.8.5		
CA ELD Connections: ELD.PI.1.1, 3, 6a, 6b, 10b, 11a; ELD.PII.1		

Discipline Specific Grade Eight – Instructional Segment 3: Electric and Magnetic Interactions and Energy

Guiding Questions:

- How do electric motors work to convert electricity into motion?
- How does a compass needle move?

Performance Expectations

Students who demonstrate understanding can:

MS-PS2-3. Ask questions about data to determine the factors that affect the strength of electric and magnetic forces. [Clarification Statement: Examples of devices that use electric and magnetic forces could include electromagnets, electric motors, or generators. Examples of data could include the effect of the number of turns of wire on the strength of an electromagnet, or the effect of increasing the number or strength of magnets on the speed of an electric motor.] [Assessment Boundary: Assessment about questions that require quantitative answers is limited to proportional reasoning and algebraic thinking.]

MS-PS2-5. Conduct an investigation and evaluate the experimental design to provide evidence that fields exist between objects exerting forces on each other even though the objects are not in contact. [Clarification Statement: Examples of this phenomenon could include the interactions of magnets, electrically-charged strips of tape, and electrically-charged pith balls. Examples of investigations could include first-hand experiences or simulations.] [Assessment Boundary: Assessment is limited to electric and magnetic fields, and is limited to qualitative evidence for the existence of fields.]

MS-PS3-2. Develop a model to describe that when the arrangement of objects interacting at a distance changes, different amounts

of potential energy are stored in the system. [Clarification Statement: Emphasis is on relative amounts of potential energy, not on calculations of potential energy. Examples of objects within systems interacting at varying distances could include: the Earth and either a roller coaster cart at varying positions on a hill or objects at varying heights on shelves, changing the direction/orientation of a magnet, and a balloon with static electrical charge being brought closer to a classmate's hair. Examples of models could include representations, diagrams, pictures, and written descriptions of systems.] [Assessment Boundary: Assessment is limited to two objects and electric, magnetic, and gravitational interactions.]

MS-ETS1-3. Analyze data from tests to determine similarities and differences among several design solutions to identify the best characteristics of each that can be combined into a new solution to better meet the criteria for success.

The bundle of performance expectations above focuses on the following elements from the NRC document *A Framework for K–12 Science Education*:

Highlighted Science and Engineering Practices	Highlighted Disciplinary Core Ideas	Highlighted Crosscutting Concepts
[SEP-1] Asking Questions and Defining Problems	PS2.B Types of Interactions	[CCC-2] Cause and Effect: Mechanism and Explanation
[SEP-2] Developing and Using Models	PS3.A Definitions of Energy	[CCC-4] Systems and System Models
[SEP-3] Planning and Carrying Out	PS3.B Relationship Between Energy and Force	[CCC-5] Energy and Matter: Flows, Cycles, and Conservation

Investigations [SEP-4] Analyzing and Interpreting Data	ETS1.B: Developing Possible Solutions ETS1.C: Optimizing the Design Solution	
<i>CA CCSC Math Connections: MP.2</i>		
<i>CA CCSC ELA/Literacy Connections: RST.6–8.1,3; WHST.6–8.1a-e, 7</i>		
<i>CA ELD Connections: ELD.PI.8.1, 3, 6a, 6b, 10b, 11a</i>		

Discipline Specific Grade Eight – Instructional Segment 4: Waves Transmitting Energy and Information
<i>Guiding Questions:</i> • How do waves interact with different objects? • How are waves used to move energy and information from place to place?
Performance Expectations
<i>Students who demonstrate understanding can:</i> MS-PS4-1. Use mathematical representations to describe a simple model for waves that includes how the amplitude of a wave is related to the energy in a wave. [Clarification Statement: Emphasis is on describing waves with both qualitative and quantitative thinking.] [Assessment Boundary: Assessment does not include electromagnetic waves and is limited to standard repeating waves.] MS-PS4-2. Develop and use a model to describe that waves are reflected, absorbed, or transmitted through various materials. [Clarification Statement: Emphasis is on both light and

mechanical waves. Examples of models could include drawings, simulations, and written descriptions.] [Assessment Boundary: Assessment is limited to qualitative applications pertaining to light and mechanical waves.]

MS-PS4-3. Integrate qualitative scientific and technical information to support the claim that digitized signals are a more reliable way to encode and transmit information than analog signals.
 [Clarification Statement: Emphasis is on a basic understanding that waves can be used for communication purposes. Examples could include using fiber optic cable to transmit light pulses, radio wave pulses in wifi devices, and conversion of stored binary patterns to make sound or text on a computer screen.]
 [Assessment Boundary: Assessment does not include binary counting. Assessment does not include the specific mechanism of any given device.]

The bundle of performance expectations above focuses on the following elements from the NRC document *A Framework for K–12 Science Education*:

Highlighted Science and Engineering Practices	Highlighted Disciplinary Core Ideas	Highlighted Crosscutting Concepts
[SEP-2] Developing and Using Models [SEP-5] Using Mathematics and Computational Thinking [SEP-7] Engaging in	PS4.A: Wave Properties PS4.B Electromagnetic Radiation	[CCC-1] Patterns [CCC-6] Structure and Function

Argument from Evidence [SEP-8] Obtaining, Evaluating, and Communicating Information	PS4.C: Information Technologies and Instrumentation	
<i>CA CCSC Math Connections:</i> 6.RP.1,3a-d; 7.RP.2a-d; 8.F.3; MP.2,MP.4		
<i>CA CCSC ELA/Literacy Connections:</i> RST.6–8.1,2,9; WHST.6–8.9; SL.8.5		
<i>CA ELD Connections:</i> ELD.PI.1.1, 3, 6a, 6b, 10b, 11a; ELD.PII.1		

Discipline Specific Grade Eight – Instructional Segment 5: Thermal Energy and Heat Flow
<i>Guiding Questions:</i> • How can we represent matter at the microscopic level? • When an object is hot, how is it different from when it is cold? • What happens when hot objects and cold objects interact? • What happens to the kinetic energy of an object when it crashes or collides with the ground and stops?
Performance Expectations
<i>Students who demonstrate understanding can:</i> MS-PS1-4. Develop a model that predicts and describes changes in particle motion, temperature, and state of a pure substance when thermal energy is added or removed. [Clarification Statement: Emphasis is on qualitative molecular-level models of solids, liquids, and gases to show that adding or removing thermal energy increases or decreases kinetic energy of the particles until a change of state occurs. Examples of models

could include drawings and diagrams. Examples of particles could include molecules or inert atoms. Examples of pure substances could include water, carbon dioxide, and helium.]

MS-PS3-3. Apply scientific principles to design, construct, and test a device that either minimizes or maximizes thermal energy transfer.* [Clarification Statement: Examples of devices could include an insulated box, a solar cooker, and a Styrofoam cup.] [Assessment Boundary: Assessment does not include calculating the total amount of thermal energy transferred.]

MS-PS3-4. Plan an investigation to determine the relationships among the energy transferred, the type of matter, the mass, and the change in the average kinetic energy of the particles as measured by the temperature of the sample. [Clarification Statement: Examples of experiments could include comparing final water temperatures after different masses of ice melted in the same volume of water with the same initial temperature, the temperature change of samples of different materials with the same mass as they cool or heat in the environment, or the same material with different masses when a specific amount of energy is added.] [Assessment Boundary: Assessment does not include calculating the total amount of thermal energy transferred.]

MS-PS3-5. Construct, use, and present arguments to support the claim that when the kinetic energy of an object changes, energy is transferred to or from the object. [Clarification Statement: Examples of empirical evidence used in arguments could include an inventory or other representation of the energy before and after the transfer in the form of temperature changes or motion of object.] [Assessment Boundary: Assessment does not include calculations of energy.] (Revisited from IS1.)

MS-ETS1-2. Evaluate competing design solutions using a systematic process to determine how well they meet the criteria and constraints of the problem.

*The performance expectations marked with an asterisk integrate traditional science content with engineering through a practice or disciplinary core idea.

The bundle of performance expectations above focuses on the following elements from the NRC document *A Framework for K–12 Science Education*:

Highlighted Science and Engineering Practices	Highlighted Disciplinary Core Ideas	Highlighted Crosscutting Concepts
[SEP-2] Developing and Using Models	PS1.A: Structure and Properties of Matter	[CCC-2] Cause and Effect: Mechanism and Explanation
[SEP-3] Planning and Carrying Out Investigations	PS3.A: Definitions of Energy	[CCC-3] Scale, Proportion, and Quantity
[SEP-4] Analyzing and Interpreting Data	PS3.B: Conservation of Energy and Energy Transfer	[CCC-5] Energy and Matter: Flows, Cycles, and Conservation
[SEP-6] Constructing Explanations (for science) and Designing Solutions (for engineering)	ETS1.A: Defining and Delimiting an Engineering Problem	
[SEP-7] Engaging in Argument from Evidence	ETS1.B: Developing Possible Solutions	

Highlighted California Environmental Principles & Concepts:

Principle IV The exchange of matter between natural systems and human

societies affects the long term functioning of both.
CA CCSC Math Connections: 6.RP.1; 6.SP.5a-d; 7.EE.3; MP.2
CA CCSC ELA/Literacy Connections: RST.6–8.1,3,9; RI.7.8,WHST.6–8.1a-e, 7,9
CA ELD Connections: ELD.PI.8.1, 3, 6a, 6b, 10b, 11a

Discipline Specific Grade Eight – Instructional Segment 6: Chemical Energy and Reactions
<i>Guiding Questions:</i> • How do car engines turn gasoline into motion? • How do people use technology change natural materials into synthetic ones?
Performance Expectations
<i>Students who demonstrate understanding can:</i> MS-PS1-1. Develop models to describe the atomic composition of simple molecules and extended structures. [Clarification Statement: Emphasis is on developing models of molecules that vary in complexity. Examples of simple molecules could include ammonia and methanol. Examples of extended structures could include sodium chloride or diamonds. Examples of molecular-level models could include drawings, 3D ball and stick structures, or computer representations showing different molecules with different types of atoms.] [Assessment Boundary: Assessment does not include valence electrons and bonding energy, discussing the ionic nature of subunits of complex structures, or a complete description of all individual atoms in a complex molecule or extended structure is not required.]

- MS-PS1-2. Analyze and interpret data on the properties of substances before and after the substances interact to determine if a chemical reaction has occurred.** [Clarification Statement: Examples of reactions could include burning sugar or steel wool, fat reacting with sodium hydroxide, and mixing zinc with hydrogenchloride.] [Assessment Boundary: Assessment is limited to analysis of the following properties: density, melting point, boiling point, solubility, flammability, and odor.]
- MS-PS1-3. Gather and make sense of information to describe that synthetic materials come from natural resources and impact society.** [Clarification Statement: Emphasis is on natural resources that undergo a chemical process to form the synthetic material. Examples of new materials could include new medicine, foods, and alternative fuels.] [Assessment Boundary: Assessment is limited to qualitative information.]
- MS-PS1-4. Develop a model that predicts and describes changes in particle motion, temperature, and state of a pure substance when thermal energy is added or removed.** [Clarification Statement: Emphasis is on qualitative molecular-level models of solids, liquids, and gases to show that adding or removing thermal energy increases or decreases kinetic energy of the particles until a change of state occurs. Examples of models could include drawings and diagrams. Examples of particles could include molecules or inert atoms. Examples of pure substances could include water, carbon dioxide, and helium.]
- MS-PS1-5. Develop and use a model to describe how the total number of atoms does not change in a chemical reaction and thus mass is conserved.** [Clarification Statement: Emphasis is on law of conservation of matter and on physical models or drawings, including digital forms that represent atoms.]

[Assessment

Boundary: Assessment does not include the use of atomic masses, balancing symbolic equations, or intermolecular forces.]

MS-PS1-6. Undertake a design project to construct, test, and modify a device that either releases or absorbs thermal energy by chemical processes.* [Clarification Statement: Emphasis is on the design, controlling the transfer of energy to the environment, and modification of a device using factors such as type and concentration of a substance. Examples of designs could involve chemical reactions such as dissolving ammonium chloride or calcium chloride.] [Assessment Boundary: Assessment is limited to the criteria of amount, time, and temperature of substance in testing the device.]

MS-ETS1-1. Define the criteria and constraints of a design problem with sufficient precision to ensure a successful solution, taking into account relevant scientific principles and potential impacts on people and the natural environment that may limit possible solutions.

MS-ETS1-2. Evaluate competing design solutions using a systematic process to determine how well they meet the criteria and constraints of the problem.

MS-ETS1-3. Analyze data from tests to determine similarities and differences among several design solutions to identify the best characteristics of each that can be combined into a new solution to better meet the criteria for success.

MS-ETS1-4. Develop a model to generate data for iterative testing and modification of a proposed object, tool, or process such that an optimal design can be achieved.

*The performance expectations marked with an asterisk integrate traditional science content with engineering through a practice or disciplinary core idea.

The bundle of performance expectations above focuses on the following elements from the NRC document *A Framework for K–12 Science Education*:

Highlighted Science and Engineering Practices	Highlighted Disciplinary Core Ideas	Highlighted Crosscutting Concepts
[SEP-2] Developing and Using Models [SEP-4] Analyzing and Interpreting Data [SEP-6] Constructing Explanations (for science) and Designing Solutions (for engineering) [SEP-8] Obtaining, Evaluating, and Communicating Information	PS1.A: Structure and Properties of Matter PS1.B: Chemical Reactions PS3.B: Conservation of Energy and Energy Transfer ETS1.A: Defining and Delimiting an Engineering Problem ETS1.B: Developing Possible Solutions ETS1.C: Optimizing the Design Solution	[CCC-1] Patterns [CCC-2] Cause and Effect: Mechanism and Explanation [CCC-3] Scale, Proportion, and Quantity [CCC-5] Energy and Matter: Flows, Cycles, and Conservation [CCC-6] Structure and Function
<i>Highlighted California Environmental Principles & Concepts:</i> Principle II The long-term functioning and health of terrestrial, freshwater, coastal and marine ecosystems are influenced by their relationships with human societies. Principle V Decisions affecting resources and natural systems are based on a wide range of considerations and decision-making processes.		
<i>CA CCSC Math Connections:</i> 6.NS.5; 6.SP.4,5a-d; 6.RP.3, 7; 7.EE.3; 7.SP7a-b; MP.2, MP.4		

CA CCSC ELA/Literacy Connections: RST.6–8.1,3,7,9; WHST.6–8.7,8,9

CA ELD Connections: ELD.PI.8.1, 3, 6a, 6b, 10b, 11a

High School Three Course Model- Chapter Seven

Living Earth – Instructional Segment 1:

Ecosystem Interactions and Energy

Guiding Questions:

- What factors affect the size of populations within an ecosystem?
- What are common threats to remaining natural ecosystems and biodiversity? How can these threats be reduced?

Performance Expectations

Students who demonstrate understanding can:

HS-LS2-1. Use mathematical and/or computational representations to support explanations of factors that affect carrying capacity of ecosystems at different scales. [Clarification Statement:

Emphasis is on quantitative analysis and comparison of the relationships among interdependent factors including boundaries, resources, climate, and competition. Examples of mathematical comparisons could include graphs, charts, histograms, and population changes gathered from simulations or historical data sets.] [Assessment Boundary: Assessment does not include deriving mathematical equations to make comparisons.]

HS-LS2-2. Use mathematical representations to support and revise explanations based on evidence about factors affecting biodiversity and populations in ecosystems of different scales.

[Clarification Statement: Examples of mathematical representations include finding the average, determining trends, and using graphical

comparisons of multiple sets of data.] *[Assessment Boundary: Assessment is limited to provided data.]*

HS-LS2-4. Use mathematical representations to support claims for the cycling of matter and flow of energy among organisms in an ecosystem. *[Clarification Statement: Emphasis is on using a mathematical model of stored energy in biomass to describe the transfer of energy from one trophic level to another and that matter and energy are conserved as matter cycles and energy flows through ecosystems. Emphasis is on atoms and molecules such as carbon, oxygen, hydrogen and nitrogen being conserved as they move through an ecosystem.] [Assessment Boundary: Assessment is limited to proportional reasoning to describe the cycling of matter and flow of energy.]*

HS-LS2-8. Evaluate the evidence for the role of group behavior on individual and species' chances to survive and reproduce. *[Clarification Statement: Emphasis is on: (1) distinguishing between group and individual behavior, (2) identifying evidence supporting the outcomes of group behavior, and (3) developing logical and reasonable arguments based on evidence. Examples of group behaviors could include flocking, schooling, herding, and cooperative behaviors such as hunting, migrating, and swarming.]*

The bundle of performance expectations above focuses on the following elements from the NRC document *A Framework for K–12 Science Education*:

Highlighted Science and Engineering Practices [SEP-2] Developing and Using Models	Highlighted Disciplinary Core Ideas LS2.A: Interdependent Relationships in Ecosystems	Highlighted Crosscutting Concepts [CCC-2] Cause and Effect
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[SEP-5] Using Mathematics and Computational Thinking [SEP-7] Engaging in Argument from Evidence	LS2.C: Ecosystem Dynamics, Functioning, and Resilience LS2.D: Social Interactions and Group Behavior	[CCC-3] Scale, Proportion, and Quantity [CCC-4] System and System Models [CCC-5] Energy and Matter: Flows, Cycles, and Conservation
Highlighted California Environmental Principles & Concepts: Principle II The long-term functioning and health of terrestrial, freshwater, coastal and marine ecosystems are influenced by their relationships with human societies. Principle IV The exchange of matter between natural systems and human societies affects the long term functioning of both.		
CA CCSS Math Connections: N-Q.1-3; S-ID.1; S-IC.1,6; MP.2, MP.4		
CA CCSS ELA/Literacy Connections: RST.9-10.8; RST.11-12.1,7,8; WHST.9-12.2a-e		
CA ELD Connections: ELD.PI.11-12.1,5,6a-b,9,10,11a		
Living Earth – Instructional Segment 2: Earth’s Atmosphere: Photosynthesis and Respiration		
<i>Guiding Questions:</i> • How do living things acquire energy and matter for life? • How do organisms store energy? • How are photosynthesis and cellular respiration connected? • How do organisms use the raw materials they ingest from the environment? • How has the cycling of energy and matter changed over Earth’s history?		
Performance Expectations		
<i>Students who demonstrate understanding can:</i>		

HS-LS1-5. Use a model to illustrate how photosynthesis transforms light energy into stored chemical energy. [Clarification Statement:

Emphasis is on illustrating inputs and outputs of matter and the transfer and transformation of energy in photosynthesis by plants and other photosynthesizing organisms. Examples of models could include diagrams, chemical equations, and conceptual models.]

[Assessment Boundary: Assessment does not include specific biochemical steps.]

HS-LS1-6. Construct and revise an explanation based on evidence for how carbon, hydrogen, and oxygen from sugar molecules may combine with other elements to form amino acids and/or other large carbon-based molecules. [Clarification Statement:

Emphasis is on using evidence from models and simulations to support explanations.] **[Assessment Boundary: Assessment does not include the details of the specific chemical reactions or identification of macromolecules.]**

HS-LS1-7. Use a model to illustrate that cellular respiration is a chemical process whereby the bonds of food molecules and oxygen molecules are broken and the bonds in new compounds are formed resulting in a net transfer of energy.[Clarification

Statement: Emphasis is on the conceptual understanding of the inputs and outputs of the process of cellular respiration.]

[Assessment Boundary: Assessment should not include identification of the steps or specific processes involved in cellular respiration.]

HS-LS2-3. Construct and revise an explanation based on evidence for the cycling of matter and flow of energy in aerobic and anaerobic conditions. [Clarification Statement: Emphasis is on conceptual understanding of the role of aerobic and anaerobic respiration in different environments.] *[Assessment Boundary: Assessment does*

not include the specific chemical processes of either aerobic or anaerobic respiration.]

HS-LS2-5. Develop a model to illustrate the role of photosynthesis and cellular respiration in the cycling of carbon among the biosphere, atmosphere, hydrosphere, and geosphere.

[Clarification Statement: Examples of models could include simulations and mathematical models.] [Assessment Boundary: Assessment does not include the specific chemical steps of photosynthesis and respiration.]

HS-ESS1-6. Apply scientific reasoning and evidence from ancient Earth materials, meteorites, and other planetary surfaces to construct an account of Earth's formation and early history. *[Clarification Statement: Emphasis is on using available evidence within the solar system to reconstruct the early history of Earth, which formed along with the rest of the solar system 4.6 billion years ago. Examples of evidence include the absolute ages of ancient materials (obtained by radiometric dating of meteorites, moon rocks, and Earth's oldest minerals), the sizes and compositions of solar system objects, and the impact cratering record of planetary surfaces.]*

HS-ESS2-6. Develop a quantitative model to describe the cycling of carbon among the hydrosphere, atmosphere, geosphere, and biosphere. *[Clarification Statement: **The carbon cycle is a property of the Earth system that arises from interactions among the hydrosphere, atmosphere, geosphere, and biosphere.** Emphasis is on modeling biogeochemical cycles that include the cycling of carbon through the ocean, atmosphere, soil, and biosphere (including humans), providing the foundation for living organisms.]*

HS-ESS2-7. Construct an argument based on evidence about the simultaneous coevolution of Earth's systems and life on Earth.

[Clarification Statement: Emphasis is on the dynamic causes, effects, and feedbacks between the biosphere and Earth's other systems, whereby geoscience factors control the evolution of life, which in turn continuously alters Earth's surface. Examples of include how photosynthetic life altered the atmosphere through the production of oxygen, which in turn increased weathering rates and allowed for the evolution of animal life; how microbial life on land increased the formation of soil, which in turn allowed for the evolution of land plants; or how the evolution of corals created reefs that altered patterns of erosion and deposition along coastlines and provided habitats for the evolution of new life forms.] **[Assessment Boundary:** Assessment does not include a comprehensive understanding of the mechanisms of how the biosphere interacts with all of Earth's other systems.]

HS-ESS3-6. Use a computational representation to illustrate the relationships among Earth systems and how those relationships are being modified due to human activity.* [Clarification Statement: Examples of Earth systems to be considered are the hydrosphere, atmosphere, cryosphere, geosphere, and/or biosphere. An example of the far-reaching impacts from a human activity is how an increase in atmospheric carbon dioxide results in an increase in photosynthetic biomass on land and an increase in ocean acidification, with resulting impacts on sea organism health and marine populations.] [Assessment Boundary: Assessment does not include running computational representations but is limited to using the published results of scientific computational models.]
(Introduced but not fully assessed until IS6)

*The performance expectations marked with an asterisk integrate traditional science content with engineering through a practice or disciplinary core idea.

The bundle of performance expectations above focuses on the following elements from the NRC document *A Framework for K–12 Science Education*:

Highlighted Science and Engineering Practices	Highlighted Disciplinary Core Ideas	Highlighted Crosscutting Concepts
[SEP-2] Developing and Using Models [SEP-5] Using Mathematical and Computational Thinking [SEP-6] Constructing Explanations (for science) and Designing Solutions (for engineering) [SEP-7] Engaging in Argument from Evidence	LS1.C: Organization for Matter and Energy Flow in Organisms LS2.B: Cycles of Matter and Energy Transfer in Ecosystems PS1.C: Nuclear Processes PS3.D: Energy in Chemical Processes ESS2.D: Weather and Climate ESS2.E: Biogeology ESS3.D: Global Climate Change	[CCC-1] Patterns [CCC-4] System and System Models [CCC-5] Energy and Matter: Flows, Cycles, and Conservation [CCC-7] Stability and Change

Highlighted California Environmental Principles & Concepts:

Principle II The long-term functioning and health of terrestrial, freshwater,

coastal and marine ecosystems are influenced by their relationships with human societies. Principle III Natural systems proceed through cycles that humans depend upon, benefit from and can alter. Principle IV The exchange of matter between natural systems and human societies affects the long term functioning of both.
CA CCSS Math Connections: N-Q.1-3; F.IF.5; S-ID.6.a-c; MP.2, MP.4
CA CCSS ELA/Literacy Connections: SL.11-12.5; RST.11-12.1, WHST.9-12.2a-e, 5, 8, 9
CA ELD Connections: ELD.PI.11-12.1, 5, 6a-b, 9, 10, 11a

Living Earth – Instructional Segment 3: Evidence of Common Ancestry and Diversity
<i>Guiding Questions:</i> • How do layers of rock form and how do they contain fossils? • Why do we see fossils across the world from each other but living organisms that are very different from each other? • What evidence shows that different species are related? • How did modern day humans evolve?
Performance Expectations
<i>Students who demonstrate understanding can:</i> HS-LS4-1. Communicate scientific information that common ancestry and biological evolution are supported by multiple lines of empirical evidence. [Clarification Statement: Emphasis is on a conceptual understanding of the role each line of evidence has relating to common ancestry and biological evolution. Examples of evidence could include similarities in DNA sequences, anatomical structures, and order of appearance of structures in embryological development.]

HS-LS4-2. Construct an explanation based on evidence that the process of evolution primarily results from four factors: (1) the potential for a species to increase in number, (2) the heritable genetic variation of individuals in a species due to mutation and sexual reproduction, (3) competition for limited resources, and (4) the proliferation of those organisms that are better able to survive and reproduce in the environment. [Clarification Statement:

Emphasis is on using evidence to explain the influence each of the four factors has on number of organisms, behaviors, morphology, or physiology in terms of ability to compete for limited resources and subsequent survival of individuals and adaptation of species. Examples of evidence could include mathematical models such as simple distribution graphs and proportional reasoning.]

[Assessment Boundary: Assessment does not include other mechanisms of evolution, such as genetic drift, gene flow through migration, and co-evolution.]

HS-LS4-4. Construct an explanation based on evidence for how natural selection leads to adaptation of populations. [Clarification Statement: Emphasis is on using data to provide evidence for how specific biotic and abiotic differences in ecosystems (such as ranges of seasonal temperature, long-term climate change, acidity, light, geographic barriers, or evolution of other organisms) contribute to a change in gene frequency over time, leading to adaptation of populations.]

HS-LS4-5. Evaluate the evidence supporting claims that changes in environmental conditions may result in: (1) increases in the number of individuals of some species, (2) the emergence of new species over time, and (3) the extinction of other species. [Clarification Statement: Emphasis is on determining cause and effect relationships for how changes to the environment such as

deforestation, fishing, application of fertilizers, drought, flood, and the rate of change of the environment affect distribution or disappearance of traits in species.]

HS-ESS1-5. Evaluate evidence of the past and current movements of continental and oceanic crust and the theory of plate tectonics to explain the ages of crustal rocks. [Clarification Statement: Emphasis is on the ability of plate tectonics to explain the ages of crustal rocks. Examples include evidence of the ages oceanic crust increasing with distance from mid-ocean ridges (a result of plate spreading) and the ages of North American continental crust increasing with distance away from a central ancient core (a result of past plate interactions).] (Introduced, but assessed in High School Chemistry in the Earth System course)

HS-ESS2-5. Plan and conduct an investigation of the properties of water and its effects on Earth materials and surface processes. [Clarification Statement: Emphasis is on mechanical and chemical investigations with water and a variety of solid materials to provide the evidence for connections between the hydrologic cycle and system interactions commonly known as the rock cycle. Examples of mechanical investigations include stream transportation and deposition using a stream table, erosion using variations in soil moisture content, or frost wedging by the expansion of water as it freezes. Examples of chemical investigations include chemical weathering and recrystallization (by testing the solubility of different materials) or melt generation (by examining how water lowers the melting temperature of most solids).]

HS-ESS3-1. Construct an explanation based on evidence for how the availability of natural resources, occurrence of natural hazards, and changes in climate have influenced human activity. [Clarification Statement: Examples of key natural resources include

access to fresh water (such as rivers, lakes, and groundwater), regions of fertile soils such as river deltas, and high concentrations of minerals and fossil fuels. Examples of natural hazards can be from interior processes (such as volcanic eruptions and earthquakes), surface processes (such as tsunamis, mass wasting, and soil erosion), and severe weather (such as hurricanes, floods, and droughts). Examples of the results of changes in climate that can affect populations or drive mass migrations include changes to sea level, regional patterns of temperature and precipitation, and the types of crops and livestock that can be raised.]

HS-ESS3-4. Evaluate or refine a technological solution that reduces impacts of human activities on natural systems.* [Clarification Statement: Examples of data on the impacts of human activities could include the quantities and types of pollutants released, changes to biomass and species diversity, or areal changes in land surface use (such as for urban development, agriculture and livestock, or surface mining). Examples for limiting future impacts could range from local efforts (such as reducing, reusing, and recycling resources) to large-scale geoengineering design solutions (such as altering global temperatures by making large changes to the atmosphere or ocean).]

HS-ETS1-3. Evaluate a solution to a complex real-world problem based on prioritized criteria and trade-offs that account for a range of constraints, including cost, safety, reliability, and aesthetics, as well as possible social, cultural, and environmental impacts.

*The performance expectations marked with an asterisk integrate traditional science content with engineering through a practice or disciplinary core idea.

The bundle of performance expectations above focuses on the following elements from the NRC document *A Framework for K–12 Science Education*:

Highlighted Science and Engineering Practices	Highlighted Disciplinary Core Ideas	Highlighted Crosscutting Concepts
[SEP-3] Planning and Carrying Out Investigations [SEP-6] Constructing Explanations (for science) and Designing Solutions (for engineering) [SEP-7] Engaging in Argument from Evidence [SEP-8] Obtaining, Evaluating, and Communicating Information	LS4.A: Evidence of Common Ancestry and Diversity LS4.B: Natural Selection LS4.C: Adaptation ESS2.B: Plate Tectonics and Large-Scale System Interactions ESS2.C: The Roles of Water in Earth's Surface Processes ESS3.A: Natural Resources ESS3.B: Natural Hazards ESS3.C: Human Impacts on Earth Systems PS1.C: Nuclear Processes ETS1.B: Developing Possible Solutions	[CCC-1] Patterns [CCC-2] Cause and Effect: Mechanism and Explanation [CCC-6] Structure and Function [CCC-7] Stability and Change

Highlighted California Environmental Principles & Concepts:
Principle I The continuation and health of individual human lives and of human communities and societies depend on the health of the natural systems that provide essential goods and ecosystem services.
Principle II The long-term functioning and health of terrestrial, freshwater, coastal and marine ecosystems are influenced by their relationships with human societies.
Principle III Natural systems proceed through cycles that humans depend upon, benefit from and can alter.
Principle IV The exchange of matter between natural systems and human societies affects the long term functioning of both.
Principle V Decisions affecting resources and natural systems are based on a wide range of considerations and decision-making processes.
CA CCSS Math Connections: MP.2; MP.4
CA CCSS ELA/Literacy Connections: SL.11-12.4; RST.11-12.1,8, WHST.9-12.2.a-e, 7, 9
CA ELD Connections: ELD.PI.11-12.1,5,6a-b, 9, 10, 11a

Living Earth – Instructional Segment 4: Inheritance of Traits
<i>Guiding Questions:</i> • How are characteristics of one generation passed to the next? • What allows traits to be transmitted from parents to offspring? • How does variation affect a population under selective pressures?
Performance Expectations
<i>Students who demonstrate understanding can:</i> HS-LS3-1. Ask questions to clarify relationships about the role of DNA and chromosomes in coding the instructions for characteristic traits passed from parents to offspring. [Assessment Boundary: Assessment does not include the phases of meiosis or the biochemical mechanism of specific steps in the process.] HS-LS3-2. Make and defend a claim based on evidence that inheritable genetic variations may result from: (1) new genetic combinations through meiosis, (2) viable errors occurring during replication, and/or (3) mutations caused by

environmental factors. [Clarification Statement: Emphasis is on using data to support arguments for the way variation occurs.]

[Assessment Boundary: Assessment does not include the phases of meiosis or the biochemical mechanism of specific steps in the process.]

HS-LS3-3. Apply concepts of statistics and probability to explain the variation and distribution of expressed traits in a population.

[Clarification Statement: Emphasis is on the use of mathematics to describe the probability of traits as it relates to genetic and environmental factors in the expression of traits.] [Assessment Boundary: Assessment does not include Hardy-Weinberg calculations.]

HS-LS4-2. Construct an explanation based on evidence that the process of evolution primarily results from four factors: (1) the potential for a species to increase in number, (2) the heritable genetic variation of individuals in a species due to mutation and sexual reproduction, (3) competition for limited resources, and (4) the proliferation of those organisms that are better able to survive and reproduce in the environment. [Clarification Statement:

Emphasis is on using evidence to explain the influence each of the four factors has on number of organisms, behaviors, morphology, or physiology in terms of ability to compete for limited resources and subsequent survival of individuals and adaptation of species. Examples of evidence could include mathematical models such as simple distribution graphs and proportional reasoning.]

[Assessment Boundary: Assessment does not include other mechanisms of evolution, such as genetic drift, gene flow through migration, and co-evolution.]

HS-LS4-3. Apply concepts of statistics and probability to support explanations that organisms with an advantageous heritable

trait tend to increase in proportion to organisms lacking this trait. [Clarification Statement: Emphasis is on analyzing shifts in numerical distribution of traits and using these shifts as evidence to support explanations.] [Assessment Boundary: Assessment is limited to basic statistical and graphical analysis. Assessment does not include allele frequency calculations.]

The bundle of performance expectations above focuses on the following elements from the NRC document *A Framework for K–12 Science Education*:

Highlighted Science and Engineering Practices	Highlighted Disciplinary Core Ideas	Highlighted Crosscutting Concepts
[SEP-1] Asking Questions and Defining Problems	LS1.A: Structure and Function	[CCC-1] Patterns
[SEP-4] Analyzing and Interpreting Data	LS3.A: Inheritance of Traits	[CCC-2] Cause and Effect: Mechanism and Explanation
[SEP-6] Constructing Explanations (for science) and Designing Solutions (for engineering)	LS3.B: Variation of Traits	[CCC-3] Scale, Proportion, and Quantity
[SEP-7] Engaging in Argument from Evidence	LS4.B: Natural Selection	
	LS4.C: Adaptation	

Highlighted California Environmental Principles & Concepts:

Principle I The continuation and health of individual human lives and of human

communities and societies depend on the health of the natural systems that provide essential goods and ecosystem services. Principle II The long-term functioning and health of terrestrial, freshwater, coastal and marine ecosystems are influenced by their relationships with human societies. Principle III Natural systems proceed through cycles that humans depend upon, benefit from and can alter. Principle IV The exchange of matter between natural systems and human societies affects the long term functioning of both.
CA CCSS Math Connections: MP.2;MP.4
CA CCSS ELA/Literacy Connections: RST.11-12.1,9, WHST.9-12.1.a-e, 2.a-e, 7, 9
CA ELD Connections: ELD.PI.11-12.1,5,6a-b, 9, 10, 11a

Living Earth – Instructional Segment 5: Structure, Function, and Growth (From Cells to Organisms)
<i>Guiding Questions:</i> • What happens if a cell in our body dies? • How does the structure of DNA affect how cells look and behave? • How do systems work in a multi-celled organism (emergent properties) and what happens if there is a change in the system? • How do organisms survive even when there are changes in their environment?
Performance Expectations
<i>Students who demonstrate understanding can:</i> HS-LS1-1. Construct an explanation based on evidence for how the structure of DNA determines the structure of proteins which carry out the essential functions of life through systems of specialized cells. [Assessment Boundary: Assessment does not include identification of specific cell or tissue types, whole body systems, specific protein structures and functions, or the biochemistry of protein synthesis.]

HS-LS1-2. Develop and use a model to illustrate the hierarchical organization of interacting systems that provide specific functions within multicellular organisms. [Clarification Statement: Emphasis is on functions at the organism system level such as nutrient uptake, water delivery, and organism movement in response to neural stimuli. An example of an interacting system could be an artery depending on the proper function of elastic tissue and smooth muscle to regulate and deliver the proper amount of blood within the circulatory system.] [Assessment Boundary: Assessment does not include interactions and functions at the molecular or chemical reaction level.]

HS-LS1-3. Plan and conduct an investigation to provide evidence that feedback mechanisms maintain homeostasis. [Clarification Statement: Examples of investigations could include heart rate response to exercise, stomate response to moisture and temperature, and root development in response to water levels.] [Assessment Boundary: Assessment does not include the cellular processes involved in the feedback mechanism.]

HS-LS1-4. Use a model to illustrate the role of cellular division (mitosis) and differentiation in producing and maintaining complex organisms. [Assessment Boundary: *Assessment does not include specific gene control mechanisms or rote memorization of the steps of mitosis.*]

The bundle of performance expectations above focuses on the following elements from the NRC document *A Framework for K–12 Science Education*:

Highlighted Science and Engineering Practices	Highlighted Disciplinary Core Ideas	Highlighted Crosscutting Concepts
[SEP-2] Developing and Using Models [SEP-3] Planning and Carrying Out Investigations [SEP-4] Analyzing and Interpreting Data [SEP-6] Constructing Explanations (for science) and Designing Solutions (for engineering)	LS1.A: Structure and Function LS1.B: Growth and Development of Organisms	[CCC-4] System and System Models [CCC-6] Structure and Function [CCC-7] Stability and Change
CA CCSS Math Connections: F-IF.7.a-e; F-BF.1a-c; MP.2; MP.4		
CA CCSS ELA/Literacy Connections: RST.11-12.1,8, WHST.9-12.2.a-e, 7, 9		
CA ELD Connections: ELD.PI.11-12.1,5,6a-b, 9, 10, 11a		

**Living Earth – Instructional Segment 6:
Ecosystem Stability & the Response to Climate Change**

Guiding Questions:

- What effects changes in ecosystems that ultimately effect populations?
- What are the changes that are happening in the climate and what effects are those having on life?
- How are human activities impacting Earth's systems and how does that affect life on Earth?
- What can humans do to mitigate their negative impact on the environment?

Performance Expectations

Students who demonstrate understanding can:

HS-LS2-6. Evaluate the claims, evidence, and reasoning that the complex interactions in ecosystems maintain relatively consistent numbers and types of organisms in stable conditions, but changing conditions may result in a new ecosystem.

[Clarification Statement: Examples of changes in ecosystem conditions could include modest biological or physical changes, such as moderate hunting or a seasonal flood; and extreme changes, such as volcanic eruption or sea level rise.]

HS-LS2-7. Design, evaluate, and refine a solution for reducing the impacts of human activities on the environment and biodiversity.* [Clarification Statement: Examples of human activities can include urbanization, building dams, and dissemination of invasive species.]

HS-LS4-5. Evaluate the evidence supporting claims that changes in environmental conditions may result in: (1) increases in the number of individuals of some species, (2) the emergence of new species over time, and (3) the extinction of other species.

[Clarification Statement: Emphasis is on determining cause and

effect relationships for how changes to the environment such as deforestation, fishing, application of fertilizers, drought, flood, and the rate of change of the environment affect distribution or disappearance of traits in species.]

HS-LS4-6. Create or revise a simulation to test a solution to mitigate adverse impacts of human activity on biodiversity.*

[Clarification Statement: Emphasis is on designing solutions for a proposed problem related to threatened or endangered species, or to genetic variation of organisms for multiple species.]

HS-ESS3-5. Analyze geoscience data and the results from global climate models to make an evidence-based forecast of the current rate of global or regional climate change and associated future impacts to Earth systems. [Clarification Statement: Examples of evidence, for both data and climate model outputs, are for climate changes (such as precipitation and temperature) and their associated impacts (such as on sea level, glacial ice volumes, or atmosphere and ocean composition).] [Assessment Boundary: Assessment is limited to one example of a climate change and its associated impacts.]

HS-ESS3-6. Use a computational representation to illustrate the relationships among Earth systems and how those relationships are being modified due to human activity. [Clarification Statement: Examples of Earth systems to be considered are the hydrosphere, atmosphere, cryosphere, geosphere, and/or biosphere. An example of the far-reaching impacts from a human activity is how an increase in atmospheric carbon dioxide results in an increase in photosynthetic biomass on land and an increase in ocean acidification, with resulting impacts on sea organism health and marine populations.] [Assessment Boundary: Assessment does not

include running computational representations but is limited to using the published results of scientific computational models.]

HS-ETS1-1. Analyze a major global challenge to specify qualitative and quantitative criteria and constraints for solutions that account for societal needs and wants.

HS-ETS1-2. Design a solution to a complex real-world problem by breaking it down into smaller, more manageable problems that can be solved through engineering.

HS-ETS1-3. Evaluate a solution to a complex real-world problem based on prioritized criteria and trade-offs that account for a range of constraints, including cost, safety, reliability, and aesthetics, as well as possible social, cultural, and environmental impacts.

HS-ETS1-4. Use a computer simulation to model the impact of proposed solutions to a complex real-world problem with numerous criteria and constraints on interactions within and between systems relevant to the problem.

*The performance expectations marked with an asterisk integrate traditional science content with engineering through a practice or disciplinary core idea.

The bundle of performance expectations above focuses on the following elements from the NRC document *A Framework for K–12 Science Education*:

Highlighted Science and Engineering Practices	Highlighted Disciplinary Core Ideas	Highlighted Crosscutting Concepts
[SEP-1] Asking Questions and Defining Problems	LS2.C: Ecosystem Dynamics, Functioning, and Resilience	[CCC-2] Cause and Effect: Mechanism and Explanation
[SEP-4] Analyzing	LS4.C: Adaptation	[CCC-4] System and System Models

and Interpreting Data [SEP-5] Using Mathematics and Computational Thinking [SEP-6] Constructing Explanations (for science) and Designing Solutions (for engineering) [SEP-7] Engaging in Argument from Evidence	LS4.D: Biodiversity and Humans ESS2.D: Weather and Climate ESS3.D: Global Climate Change ETS1.A: Defining and Delimiting Engineering Problems ETS1.B: Developing Possible Solutions ETS1.C: Optimizing the Design Solution	[CCC-7] Stability and Change ----- Influence of Science, Engineering, and Technology on Society and the Natural World
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Highlighted California Environmental Principles & Concepts:

- Principle I The continuation and health of individual human lives and of human communities and societies depend on the health of the natural systems that provide essential goods and ecosystem services.
- Principle II The long-term functioning and health of terrestrial, freshwater, coastal and marine ecosystems are influenced by their relationships with human societies.
- Principle III Natural systems proceed through cycles that humans depend upon, benefit from and can alter.
- Principle IV The exchange of matter between natural systems and human societies affects the long term functioning of both.
- Principle V Decisions affecting resources and natural systems are based on a wide range of considerations and decision-making processes.

CA CCSS Math Connections: MP.2; N-Q.1-3; S-ID.1; S-IC.1, 6

CA CCSS ELA/Literacy Connections: RST.9-10.8; RST.11-12.1, 2, 7, 8;
WHST.9-12.2.a-e, 7, 8, 9

CA ELD Connections: ELD.PI.11-12.1, 5, 6a-b, 9, 10, 11a

Chemistry in the Earth System – Instructional Segment 1: Combustion
<i>Guiding Questions:</i> • What is energy, how is it measured, and how does it flow within a system? • What mechanisms allow us to utilize the energy of our foods and fuels?
Performance Expectations
<i>Students who demonstrate understanding can:</i> HS-PS1-3. Plan and conduct an investigation to gather evidence to compare the structure of substances at the bulk scale to infer the strength of electrical forces between particles. [Clarification Statement: Emphasis is on understanding the strengths of forces between particles, not on naming specific intermolecular forces (such as dipole-dipole). Examples of particles could include ions, atoms, molecules, and networked materials (such as graphite). Examples of bulk properties of substances could include the melting point and boiling point, vapor pressure, and surface tension.] [Assessment Boundary: Assessment does not include Raoult's law calculations of vapor pressure.] (Introduced, but not assessed until IS3) HS-PS1-4. Develop a model to illustrate that the release or absorption of energy from a chemical reaction system depends upon the changes in total bond energy. [Clarification Statement: Emphasis is on the idea that a chemical reaction is a system that affects the energy change. Examples of models could include molecular-level drawings and diagrams of reactions, graphs showing the relative energies of reactants and products, and representations showing energy is conserved.] [Assessment Boundary: Assessment does not include calculating the total bond energy changes during a chemical reaction from the bond energies of reactants and products.] (Introduced, but not assessed until IS4) HS-PS1-7. Use mathematical representations to support the claim that atoms, and therefore mass, are conserved during a chemical reaction. [Clarification Statement: Emphasis is on using mathematical ideas to communicate the proportional relationships between masses of atoms in the reactants and the products, and the translation of

these relationships to the macroscopic scale using the mole as the conversion from the atomic to the macroscopic scale. Emphasis is on assessing students' use of mathematical thinking and not on memorization and rote application of problem-solving techniques.] [Assessment Boundary: Assessment does not include complex chemical reactions.] (Introduced, but not assessed until IS6. Revisited in IS7.)

HS-PS3-1. Create a computational model to calculate the change in the energy of one component in a system when the change in energy of the other component(s) and energy flows in and out of the system are known. [Clarification Statement: Emphasis is on explaining the meaning of mathematical expressions used in the model.] [Assessment Boundary: Assessment is limited to basic algebraic expressions or computations; to systems of two or three components; and to thermal energy, kinetic energy, and/or the energies in gravitational, magnetic, or electric fields.] (Introduced, but not assessed until IS2)

The bundle of performance expectations above focuses on the following elements from the NRC document *A Framework for K–12 Science Education*:

Highlighted Science and Engineering Practices	Highlighted Disciplinary Core Ideas	Highlighted Crosscutting Concepts
[SEP-2] Developing and Using Models	PS1.A: Structure and Properties of Matter	[CCC-1] Patterns
[SEP-3] Planning and Carrying out Investigations	PS1.B: Chemical Reactions	[CCC-4] System and System Models
[SEP-5] Using Mathematical and Computational Thinking	PS3.A: Definitions of Energy	[CCC-5] Energy and Matter: Flows, Cycles, and Conservation
	PS3.B: Conservation of Energy and Energy Transfer	
	PS3.D: Energy and Chemical Processes in Everyday Life	

CA CCSS Math Connections: MP.4; N-Q.1-3;
CA CCSS ELA/Literacy Connections: SL.11-12.5; RST.11-12.1; WHST.9-12.7, 8, 9
CA ELD Connections: ELD.PI.11-12.1, 5, 6a-b, 9, 10, 11a

Chemistry in the Earth System – Instructional Segment 2: Heat and Energy in the Earth System
<i>Guiding Questions:</i> • How is energy transferred and conserved? • How can energy be harnessed to perform useful tasks?
Performance Expectations
<i>Students who demonstrate understanding can:</i> HS-PS3-1. Create a computational model to calculate the change in the energy of one component in a system when the change in energy of the other component(s) and energy flows in and out of the system are known. [Clarification Statement: Emphasis is on explaining the meaning of mathematical expressions used in the model.] [Assessment Boundary: Assessment is limited to basic algebraic expressions or computations; to systems of two or three components; and to thermal energy, kinetic energy, and/or the energies in gravitational, magnetic, or electric fields.] HS-PS3-2. Develop and use models to illustrate that energy at the macroscopic scale can be accounted for as a combination of energy associated with the motions of particles (objects) and energy associated with the relative position of particles (objects). [Clarification Statement: Examples of phenomena at the macroscopic scale could include the conversion of kinetic energy to thermal energy, the energy stored due to position of an object above the earth, and the energy stored between two electrically-charged plates. Examples of models could include diagrams, drawings, descriptions, and computer simulations.] HS-PS3-4. Plan and conduct an investigation to provide evidence that the transfer of thermal energy when two components of different temperature are combined within a closed system results in a more uniform energy distribution among the components in the system (second law of thermodynamics). [Clarification Statement: Emphasis is on analyzing data from student investigations and using mathematical thinking to describe the energy changes both quantitatively and conceptually. Examples of investigations could include mixing liquids at different initial temperatures or adding objects

at different temperatures to water.] [Assessment Boundary: Assessment is limited to investigations based on materials and tools provided to students.]

HS-ESS2-3. Develop a model based on evidence of Earth’s interior to describe the cycling of matter by thermal convection.

[Clarification Statement: Emphasis is on both a one-dimensional model of Earth, with radial layers determined by density, and a three-dimensional model, which is controlled by mantle convection and the resulting plate tectonics. Examples of evidence include maps of Earth’s three-dimensional structure obtained from seismic waves, records of the rate of change of Earth’s magnetic field (as constraints on convection in the outer core), and identification of the composition of Earth’s layers from high-pressure laboratory experiments.]

HS-ETS1-4. Use a computer simulation to model the impact of proposed solutions to a complex real-world problem with numerous criteria and constraints on interactions within and between systems relevant to the problem.

The bundle of performance expectations above focuses on the following elements from the NRC document *A Framework for K–12 Science Education*:

Highlighted Science and Engineering Practices	Highlighted Disciplinary Core Ideas	Highlighted Crosscutting Concepts
[SEP-2] Developing and Using Models	PS3.A: Definitions of Energy	[CCC-4] Systems and System Models
[SEP-3] Planning and Carrying Out Investigations	PS3.B: Conservation of Energy and Energy Transfer	[CCC-5] Energy and Matter: Flows, Cycles, and Conservation
[SEP-5] Using mathematics and Computational Thinking	PS3.D: Energy in Chemical Processes	
	PS4.A: Wave Properties	

[SEP-6] Constructing Explanations (for science) and Designing Solutions (for engineering)	ESS2.A: Earth Materials and Systems ESS2.B: Plate Tectonics and Large-Scale System Interactions ETS1.B: Developing Possible Solutions	
CA CCSS Math Connections: N-Q.1-3; MP.2; MP.4		
CA CCSS ELA/Literacy Connections: SL.11-12.4, 5; RST.11-12.1, 2, 8, WHST.9-12.7, 8, 9		
CA ELD Connections: ELD.PI.11-12.1, 5, 6a-b, 9, 10, 11a		

Chemistry in the Earth System – Instructional Segment 3: Atoms, Elements, and Molecules
<i>Guiding Questions:</i> • What is inside atoms and how does this affect how they interact? • What models can we use to predict the outcomes of chemical reactions?
Performance Expectations
<i>Students who demonstrate understanding can:</i> HS-PS1-1. Use the periodic table as a model to predict the relative properties of elements based on the patterns of electrons in the outermost energy level of atoms. [Clarification Statement: Examples of properties that could be predicted from patterns could include reactivity of metals, types of bonds formed, numbers of bonds formed, and reactions with oxygen.] [Assessment Boundary: Assessment is limited to main group elements. Assessment does not include quantitative understanding of ionization energy beyond relative trends.] HS-PS1-2 Construct and revise an explanation for the outcome of a simple chemical reaction based on the outermost electron states of atoms, trends in the periodic table, and knowledge of the

patterns of chemical properties. [Clarification Statement: Examples of chemical reactions could include the reaction of sodium and chlorine, of carbon and oxygen, or of carbon and hydrogen.] [Assessment Boundary: Assessment is limited to chemical reactions involving main group elements and combustion reactions.] HS-PS1-7 Use mathematical representations to support the claim that atoms, and therefore mass, are conserved during a chemical reaction. [Clarification Statement: Emphasis is on using mathematical ideas to communicate the proportional relationships between masses of atoms in the reactants and the products, and the translation of these relationships to the macroscopic scale using the mole as the conversion from the atomic to the macroscopic scale. Emphasis is on assessing students' use of mathematical thinking and not on memorization and rote application of problem- solving techniques.] [Assessment Boundary: Assessment does not include complex chemical reactions.] (Introduced here and revisited again in IS4 and IS6)		
The bundle of performance expectations above focuses on the following elements from the NRC document <i>A Framework for K–12 Science Education</i>:		
Highlighted Science and Engineering Practices [SEP-2] Developing and Using Models [SEP-5] Using Mathematics and Computational Thinking [SEP-6] Constructing Explanations (for	Highlighted Disciplinary Core Ideas PS1.A: Structure and Properties of matter PS1.B: Chemical Reactions	Highlighted Crosscutting Concepts [CCC-1] Patterns [CCC-2] Cause and Effect [CCC-5] Energy and Matter: Flows, Cycles, and Conservation

science) and Designing Solutions (for engineering) [SEP-7] Engaging in Argument from Evidence		
CA CCSS Math Connections: N-Q.1-3; MP.2;		
CA CCSS ELA/Literacy Connections: SL.11-12.4; RST.9-10.7, WHST.11-12.2, 5		
CA ELD Connections: ELD.PI.11-12.1, 5, 6a-b, 9, 10, 11a		

Chemistry in the Earth System – Instructional Segment 4: Chemical Reactions
<i>Guiding Questions:</i> • What holds atoms together in molecules? • How do chemical reactions absorb and release energy?
Performance Expectations
<i>Students who demonstrate understanding can:</i> HS-PS1-3 Plan and conduct an investigation to gather evidence to compare the structure of substances at the bulk scale to infer the strength of electrical forces between particles. <i>[Clarification Statement: Emphasis is on understanding the strengths of forces between particles, not on naming specific intermolecular forces (such as dipole-dipole). Examples of particles could include ions, atoms, molecules, and networked materials (such as graphite). Examples of bulk properties of substances could include the melting point and boiling point, vapor pressure, and surface tension.] [Assessment Boundary: Assessment does not include Raoult’s law calculations of vapor pressure.]</i> HS-PS1-4 Develop a model to illustrate that the release or absorption of energy from a chemical reaction system depends upon the changes in total bond energy. <i>[Clarification Statement: Emphasis is</i>

on the idea that a chemical reaction is a system that affects the energy change. Examples of models could include molecular-level drawings and diagrams of reactions, graphs showing the relative energies of reactants and products, and representations showing energy is conserved.] [Assessment Boundary: Assessment does not include calculating the total bond energy changes during a chemical reaction from the bond energies of reactants and products.]

HS-PS1-5. Apply scientific principles and evidence to provide an explanation about the effects of changing the temperature or concentration of the reacting particles on the rate at which a reaction occurs. [Clarification Statement: Emphasis is on student reasoning that focuses on the number and energy of collisions between molecules.] [Assessment Boundary: Assessment is limited to simple reactions in which there are only two reactants; evidence from temperature, concentration, and rate data; and qualitative relationships between rate and temperature.]

HS-PS1-7 Use mathematical representations to support the claim that atoms, and therefore mass, are conserved during a chemical reaction. [Clarification Statement: Emphasis is on using mathematical ideas to communicate the proportional relationships between masses of atoms in the reactants and the products, and the translation of these relationships to the macroscopic scale using the mole as the conversion from the atomic to the macroscopic scale. Emphasis is on assessing students' use of mathematical thinking and not on memorization and rote application of problem-solving techniques.] [Assessment Boundary: Assessment does not include complex chemical reactions.]

(Introduced in IS3 and revisited again in IS6)

HS-PS2-4 Use mathematical representations of Newton's Law of Gravitation and Coulomb's Law to describe and predict the gravitational and electrostatic forces between objects. [Clarification Statement: Emphasis is on both quantitative and conceptual descriptions of gravitational and electric fields.] [Assessment Boundary: Assessment is limited to systems with two objects.]

HS-PS3-5 Develop and use a model of two objects interacting through electric or magnetic fields to illustrate the forces between objects and the changes in energy of the objects due to the interaction. [Clarification Statement: Examples of models could include drawings, diagrams, and texts, such as drawings of what happens when two charges of opposite polarity are near each other.] [Assessment Boundary: Assessment is limited to systems containing two objects.]

The bundle of performance expectations above focuses on the following elements from the NRC document *A Framework for K–12 Science Education*:

Highlighted Science and Engineering Practices	Highlighted Disciplinary Core Ideas	Highlighted Crosscutting Concepts
[SEP-2] Developing and Using Models [SEP-3] Planning and Carrying Out Investigations [SEP-5] Using Mathematical and Computational Thinking [SEP-6] Constructing Explanations (for science) and Designing Solutions (for engineering)	PS1.A: Structure and Properties of Matter PS1.B: Chemical reactions PS2.B: Types of Interactions PS3.C: Relationship Between Energy and Forces	[CCC-1] Patterns [CCC-2] Cause and Effect [CCC-5] Energy and Matter: Flows, Cycles, and Conservation
CA CCSS Math Connections: A-SSE.1a-b, 3a-c; N-Q.1-3; MP.2; MP.4		
CA CCSS ELA/Literacy Connections: SL.11-12.5; RST.11-12.1, WHST.11-12.7, 8, 9		
CA ELD Connections: ELD.PI.11-12.1,5,6a-b, 9, 10, 11a		

Chemistry in the Earth System – Instructional Segment 5: Chemistry of Climate Change
<i>Guiding Questions:</i> • What regulates weather and climate? • What effects are humans having on the climate?
Performance Expectations
<i>Students who demonstrate understanding can:</i> HS-ESS2-2. Analyze geoscience data to make the claim that one change to Earth's surface can create feedbacks that cause changes to other Earth systems. [Clarification Statement: Examples should include climate feedbacks, such as how an increase in greenhouse gases causes a rise in global temperatures that melts glacial ice, which reduces the amount of sunlight reflected from Earth's surface, increasing surface temperatures and further reducing the amount of ice. Examples could also be taken from other system interactions, such as how the loss of ground vegetation causes an increase in water runoff and soil erosion; how dammed rivers increase groundwater recharge, decrease sediment transport, and increase coastal erosion; or how the loss of wetlands causes a decrease in local humidity that further reduces the wetland extent.] HS-ESS2-4. Use a model to describe how variations in the flow of energy into and out of Earth's systems result in changes in climate. [Clarification Statement: Examples of the causes of climate change differ by timescale, over 1-10 years: large volcanic eruption, ocean circulation; 10-100s of years: changes in human activity, ocean circulation, solar output; 10-100s of thousands of years: changes to Earth's orbit and the orientation of its axis; and 10-100s of millions of years: long-term changes in atmospheric composition.] [Assessment Boundary: Assessment of the results of changes in climate is limited to changes in surface temperatures, precipitation patterns, glacial ice volumes, sea levels, and biosphere distribution.] HS-ESS2-6. Develop a quantitative model to describe the cycling of carbon among the hydrosphere, atmosphere, geosphere, and biosphere. [Clarification Statement: The carbon cycle is a property of the Earth system that arises from interactions among the

hydrosphere, atmosphere, geosphere, and biosphere. Emphasis is on modeling biogeochemical cycles that include the cycling of carbon through the ocean, atmosphere, soil, and biosphere (including humans), providing the foundation for living organisms.]

HS-ESS3-2. Evaluate competing design solutions for developing, managing, and utilizing energy and mineral resources based on cost-benefit ratios.* **[Clarification Statement: Emphasis is on the conservation, recycling, and reuse of resources (such as minerals and metals) where possible, and on minimizing impacts where it is not. Examples include developing best practices for agricultural soil use, mining (for coal, tar sands, and oil shales), and pumping (for petroleum and natural gas). Science knowledge indicates what can happen in natural systems—not what should happen.]**

HS-ESS3-5. Analyze geoscience data and the results from global climate models to make an evidence-based forecast of the current rate of global or regional climate change and associated future impacts to Earth systems. **[Clarification Statement: Examples of evidence, for both data and climate model outputs, are for climate changes (such as precipitation and temperature) and their associated impacts (such as on sea level, glacial ice volumes, or atmosphere and ocean composition).]** **[Assessment Boundary: Assessment is limited to one example of a climate change and its associated impacts.]**

HS-ESS3-6. Use a computational representation to illustrate the relationships among Earth systems and how those relationships are being modified due to human activity.* **[Clarification Statement: Examples of Earth systems to be considered are the hydrosphere, atmosphere, cryosphere, geosphere, and/or biosphere. An example of the far-reaching impacts from a human activity is how an increase in atmospheric carbon dioxide results in an increase in photosynthetic biomass on land and an increase in ocean acidification, with resulting impacts on sea organism health and marine populations.]** **[Assessment Boundary: Assessment does not include running computational representations but is limited to using the published results of scientific computational models.]**

* The performance expectations marked with an asterisk integrate traditional science content with engineering through a practice or disciplinary core idea.

The bundle of performance expectations above focuses on the following elements from the NRC document *A Framework for K–12 Science Education*:

Highlighted Science and Engineering Practices	Highlighted Disciplinary Core Ideas	Highlighted Crosscutting Concepts
[SEP-2] Developing and Using Models	ESS1.B: Earth and the Solar System	[CCC-2] Cause and Effect
[SEP-4] Analyzing and Interpreting Data	ESS2.A: Earth Materials and Systems	[CCC-4] System and System Models
[SEP-5] Using Mathematics and Computational Thinking	ESS2.D: Weather and Climate	[CCC-5] Energy and Matter: Flows, Cycles, and Conservation
[SEP-7] Engaging in Argument from Evidence	PS3.B: Conservation of Energy and Energy Transfer	[CCC-7] Stability and Change
	PS3.D: Energy and Chemical Processes in Everyday Life	
	PS4.B: Electromagnetic Radiation	
	ESS3.A: Natural Resources	
	ESS3.D: Global Climate Change	
<i>Highlighted California Environmental Principles & Concepts:</i>		

Principle III Natural systems proceed through cycles that humans depend upon, benefit from and can alter. Principle V Decisions affecting resources and natural systems are complex and involve many factors.
CA CCSS Math Connections: MP. 1, MP. 2, MP. 3, MP.4; N-Q.1; LE.1b,c; S-ID.6,7
CA CCSS ELA/Literacy Connections: SL.9-10.1c-d; SL.11-12.1c-d; WHST.9-10.4, 6, 9, 10; RST.9-10.1, 7, 9.
CA ELD Connections: ELD.PI.9-10.1,2,3,6a-b,11a., ELD.PII.9-10.1
Chemistry in the Earth System – Instructional Segment 6: The Dynamics of Chemical Reactions and Ocean Acidification
<i>Guiding Questions:</i> • How can you alter chemical equilibrium and reaction rates? • How can you predict the relative quantities of products in a chemical reaction?
Performance Expectations
<i>Students who demonstrate understanding can:</i> HS-PS1-5. Apply scientific principles and evidence to provide an explanation about the effects of changing the temperature or concentration of the reacting particles on the rate at which a reaction occurs. [Clarification Statement: Emphasis is on student reasoning that focuses on the number and energy of collisions between molecules.] [Assessment Boundary: Assessment is limited to simple reactions in which there are only two reactants; evidence from temperature, concentration, and rate data; and qualitative relationships between rate and temperature.] HS-PS1-6 Refine the design of a chemical system by specifying a change in conditions that would produce increased amounts of products at equilibrium.* [Clarification Statement: Emphasis is on the application of Le Chatelier’s Principle and on refining designs of chemical reaction systems, including descriptions of the connection between changes made at the macroscopic level and what happens at the molecular level. Examples of designs could include different ways to increase product formation including adding reactants or removing products.] [Assessment Boundary: Assessment is limited to specifying the change in only one variable at a time. Assessment does not include calculating equilibrium constants and concentrations.]

HS-PS1-7 Use mathematical representations to support the claim that atoms, and therefore mass, are conserved during a chemical reaction. [Clarification Statement: Emphasis is on using mathematical ideas to communicate the proportional relationships between masses of atoms in the reactants and the products, and the translation of these relationships to the macroscopic scale using the mole as the conversion from the atomic to the macroscopic scale. Emphasis is on assessing students' use of mathematical thinking and not on memorization and rote application of problem-solving techniques.]
[Assessment Boundary: Assessment does not include complex chemical reactions.]

(Revisited from IS3 and IS4)

HS-ESS2-2. Analyze geoscience data to make the claim that one change to Earth's surface can create feedbacks that cause changes to other Earth systems. [Clarification Statement: Examples should include climate feedbacks, such as how an increase in greenhouse gases causes a rise in global temperatures that melts glacial ice, which reduces the amount of sunlight reflected from Earth's surface, increasing surface temperatures and further reducing the amount of ice. Examples could also be taken from other system interactions, such as how the loss of ground vegetation causes an increase in water runoff and soil erosion; how dammed rivers increase groundwater recharge, decrease sediment transport, and increase coastal erosion; or how the loss of wetlands causes a decrease in local humidity that further reduces the wetland extent.]

HS-ESS2-6. Develop a quantitative model to describe the cycling of carbon among the hydrosphere, atmosphere, geosphere, and biosphere. [Clarification Statement: **The carbon cycle is a property of the Earth system that arises from interactions among the hydrosphere, atmosphere, geosphere, and biosphere.** Emphasis is on modeling biogeochemical cycles that include the cycling of carbon through the ocean, atmosphere, soil, and biosphere (including humans), providing the foundation for living organisms.]

*The performance expectations marked with an asterisk integrate traditional science content with engineering through a practice or disciplinary core idea.

The bundle of performance expectations above focuses on the following elements from the NRC document *A Framework for K–12 Science Education*:

Highlighted Science and Engineering Practices [SEP-2] Developing and Using Models [SEP-4] Analyzing and Interpreting Data [SEP-5] Using Mathematics and Computational Thinking [SEP-6] Constructing Explanations (for science) and Designing Solutions (for engineering) [SEP-7] Engaging in Argument from Evidence	Highlighted Disciplinary Core Ideas PS1.B: Chemical reactions ESS2.A: Earth Materials and Systems ESS2.D: Weather and Climate ETS1.C: Optimizing the Design Solution	Highlighted Crosscutting Concepts [CCC-1] Patterns [CCC-5] Energy and Matter: Flows, Cycles, and Conservation [CCC-7] Stability and Change
Highlighted California Environmental Principles & Concepts:		
Principle III Natural systems proceed through cycles that humans depend upon, benefit from and can alter.		
Principle IV The exchange of matter between natural systems and human societies affects the long term functioning of both.		
CA CCSS Math Connections: N-Q.1-3; MP.2; MP.4		
CA CCSS ELA/Literacy Connections: SL.11-12.5; RST.11-12.1, 2		
CA ELD Connections: ELD.PI.11-12.1, 5, 6a-b, 9, 10, 11a		

Physics of the Universe – Instructional Segment 1: Forces and Motion
<i>Guiding Questions</i> • How can Newton’s Laws be used to explain how and why things move? • How can mathematical models of Newton’s Laws be used to test and improve engineering designs?
Performance Expectations
<i>Students who demonstrate understanding can:</i> HS-PS2-1. Analyze data to support the claim that Newton’s second law of motion describes the mathematical relationship among the net force on a macroscopic object, its mass, and its acceleration. [Clarification Statement: Examples of data could include tables or graphs of position or velocity as a function of time for objects subject to a net unbalanced force, such as a falling object, an object rolling down a ramp, or a moving object being pulled by a constant force.] [Assessment Boundary: Assessment is limited to one-dimensional motion and to macroscopic objects moving at non-relativistic speeds.] HS-PS2-2. Use mathematical representations to support the claim that the total momentum of a system of objects is conserved when there is no net force on the system. [Clarification Statement: Emphasis is on the quantitative conservation of momentum in interactions and the qualitative meaning of this principle.] [Assessment Boundary: Assessment is limited to systems of two macroscopic bodies moving in one dimension.] HS-PS2-3. Apply scientific and engineering ideas to design, evaluate, and refine a device that minimizes the force on a macroscopic object during a collision.* [Clarification Statement: Examples of evaluation and refinement could include determining the success of the device at protecting an object from damage and modifying the design to improve it. Examples of a device could include a football helmet or a parachute.] [Assessment Boundary: Assessment is limited to qualitative evaluations and/or algebraic manipulations.] HS-ETS1-1. Analyze a major global challenge to specify qualitative and quantitative criteria and constraints for solutions that account for societal needs and wants. HS-ETS1-2. Design a solution to a complex real-world problem by breaking it down into smaller, more manageable problems that can be solved through engineering. HS-ETS1-3. Evaluate a solution to a complex real-world problem based on prioritized criteria and trade-offs that account for a range of

constraints, including cost, safety, reliability, and aesthetics as well as possible social, cultural, and environmental impacts. HS-ETS1-4. Use a computer simulation to model the impact of proposed solutions to a complex real-world problem with numerous criteria and constraints on interactions within and between systems relevant to the problem. *The performance expectations marked with an asterisk integrate traditional science content with engineering through a practice or disciplinary core idea.		
The bundle of performance expectations above focuses on the following elements from the NRC document <i>A Framework for K–12 Science Education</i>:		
Highlighted Science and Engineering Practices	Highlighted Disciplinary Core Ideas	Highlighted Crosscutting Concepts
[SEP-1] Asking Questions and Defining Problems	PS2.A : Forces and Motion	[CCC-2] Cause and Effect
[SEP-4] Analyzing and Interpreting Data	PS2.B: Types of Interactions	[CCC-4] System and System Models
[SEP-5] Using mathematics and Computational Thinking	ETS1.A: Defining and Delimiting Engineering Problems	
[SEP-6] Constructing Explanations (for science) and Designing Solutions (for engineering)	ETS1.B: Developing Possible Solutions	
	ETS1.C: Optimizing the Design Solution	

Highlighted California Environmental Principles & Concepts: Principle V Decisions affecting resources and natural systems are based on a wide range of considerations and decision-making processes.
CA CCSS Math Connections: N-Q.1-3; A-SSE.1a-b, 3a-c; A-CED.1, 2, 4; F-IF.7.a-e; S-ID.1; MP.2; MP.4
CA CCSS ELA/Literacy Connections: SL.11-12.4,5; RST.11-12.1, 7, 8, WHST.9-12.9
CA ELD Connections: ELD.PI.11-12.1, 5, 6a-b, 9, 10, 11a

Physics in the Universe – Instructional Segment 2: Forces at a Distance
<i>Guiding Questions</i> • How can different objects interact when they are not even touching? • How do interactions between matter at the microscopic scale affect the macroscopic properties of matter that we observe? • How do satellites stay in orbit?
Performance Expectations
<i>Students who demonstrate understanding can:</i> HS-PS2-4 Use mathematical representations of Newton’s Law of Gravitation and Coulomb’s Law to describe and predict the gravitational and electrostatic forces between objects. [Clarification Statement: Emphasis is on both quantitative and conceptual descriptions of gravitational and electric fields.] [Assessment Boundary: Assessment is limited to systems with two objects.] HS-PS2-6 Communicate scientific and technical information about why the molecular-level structure is important in the functioning of designed materials.* [Clarification Statement: Emphasis is on the attractive and repulsive forces that determine the functioning of the material. Examples could include why electrically conductive materials are often made of metal, flexible but durable materials are made up of long chained molecules, and pharmaceuticals are designed to interact with specific receptors.] [Assessment Boundary: Assessment is limited to provided molecular structures of specific designed materials.] HS-ESS1-4. Use mathematical or computational representations to predict the motion of orbiting objects in the solar system. [Clarification Statement: Emphasis is on Newtonian gravitational laws governing orbital motions, which apply to human-made satellites as

well as planets and moons.] [Assessment Boundary: Mathematical representations for the gravitational attraction of bodies and Kepler's Laws of orbital motions should not deal with more than two bodies, nor involve calculus.] *The performance expectations marked with an asterisk integrate traditional science content with engineering through a practice or disciplinary core idea. The bundle of performance expectations above focuses on the following elements from the NRC document <i>A Framework for K–12 Science Education</i>:		
Highlighted Science and Engineering Practices [SEP-5] Using mathematics and Computational Thinking [SEP-8] Obtaining, Evaluating, and Communicating Information	Highlighted Disciplinary Core Ideas PS2.B: Types of Interactions ESS1.B: Earth and the Solar System	Highlighted Crosscutting Concepts [CCC-1] Patterns [CCC-2] Cause and Effect: Mechanism and Explanation [CCC-3] Scale, Proportion, and Quantity [CCC-6] Structure and Function
CA CCSS Math Connections: N-Q.1-3; A-SSE.1a-b, 3a-c; A-CDE.2; A-CDE.4; MP.2; MP.4		
CA CCSS ELA/Literacy Connections: RST.11-12.1, WHST.9-12.2.a-e		
CA ELD Connections: ELD.PI.11-12.1,5, 6a-b, 9, 10, 11a		

Physics in the Universe – Instructional Segment 3:

Energy Conversion and Renewable Energy
<i>Guiding Questions</i> • How do power plants generate electricity? • What engineering designs can help increase the efficiency of our electricity production and reduce the negative impacts of using fossil fuels?
Performance Expectations
<i>tudents who demonstrate understanding can:</i> HS-PS2-5 Plan and conduct an investigation to provide evidence that an electric current can produce a magnetic field and that a changing magnetic field can produce an electric current. [Assessment Boundary: Assessment is limited to designing and conducting investigations with provided materials and tools.] HS-PS3-1. Create a computational model to calculate the change in the energy of one component in a system when the change in energy of the other component(s) and energy flows in and out of the system are known. [Clarification Statement: Emphasis is on explaining the meaning of mathematical expressions used in the model.] [Assessment Boundary: Assessment is limited to basic algebraic expressions or computations; to systems of two or three components; and to thermal energy, kinetic energy, and/or the energies in gravitational, magnetic, or electric fields.] HS-PS3-2 Develop and use models to illustrate that energy at the macroscopic scale can be accounted for as a combination of energy associated with the motions of particles (objects) and energy associated with the relative position of particles (objects). [Clarification Statement: Examples of phenomena at the macroscopic scale could include the conversion of kinetic energy to thermal energy, the energy stored due to position of an object above the earth, and the energy stored between two electrically-charged plates. Examples of models could include diagrams, drawings, descriptions, and computer simulations.] HS-PS3-3. Design, build, and refine a device that works within given constraints to convert one form of energy into another form of energy.* [Clarification Statement: Emphasis is on both qualitative and quantitative evaluations of devices. Examples of devices could include Rube Goldberg devices, wind turbines, solar cells, solar ovens, and generators. Examples of constraints could include use of renewable energy forms and efficiency.] [Assessment Boundary: Assessment for quantitative evaluations is limited to total output for a given input.]

Assessment is limited to devices constructed with materials provided to students.]

HS-PS3-5 Develop and use a model of two objects interacting through electric or magnetic fields to illustrate the forces between objects and the changes in energy of the objects due to the interaction.

[Clarification Statement: Examples of models could include drawings, diagrams, and texts, such as drawings of what happens when two charges of opposite polarity are near each other.] [Assessment Boundary: Assessment is limited to systems containing two objects.]

HS-PS4-5. Communicate technical information about how some technological devices use the principles of wave behavior and wave interactions with matter to transmit and capture information and energy.*

[Clarification Statement: Examples could include solar cells capturing light and converting it to electricity; medical imaging; and communications technology.] [Assessment Boundary: Assessments are limited to qualitative information. Assessments do not include band theory.]

HS-ESS3-2. Evaluate competing design solutions for developing, managing, and utilizing energy and mineral resources based on cost-benefit ratios.*

[Clarification Statement: Emphasis is on the conservation, recycling, and reuse of resources (such as minerals and metals) where possible, and on minimizing impacts where it is not. Examples include developing best practices for agricultural soil use, mining (for coal, tar sands, and oil shales), and pumping (for petroleum and natural gas). Science knowledge indicates what can happen in natural systems—not what should happen.]

HS-ESS3-3. Create a computational simulation to illustrate the relationships among management of natural resources, the sustainability of human populations, and biodiversity.

[Clarification Statement: Examples of factors that affect the management of natural resources include costs of resource extraction and waste management, per-capita consumption, and the development of new technologies. Examples of factors that affect human sustainability include agricultural efficiency, levels of conservation, and urban planning.] [Assessment Boundary: Assessment for computational simulations is limited to using provided multi-parameter programs or constructing simplified spreadsheet calculations.]

HS-ETS1-1. Analyze a major global challenge to specify qualitative and quantitative criteria and constraints for solutions that account for societal needs and wants.

HS-ETS1-2. Design a solution to a complex real-world problem by breaking it down into smaller, more manageable problems that can be solved through engineering.

HS-ETS1-3. Evaluate a solution to a complex real-world problem based on prioritized criteria and trade-offs that account for a range of constraints, including cost, safety, reliability, and aesthetics as well as possible social, cultural, and environmental impacts.

HS-ETS1-4. Use a computer simulation to model the impact of proposed solutions to a complex real-world problem with numerous criteria and constraints on interactions within and between systems relevant to the problem.

*The performance expectations marked with an asterisk integrate traditional science content with engineering through a practice or disciplinary core idea.

The bundle of performance expectations above focuses on the following elements from the NRC document *A Framework for K–12 Science Education*:

Highlighted Science and Engineering Practices	Highlighted Disciplinary Core Ideas	Highlighted Crosscutting Concepts
[SEP-1] Asking Questions and Defining Problems	PS2.B: Types of Interactions	[CCC-2] Cause and Effect: Mechanism and Explanation
[SEP-2] Developing and Using Models	PS3.A: Definitions of Energy	[CCC-4] System and System Models
[SEP-3] Planning and Carrying Out Investigations	PS3.B: Conservation of Energy and Energy Transfer	[CCC-5] Energy and Matter: Flows, Cycles, and Conservation
[SEP-5] Using mathematics and Computational Thinking	PS3.C: Relationship Between Energy and Forces	

[SEP-6] Constructing Explanations (for science) and Designing Solutions (for engineering)	PS3.D: Energy in Chemical Processes and Everyday Life ETS1.A: Defining and Delimiting Engineering Problems ETS1.B: Developing Possible Solutions ETS1.C: Optimizing the Design Solution	
<i>Highlighted California Environmental Principles & Concepts:</i> Principle I The continuation and health of individual human lives and of human communities and societies depend on the health of the natural systems that provide essential goods and ecosystem services. Principle II The long-term functioning and health of terrestrial, freshwater, coastal and marine ecosystems are influenced by their relationships with human societies. Principle III Natural systems proceed through cycles that humans depend upon, benefit from and can alter. Principle IV The exchange of matter between natural systems and human societies affects the long term functioning of both. Principle V Decisions affecting resources and natural systems are based on a wide range of considerations and decision-making processes. <i>CA CCSS Math Connections:</i> N-Q.1-3; MP.2; MP.4 <i>CA CCSS ELA/Literacy Connections:</i> SL.11-12.5; RST.11-12.1, 8, WHST.9-12.2.a-e, 7, 8, 9 <i>CA ELD Connections:</i> ELD.PI.11-12.1,5,6a-b, 9, 10, 11a		

**Physics in the Universe – Instructional Segment 4:
Nuclear Processes and Earth History**

Guiding Questions

- *What does $E=mc^2$ mean?*
- *How do nuclear reactions illustrate conservation of energy and mass?*
- *How do we determine the age of rocks and other geologic features?*

Performance Expectations

Students who demonstrate understanding can:

HS-ESS1-5. Evaluate evidence of the past and current movements of continental and oceanic crust and the theory of plate tectonics to explain the ages of crustal rocks. [Clarification Statement: Emphasis is on the ability of plate tectonics to explain the ages of crustal rocks. Examples include evidence of the ages oceanic crust increasing with distance from mid-ocean ridges (a result of plate spreading) and the ages of North American continental crust increasing with distance away from a central ancient core (a result of past plate interactions).] (Also addressed in the High School Chemistry in the Earth System course)

HS-ESS1-6. Apply scientific reasoning and evidence from ancient Earth materials, meteorites, and other planetary surfaces to construct an account of Earth’s formation and early history. [Clarification Statement: Emphasis is on using available evidence within the solar system to reconstruct the early history of Earth, which formed along with the rest of the solar system 4.6 billion years ago. Examples of evidence include the absolute ages of ancient materials (obtained by radiometric dating of meteorites, moon rocks, and Earth’s oldest minerals), the sizes and compositions of solar system objects, and the impact cratering record of planetary surfaces.] (Also addressed in the High School Living Earth course)

HS-ESS2-1. Develop a model to illustrate how Earth’s internal and surface processes operate at different spatial and temporal scales to form continental and ocean-floor features. [Clarification Statement: Emphasis is on how the appearance of land features (such as mountains, valleys, and plateaus) and sea-floor features (such as trenches, ridges, and seamounts) are a result of both constructive forces (such as volcanism, tectonic uplift, and orogeny) and destructive mechanisms (such as weathering, mass wasting, and coastal erosion).] [Assessment Boundary: Assessment does not include memorization of the details of the formation of specific geographic features of Earth’s surface.] (Also addressed in the High School Living Earth course)

HS-PS1-8 Develop models to illustrate the changes in the composition of the nucleus of the atom and the energy released during the processes of fission, fusion, and radioactive decay. [Clarification Statement: Emphasis is on simple qualitative models, such as pictures or diagrams, and on the scale of energy released in nuclear processes relative to other kinds of transformations.] [Assessment Boundary: Assessment does not include quantitative calculation of energy released. Assessment is limited to alpha, beta, and gamma radioactive decays.]

The bundle of performance expectations above focuses on the following elements from the NRC document *A Framework for K–12 Science Education*:

Highlighted Science and Engineering Practices	Highlighted Disciplinary Core Ideas	Highlighted Crosscutting Concepts
[SEP-2] Developing and Using Models	ESS1.C: The History of Planet Earth	[CCC-1] Patterns
[SEP-6] Constructing Explanations (for science) and Designing Solutions (for engineering)	ESS2.A: Earth Materials and Systems	[CCC-5] Energy and Matter: Flows, Cycles, and Conservation
[SEP-7] Engaging in Argument from Evidence	ESS2.B: Plate Tectonics and Large-Scale System Interactions	[CCC-7] Stability and Change
	PS1.A Structure and Properties of Matter	
	PS1.C: Nuclear Processes	

Highlighted California Environmental Principles & Concepts:

Principle I The continuation and health of individual human lives and of human

communities and societies depend on the health of the natural systems that provide essential goods and ecosystem services. Principle II The long-term functioning and health of terrestrial, freshwater, coastal and marine ecosystems are influenced by their relationships with human societies. Principle III Natural systems proceed through cycles that humans depend upon, benefit from and can alter. Principle IV The exchange of matter between natural systems and human societies affects the long term functioning of both. Principle V Decisions affecting resources and natural systems are based on a wide range of considerations and decision-making processes.
CA CCSS Math Connections: MP.2; MP.4
CA CCSS ELA/Literacy Connections: SL.11-12.4; RST.11-12.1,8, WHST.9-12.2.a-e, 7, 9
CA ELD Connections: ELD.PI.11-12.1,5,6a-b, 9, 10, 11a

Physics in the Universe – Instructional Segment 5: Waves and Electromagnetic Radiation
<i>Guiding Questions</i> • How do we know what is inside the Earth? • Why do people get sunburned by UV light? • How do can we transmit information over wires and wirelessly?
Performance Expectations
<i>Students who demonstrate understanding can:</i> HS-ESS2-1. Develop a model to illustrate how Earth’s internal and surface processes operate at different spatial and temporal scales to form continental and ocean-floor features. [Clarification Statement: Emphasis is on how the appearance of land features (such as mountains, valleys, and plateaus) and sea-floor features (such as trenches, ridges, and seamounts) are a result of both constructive forces (such as volcanism, tectonic uplift, and orogeny) and destructive mechanisms (such as weathering, mass wasting, and coastal erosion).] [Assessment Boundary: Assessment does not include memorization of the details of the formation of specific geographic features of Earth’s surface.] (Introduced in IS4) HS-PS4-1. Use mathematical representations to support a claim regarding relationships among the frequency, wavelength, and speed of waves traveling in various media. [Clarification Statement: Examples of data could include electromagnetic radiation traveling in a vacuum and glass, sound waves traveling through air and water,

and seismic waves traveling through the Earth.] [Assessment Boundary: Assessment is limited to algebraic relationships and describing those relationships qualitatively.]

HS-PS4-2. Evaluate questions about the advantages of using a digital transmission and storage of information. [Clarification Statement: Examples of advantages could include that digital information is stable because it can be stored reliably in computer memory, transferred easily, and copied and shared rapidly. Disadvantages could include issues of easy deletion, security, and theft.]

HS-PS4-3. Evaluate the claims, evidence, and reasoning behind the idea that electromagnetic radiation can be described either by a wave model or a particle model, and that for some situations one model is more useful than the other. [Clarification Statement: Emphasis is on how the experimental evidence supports the claim and how a theory is generally modified in light of new evidence. Examples of a phenomenon could include resonance, interference, diffraction, and photoelectric effect.] [Assessment Boundary: Assessment does not include using quantum theory.]

HS-PS4-4. Evaluate the validity and reliability of claims in published materials of the effects that different frequencies of electromagnetic radiation have when absorbed by matter. [Clarification Statement: Emphasis is on the idea that photons associated with different frequencies of light have different energies, and the damage to living tissue from electromagnetic radiation depends on the energy of the radiation. Examples of published materials could include trade books, magazines, web resources, videos, and other passages that may reflect bias.] [Assessment Boundary: Assessment is limited to qualitative descriptions.]

HS-PS4-5. Communicate technical information about how some technological devices use the principles of wave behavior and wave interactions with matter to transmit and capture information and energy.* [Clarification Statement: Examples could include solar cells capturing light and converting it to electricity; medical imaging; and communications technology.] [Assessment Boundary: Assessments are limited to qualitative information. Assessments do not include band theory.]

*The performance expectations marked with an asterisk integrate traditional science content with engineering through a practice or disciplinary core idea.

The bundle of performance expectations above focuses on the following elements from the NRC document *A Framework for K–12 Science Education*:

Highlighted Science and Engineering Practices	Highlighted Disciplinary Core Ideas	Highlighted Crosscutting Concepts
[SEP-1] Asking Questions and Defining Problems	ESS2.A: Earth Materials and Systems	[CCC-2 Cause and Effect: Mechanism and Explanation
[SEP-2] Developing and Using Models	ESS2.B: Plate Tectonics and Large- Scale System Interactions	[CCC-4] Systems and System Models
[SEP-5] Using mathematics and Computational Thinking	PS3.D: Energy in Chemical Reactions	[CCC-7] Stability and Change
[SEP-7] Engaging in Argument from Evidence	PS4.A: Wave Properties	
[SEP-7] Engaging in Argument from Evidence	PS4.B: Electromagnetic Radiation	
[SEP-8] Obtaining, Evaluating, and Communicating Information	PS4.C: Information Technologies and Instrumentation	
CA CCSS Math Connections: A-SSE.1a-b,3a-c; A-CDE.4; N-Q.1-3; MP.2; MP.4		
CA CCSS ELA/Literacy Connections: SL.11-12.5; RST.9-10.8; RST.11-12.1, 7, 8; WHST.9-12.2.a-e, 8		
CA ELD Connections: ELD.PI.11-12.1, 5, 6a-b, 9, 10, 11a		

**Physics in the Universe – Instructional Segment 6:
Stars and the Origins of the Universe**

Guiding Questions:

- How do we know what are stars made out of?
- What fuels our Sun? Will it ever run out of that fuel?
- Do other stars work the same way as our Sun?
- How do patterns in motion of the stars tell us about the origin of our Universe?

Performance Expectations

Students who demonstrate understanding can:

HS-ESS1-1. Develop a model based on evidence to illustrate the life span of the sun and the role of nuclear fusion in the sun's core to release energy in the form of radiation. [Clarification Statement: Emphasis is on the energy transfer mechanisms that allow energy from nuclear fusion in the sun's core to reach Earth. Examples of evidence for the model include observations of the masses and lifetimes of other stars, as well as the ways that the sun's radiation varies due to sudden solar flares ("space weather"), the 11-year sunspot cycle, and non-cyclic variations over centuries.] [Assessment Boundary: Assessment does not include details of the atomic and sub-atomic processes involved with the sun's nuclear fusion.]

HS-ESS1-2. Construct an explanation of the Big Bang theory based on astronomical evidence of light spectra, motion of distant galaxies, and composition of matter in the universe. [Clarification Statement: Emphasis is on the astronomical evidence of the red shift of light from galaxies as an indication that the universe is currently expanding, the cosmic microwave background as the remnant radiation from the Big Bang, and the observed composition of ordinary matter of the universe, primarily found in stars and interstellar gases (from the spectra of electromagnetic radiation from stars), which matches that predicted by the Big Bang theory (3/4 hydrogen and 1/4 helium).]

HS-ESS1-3. Communicate scientific ideas about the way stars, over their life cycle, produce elements. [Clarification Statement: Emphasis is on the way nucleosynthesis, and therefore the different elements created, varies as a function of the mass of a star and the stage of its lifetime.] [Assessment Boundary: Details of the many different nucleosynthesis pathways for stars of differing masses are not assessed.]

The bundle of performance expectations above focuses on the following elements from the NRC document *A Framework for K–12 Science Education*:

Highlighted Science and Engineering Practices	Highlighted Disciplinary Core Ideas	Highlighted Crosscutting Concepts
[SEP-2] Developing and Using Models [SEP-6] Constructing Explanations (for science) and Designing Solutions (for engineering) [SEP-8] Obtaining, Evaluating, and Communicating Information⁶	ESS1.A: The Universe and Its Stars PS3.D: Energy in Chemical Processes and Everyday Life PS4.B Electromagnetic Radiation	[CCC-3] Scale, Proportion, and Quantity [CCC-5] Energy and Matter: Flows, Cycles, and Conservation
CA CCSS Math Connections: N-Q.1-3; A-SSE.1a-b; A-CED.2,4; MP.2; MP.4		
CA CCSS ELA/Literacy Connections: SL.11-12.4; RST.11-12.1; WHST.9-12.2.a-e		
CA ELD Connections: ELD.PI.11-12.1,5,6a-b, 9, 10, 11a		

High School Four-Course Model

High School Four-Course Model Life Science/Biology Instructional Segment 1: Structure and Function
<i>Guiding Questions:</i> • How does the structure of DNA affect how cells look and behave? • How do systems work in a multi-celled organism and what happens if there is a change in the system? • How do organisms survive even when there are changes in their environment?
Performance Expectations
<i>Students who demonstrate understanding can do the following:</i> HS-LS1-1. Construct an explanation based on evidence for how the structure of DNA determines the structure of proteins which carry out the essential functions of life through systems of specialized cells. [Assessment Boundary: Assessment does not include identification of specific cell or tissue types, whole body systems, specific protein structures and functions, or the biochemistry of protein synthesis.] HS-LS1-2. Develop and use a model to illustrate the hierarchical organization of interacting systems that provide specific functions within multicellular organisms. [Clarification Statement: Emphasis is on functions at the organism system level such as nutrient uptake, water delivery, and organism movement in response to neural stimuli. An example of an interacting system could be an artery depending on the proper function of elastic tissue and smooth muscle to regulate and deliver the proper amount of blood within the circulatory system.] [Assessment Boundary: Assessment does not include interactions and functions at the molecular or chemical reaction level.] HS-LS1-3. Plan and conduct an investigation to provide evidence that feedback mechanisms maintain homeostasis. [Clarification Statement: Examples of investigations could include heart rate response to exercise, stomate response to moisture and temperature, and root development in response to water levels.] [Assessment Boundary: Assessment does not include the cellular processes involved in the feedback mechanism.]
The bundle of performance expectations above focuses on the following elements from the NRC document <i>A Framework for K–12 Science Education</i>:

Highlighted Science and Engineering Practices	Highlighted Disciplinary Core Ideas	Highlighted Crosscutting Concepts
[SEP-2] Developing and Using Models [SEP-3] Planning and Carrying Out Investigations [SEP-6] Constructing Explanations (for science) and Designing Solutions (for engineering)	LS1.A: Structure and Function	[CCC-4] Systems and System Models [CCC-6] Structure and Function [CCC-7] Stability and Change
CA CCSS for ELA/Literacy Connections: RST.11–12.1; WHST.11–12.2a–e, 7, 8, 9; SL.11–12.5 CA ELD Standards Connections: ELD.PI.9–10.1, 5, 6a–b, 9, 10, 11a		

High School Four-Course Model Life Science/Biology Instructional Segment 2: Growth and Development of Organisms
<i>Guiding Questions:</i> • How do organisms grow? • How do cells create exact copies of themselves?
Performance Expectations
<i>Students who demonstrate understanding can do the following:</i> HS-LS1-4. Use a model to illustrate the role of cellular division (mitosis) and differentiation in producing and maintaining complex organisms. [Assessment Boundary: Assessment does not include specific gene control mechanisms or rote memorization of the steps of mitosis.] HS-LS3-1. Ask questions to clarify relationships about the role of DNA and chromosomes in coding the instructions for characteristic traits passed from parents to offspring. [Assessment Boundary: Assessment does not include the phases of mitosis or the biochemical mechanism of specific steps in the process.] (Introduced but not assessed until IS8)

The bundle of performance expectations above focuses on the following elements from the NRC document *A Framework for K–12 Science Education*:

Highlighted Science and Engineering Practices	Highlighted Disciplinary Core Ideas	Highlighted Crosscutting Concepts
[SEP-1] Asking Questions and Defining Problems [SEP-2] Developing and Using Models	LS1.A: Structure and Function LS1.B: Growth and Development of Organisms LS3.A: Inheritance of Traits	[CCC-2] Cause and Effect [CCC-4] Systems and System Models
CA CCSS Math Connections: F-IF.7.a–e; F-BF.1a–c; MP.2, MP.4		
CA CCSS for ELA/Literacy Connections: RST.11–12.1, 9; SL.11–12.5		
CA ELD Standards Connections: ELD.PI.11–12.1, 5, 6a–b, 9, 10, 11a		

High School Four-Course Model Life Science/Biology Instructional Segment 3: Organization for Matter and Energy Flow in Organisms
<i>Guiding Questions:</i> • How do living things acquire energy and matter for life? • How do organisms store energy? • How are photosynthesis and cellular respiration connected? • How do organisms use the raw materials they ingest from the environment?
Performance Expectations
<i>Students who demonstrate understanding can do the following:</i> HS-LS1-5. Use a model to illustrate how photosynthesis transforms light energy into stored chemical energy. [Clarification Statement: Emphasis is on illustrating inputs and outputs of matter and the transfer and transformation of energy in photosynthesis by plants and other photosynthesizing organisms. Examples of models could include diagrams, chemical equations, and conceptual models.]

[Assessment Boundary: Assessment does not include specific biochemical steps.]

HS-LS1-6. Construct and revise an explanation based on evidence for how carbon, hydrogen, and oxygen from sugar molecules may combine with other elements to form amino acids and/or other large carbon-based molecules. [Clarification Statement: Emphasis is on using evidence from models and simulations to support explanations.] [Assessment Boundary: Assessment does not include the details of the specific chemical reactions or identification of macromolecules.]

HS-LS1-7. Use a model to illustrate that cellular respiration is a chemical process whereby the bonds of food molecules and oxygen molecules are broken and the bonds in new compounds are formed resulting in a net transfer of energy. [Clarification Statement: Emphasis is on the conceptual understanding of the inputs and outputs of the process of cellular respiration.] [Assessment Boundary: Assessment should not include identification of the steps or specific processes involved in cellular respiration.]

The bundle of performance expectations above focuses on the following elements from the NRC document *A Framework for K–12 Science Education*:

Highlighted Science and Engineering Practices	Highlighted Disciplinary Core Ideas	Highlighted Crosscutting Concepts
[SEP-2] Developing and Using Models [SEP-6] Constructing Explanations (for science) and Designing Solutions (for engineering)	LS1.C Organization for Matter and Energy Flow in Organisms	[CCC-5] Energy and Matter: Flows, Cycles, and Conservation
CA CCSS for ELA/Literacy Connections: RST.11–12.1; WHST.9–12.2.a–e, 5, 9; SL.11–12.5		
CA ELD Standards Connections: ELD.PI.11–12.1, 5, 6a–b, 9, 10, 11a		

High School Four-Course Model Life Science/Biology – Instructional Segment 4: Interdependent Relationships in Ecosystems		
<i>Guiding Questions:</i> • How and why do populations change over time? • How do populations change when their resources become scarce?		
Performance Expectations		
<i>Students who demonstrate understanding can do the following:</i> HS-LS2-1. Use mathematical and/or computational representations to support explanations of factors that affect carrying capacity of ecosystems at different scales. [Clarification Statement: Emphasis is on quantitative analysis and comparison of the relationships among interdependent factors including boundaries, resources, climate, and competition. Examples of mathematical comparisons could include graphs, charts, histograms, and population changes gathered from simulations or historical data sets.] [Assessment Boundary: Assessment does not include deriving mathematical equations to make comparisons.] HS-LS2-2. Use mathematical representations to support and revise explanations based on evidence about factors affecting biodiversity and populations in ecosystems of different scales. [Clarification Statement: Examples of mathematical representations include finding the average, determining trends, and using graphical comparisons of multiple sets of data.] [Assessment Boundary: Assessment is limited to provide data.]		
The bundle of performance expectations above focuses on the following elements from the NRC document <i>A Framework for K–12 Science Education</i>:		
Highlighted Science and Engineering Practices [SEP-5] Using Mathematics and Computational Thinking	Highlighted Disciplinary Core Ideas LS2.A: Interdependent Relationships in Ecosystems LS2.C: Ecosystem Dynamics, Functioning, and Resilience	Highlighted Crosscutting Concepts [CCC-3] Scale, Proportion, and Quantity [CCC-5] Energy and Matter: Flows, Cycles, and Conservation
Highlighted California Environmental Principles & Concepts: Principle II The long-term functioning and health of terrestrial, freshwater, coastal, and marine ecosystems are influenced by their relationships with human societies.		

Principle IV The exchange of matter between natural systems and human societies affects the long-term functioning of both.
CA CCSS Math Connections: N-Q.1–3; MP.2, MP.4
CA CCSS for ELA/Literacy Connections: RST.11–12.7; WHST.11–12.2a–e, 9
CA ELD Standards Connections: ELD.PI.11–12.1, 5, 6a–b, 9, 10, 11a

High School Four-Course Model Life Science/Biology Instructional Segment 5: Cycles of Matter and Energy Transfer in Ecosystems
<i>Guiding Questions:</i> • Why is the cycling of matter and energy important? • How are matter and energy linked in ecosystems?
Performance Expectations
<i>Students who demonstrate understanding can do the following:</i> HS-LS2-3. Construct and revise an explanation based on evidence for the cycling of matter and flow of energy in aerobic and anaerobic conditions. [Clarification Statement: Emphasis is on conceptual understanding of the role of aerobic and anaerobic respiration in different environments.] [Assessment Boundary: Assessment does not include the specific chemical processes of either aerobic or anaerobic respiration.] HS-LS2-4. Use mathematical representations to support claims for the cycling of matter and flow of energy among organisms in an ecosystem. [Clarification Statement: Emphasis is on using a mathematical model of stored energy in biomass to describe the transfer of energy from one trophic level to another and that matter and energy are conserved as matter cycles and energy flows through ecosystems. Emphasis is on atoms and molecules such as carbon, oxygen, hydrogen, and nitrogen being conserved as they move through an ecosystem.] [Assessment Boundary: Assessment is limited to proportional reasoning to describe the cycling of matter and flow of energy.] HS-LS2-5. Develop a model to illustrate the role of photosynthesis and cellular respiration in the cycling of carbon among the biosphere, atmosphere, hydrosphere, and geosphere. [Clarification Statement: Examples of models could include simulations and mathematical models.] [Assessment Boundary: Assessment does not include the specific chemical steps of photosynthesis and respiration.]
The bundle of performance expectations above focuses on the following elements from the NRC document <i>A Framework for K–12 Science Education</i> :

Highlighted Science and Engineering Practices	Highlighted Disciplinary Core Ideas	Highlighted Crosscutting Concepts
[SEP-2] Developing and Using Models [SEP-5] Using Mathematical and Computational Thinking [SEP-6] Constructing Explanations (for science) and Designing Solutions (for engineering)	LS2.B: Cycles of Matter and Energy in Ecosystems PS3.D: Energy in Chemical Processes	[CCC-4] Systems and System Models [CCC-5] Energy and Matter: Flows, Cycles, and Conservation
<i>Highlighted California Environmental Principles & Concepts:</i> Principle II The long-term functioning and health of terrestrial, freshwater, coastal, and marine ecosystems are influenced by their relationships with human societies. Principle IV The exchange of matter between natural systems and human societies affects the long-term functioning of both.		
CA CCSS Math Connections: N-Q.1–3; MP.2, MP.4		
CA CCSS for ELA/Literacy Connections: RST.11–12.1, WHST.11–12.2a–e, 5		
CA ELD Standards Connections: ELD.PI.11–12.1, 5, 6a–b, 9, 10, 11a		

High School Four-Course Model Life Science/Biology Instructional Segment 6: Ecosystem Dynamics, Functioning, and Resilience
<i>Guiding Questions:</i> • What types of interactions cause changes in ecosystems that ultimately affect populations? • To what extent can humans “undo” their negative impact on the environment?
Performance Expectations
<i>Students who demonstrate understanding can do the following:</i> HS-LS2-6. Evaluate the claims, evidence, and reasoning that the complex interactions in ecosystems maintain relatively consistent

numbers and types of organisms in stable conditions, but changing conditions may result in a new ecosystem.

[Clarification Statement: Examples of changes in ecosystem conditions could include modest biological or physical changes, such as moderate hunting or a seasonal flood; and extreme changes, such as volcanic eruption or sea level rise.]

HS-LS2-7. Design, evaluate, and refine a solution for reducing the impacts of human activities on the environment and biodiversity.* [Clarification Statement: Examples of human activities can include urbanization, building dams, and dissemination of invasive species.]

*The performance expectations marked with an asterisk integrate traditional science content with engineering through a practice or disciplinary core idea.

The bundle of performance expectations above focuses on the following elements from the NRC document *A Framework for K–12 Science Education*:

Highlighted Science and Engineering Practices	Highlighted Disciplinary Core Ideas	Highlighted Crosscutting Concepts
[SEP-6] Constructing Explanations (for science) and Designing Solutions (for engineering) [SEP-7] Engaging in Argument from Evidence	LS2.C: Ecosystem Dynamics, Functioning, and Resilience LS4.D: Biodiversity and Humans ETS1.B: Developing Possible Solutions	[CCC-7] Stability and Change

Highlighted California Environmental Principles & Concepts:

Principle II The long-term functioning and health of terrestrial, freshwater, coastal, and marine ecosystems are influenced by their relationships with human societies.

Principle IV The exchange of matter between natural systems and human societies affects the long-term functioning of both.

CA CCSS Math Connections: N-Q.1–3; S-ID.1; S-IC.1, 6; MP.2, MP.4

CA CCSS for ELA/Literacy Connections: RST.9–10.8, RST.11–12.7, 8; WHST.11–12.7

CA ELD Standards Connections: ELD.PI.11–12.1, 5, 6a–b, 9, 10, 11a

High School Four-Course Model Life Science/Biology Instructional Segment 7: Social Interactions and Group Behavior		
<i>Guiding Questions:</i> - How do populations ensure that their gene pool gets passed on? - What affects a population's chance of survival?		
Performance Expectations		
<i>Students who demonstrate understanding can do the following:</i> HS-LS2-8. Evaluate the evidence for the role of group behavior on individual and species' chances to survive and reproduce. <i>[Clarification Statement: Emphasis is on (1) distinguishing between group and individual behavior, (2) identifying evidence supporting the outcomes of group behavior, and (3) developing logical and reasonable arguments based on evidence. Examples of group behaviors could include flocking, schooling, herding, and cooperative behaviors such as hunting, migrating, and swarming.]</i>		
The bundle of performance expectations above focuses on the following elements from the NRC document <i>A Framework for K–12 Science Education</i> :		
Highlighted Science and Engineering Practices [SEP-7] Engaging in Argument from Evidence	Highlighted Disciplinary Core Ideas LS2.D: Social Interactions and Group Behavior	Highlighted Crosscutting Concepts [CCC-2] Cause and Effect: Mechanism and Explanation
CA CCSS for ELA/Literacy Connections: RST.9–10.8; RST.11–12.1, 7, 8, 9		
CA ELD Standards Connections: ELD.PI.11–12.1, 5, 6a–b, 9, 10, 11a		

High School Four-Course Model Life Science/Biology Instructional Segment 8: Inheritance of Traits
<i>Guiding Questions:</i> - How are characteristics of one generation passed to the next? - What allows traits to be transmitted from parents to offspring?
Performance Expectations
<i>Students who demonstrate understanding can do the following:</i> HS-LS3-1. Ask questions to clarify relationships about the role of DNA and chromosomes in coding the instructions for characteristic traits passed from parents to offspring. <i>[Assessment Boundary:</i>

Assessment does not include the phases of meiosis or the biochemical mechanism of specific steps in the process.]		
The bundle of performance expectations above focuses on the following elements from the NRC document <i>A Framework for K–12 Science Education</i> :		
Highlighted Science and Engineering Practices	Highlighted Disciplinary Core Ideas	Highlighted Crosscutting Concepts
[SEP-1] Asking Questions and Defining Problems	LS1.A: Structure and Function LS3.A: Inheritance of Traits	[CCC-2] Cause and Effect
CA CCSS for ELA/Literacy Connections: RST.11–12.1, 9		
CA ELD Standards Connections: ELD.PI.11–12.1, 5, 6a–b, 9, 10, 11a		

High School Four-Course Model Life Science/Biology Instructional Segment 9: Variation of Traits
<i>Guiding Questions:</i> • How do genes determine traits? • What is the chance of a trait being passed from one generation to another? • What happens if there is a mutation in a gene? • What contributes to phenotypes?
Performance Expectations
<i>Students who demonstrate understanding can do the following:</i> HS-LS3-2. Make and defend a claim based on evidence that inheritable genetic variations may result from (1) new genetic combinations through meiosis, (2) viable errors occurring during replication, and/or (3) mutations caused by environmental factors. [Clarification Statement: Emphasis is on using data to support arguments for the way variation occurs.] [Assessment Boundary: Assessment does not include the phases of meiosis or the biochemical mechanism of specific steps in the process.] HS-LS3-3. Apply concepts of statistics and probability to explain the variation and distribution of expressed traits in a population. [Clarification Statement: Emphasis is on the use of mathematics to describe the probability of traits as it relates to genetic and environmental factors in the expression of traits.] [Assessment

Boundary: Assessment does not include Hardy-Weinberg calculations.]		
The bundle of performance expectations above focuses on the following elements from the NRC document <i>A Framework for K–12 Science Education</i> :		
Highlighted Science and Engineering Practices [SEP-4] Analyzing and Interpreting Data [SEP-7] Engaging in Argument from Evidence	Highlighted Disciplinary Core Ideas LS3.B: Variation of Traits	Highlighted Crosscutting Concepts [CCC-2] Cause and Effect [CCC-3] Scale, Proportion, and Quantity
Highlighted California Environmental Principles & Concepts: Principle III Natural systems proceed through cycles that humans depend upon, benefit from, and can alter. Principle IV The exchange of matter between natural systems and human societies affects the long-term functioning of both.		
CA CCSS Math Connections: MP.2		
CA CCSS for ELA/Literacy Connections: RST.11–12.1; WHST.11–12.1.a–e		
CA ELD Standards Connections: ELD.PI.11–12.1, 5, 6a–b, 9, 10, 11a		

High School Four-Course Model Life Science/Biology Instructional Segment 10: Evidence of Common Ancestry and Diversity
<i>Guiding Questions:</i> - What evidence shows that different species are related? - How did modern-day humans evolve?
Performance Expectations
<i>Students who demonstrate understanding can do the following:</i> HS-LS4-1. Communicate scientific information that common ancestry and biological evolution are supported by multiple lines of empirical evidence. [Clarification Statement: Emphasis is on a conceptual understanding of the role each line of evidence has relating to common ancestry and biological evolution. Examples of evidence could include similarities in DNA sequences, anatomical structures, and order of appearance of structures in embryological development.]
The bundle of performance expectations above focuses on the following elements from the NRC document <i>A Framework for K–12 Science Education</i> :

Highlighted Science and Engineering Practices	Highlighted Disciplinary Core Ideas	Highlighted Crosscutting Concepts
[SEP-8] Obtaining, Evaluating, and Communicating Information	LS4.A: Evidence of Common Ancestry and Diversity	[CCC-1] Patterns
CA CCSS Math Connections: MP.2		
CA CCSS for ELA/Literacy Connections: RST.11–12.1; WHST.11–12.2, 9.a–e; SL.11–12.4		
CA ELD Standards Connections: ELD.PI.11–12.1, 5, 6a–b, 9, 10, 11a		

High School Four-Course Model Life Science/Biology Instructional Segment 11: Natural Selection
<i>Guiding Questions:</i> • What processes influence natural selection? • What evidence did Darwin provide that became the foundation for the study of evolution?
Performance Expectations
<i>Students who demonstrate understanding can do the following:</i> HS-LS4-2. Construct an explanation based on evidence that the process of evolution primarily results from four factors (1) the potential for a species to increase in number, (2) the heritable genetic variation of individuals in a species due to mutation and sexual reproduction, (3) competition for limited resources, and (4) the proliferation of those organisms that are better able to survive and reproduce in the environment. [Clarification Statement: Emphasis is on using evidence to explain the influence each of the four factors has on number of organisms, behaviors, morphology, or physiology in terms of ability to compete for limited resources and subsequent survival of individuals and adaptation of species. Examples of evidence could include mathematical models such as simple distribution graphs and proportional reasoning.] [Assessment Boundary: Assessment does not include other mechanisms of evolution, such as genetic drift, gene flow through migration, and co-evolution.] HS-LS4-3. Apply concepts of statistics and probability to support explanations that organisms with an advantageous heritable trait tend to increase in proportion to organisms lacking this trait. [Clarification Statement: Emphasis is on analyzing shifts in numerical distribution of traits and using these shifts as evidence to

support explanations.] [Assessment Boundary: Assessment is limited to basic statistical and graphical analysis. Assessment does not include allele frequency calculations.]		
The bundle of performance expectations above focuses on the following elements from the NRC document <i>A Framework for K–12 Science Education</i> :		
Highlighted Science and Engineering Practices	Highlighted Disciplinary Core Ideas	Highlighted Crosscutting Concepts
[SEP-4] Analyzing and Interpreting Data [SEP-6] Constructing Explanations (for science) and Designing Solutions (for engineering)	LS4.B: Natural Selection LS4.C: Adaptation	[CCC-1] Patterns [CCC-2] Cause and Effect
Highlighted California Environmental Principles & Concepts: Principle I The continuation and health of individual human lives and of human communities and societies depend on the health of the natural systems that provide essential goods and ecosystem services. Principle II The long-term functioning and health of terrestrial, freshwater, coastal, and marine ecosystems are influenced by their relationships with human societies.		
CA CCSS Math Connections: MP.2, MP.3		
CA CCSS for ELA/Literacy Connections: RST.11–12.1, WHST.11–12.2, 9.a–e; SL.11–12.4		
CA ELD Standards Connections: ELD.PI.11–12.1, 5, 6a–b, 9, 10, 11a		

High School Four-Course Model Life Science/Biology Instructional Segment 12: Adaptation and Biodiversity
<i>Guiding Questions:</i> • How does natural selection lead to adaptation in populations? • How do changes in ecosystems influence populations?
Performance Expectations
<i>Students who demonstrate understanding can do the following:</i> HS-LS4-2. Construct an explanation based on evidence that the process of evolution primarily results from four factors: (1) the potential

for a species to increase in number, (2) the heritable genetic variation of individuals in a species due to mutation and sexual reproduction, (3) competition for limited resources, and (4) the proliferation of those organisms that are better able to survive and reproduce in the environment. [Clarification Statement: Emphasis is on using evidence to explain the influence each of the four factors has on number of organisms, behaviors, morphology, or physiology in terms of ability to compete for limited resources and subsequent survival of individuals and adaptation of species. Examples of evidence could include mathematical models such as simple distribution graphs and proportional reasoning.] [Assessment Boundary: Assessment does not include other mechanisms of evolution, such as genetic drift, gene flow through migration, and co-evolution.]

HS-LS4-3. Apply concepts of statistics and probability to support explanations that organisms with an advantageous heritable trait tend to increase in proportion to organisms lacking this trait. [Clarification Statement: Emphasis is on analyzing shifts in numerical distribution of traits and using these shifts as evidence to support explanations.] [Assessment Boundary: Assessment is limited to basic statistical and graphical analysis. Assessment does not include allele frequency calculations.]

HS-LS4-4. Construct an explanation based on evidence for how natural selection leads to adaptation of populations. [Clarification Statement: Emphasis is on using data to provide evidence for how specific biotic and abiotic differences in ecosystems (such as ranges of seasonal temperature, long-term climate change, acidity, light, geographic barriers, or evolution of other organisms) contribute to a change in gene frequency over time, leading to adaptation of populations.]

HS-LS4-5. Evaluate the evidence supporting claims that changes in environmental conditions may result in (1) increases in the number of individuals of some species, (2) the emergence of new species over time, and (3) the extinction of other species. [Clarification Statement: Emphasis is on determining cause and effect relationships for how changes to the environment such as deforestation, fishing, application of fertilizers, drought, flood, and the rate of change of the environment affect distribution or disappearance of traits in species.]

HS-LS4-6. Create or revise a simulation to test a solution to mitigate adverse impacts of human activity on biodiversity.*[Clarification Statement: Emphasis is on designing solutions for a proposed problem related to threatened or endangered species, or to genetic variation of organisms for multiple species.]

*The performance expectations marked with an asterisk integrate traditional science content with engineering through a practice or disciplinary core idea.

The bundle of performance expectations above focuses on the following elements from the NRC document *A Framework for K–12 Science Education*:

Highlighted Science and Engineering Practices	Highlighted Disciplinary Core Ideas	Highlighted Crosscutting Concepts
[SEP-4] Analyzing and Interpreting Data [SEP-5] Using mathematics and Computational Thinking [SEP-6] Constructing Explanations (for science) and Designing Solutions (for engineering) [SEP-7] Engaging in Argument from Evidence	LS4.B: Natural Selection LS4.C: Adaptation LS4.D: Biodiversity and Humans ETS1.B: Developing Possible Solutions	[CCC-1] Patterns [CCC-2] Cause and Effect
Highlighted California Environmental Principles & Concepts: Principle I The continuation and health of individual human lives and of human communities and societies depend on the health of the natural systems that provide essential goods and ecosystem services. Principle II The long-term functioning and health of terrestrial, freshwater, coastal, and marine ecosystems are influenced by their relationships with human societies.		
CA CCSS Math Connections: MP.2, MP.4		
CA CCSS for ELA/Literacy Connections: RST.11–12.1, 8; WHST.11–12.2; WHST.11–12.9a–e; SL.11–12.4		
CA ELD Standards Connections: ELD.PI.11–12.1, 5, 6a–b, 9, 10, 11a		

Chemistry Instructional Segment 1: Properties of Matter		
Guiding Questions: • How can we describe and measure the bulk properties of matter? • What causes different materials to have different bulk properties? • How can we infer the structure of matter at the atomic scale from properties of matter observed at the bulk scale?		
Performance Expectations		
<i>Students who demonstrate understanding can do the following:</i> HS-PS1-3 Plan and conduct an investigation to gather evidence to compare the structure of substances at the bulk scale to infer the strength of electrical forces between particles. [Clarification Statement: Emphasis is on understanding the strengths of forces between particles, not on naming specific intermolecular forces (such as dipole-dipole). Examples of particles could include ions, atoms, molecules, and networked materials (such as graphite). Examples of bulk properties of substances could include the melting point and boiling point, vapor pressure, and surface tension.] [Assessment Boundary: Assessment does not include Raoult's law calculations of vapor pressure.] (Introduced here but assessed in IS3)		
The bundle of performance expectations above focuses on the following elements from the NRC document <i>A Framework for K–12 Science Education</i> :		
Highlighted Science and Engineering Practices [SEP-3] Planning and Carrying Out Investigations	Highlighted Disciplinary Core Ideas PS1.A: Structure and Properties of Matter PS2.B: Types of Interactions	Highlighted Crosscutting Concepts [CCC-1] Patterns [CCC-2] Cause and Effect: Mechanism and Explanation [CCC-4] Systems and System Models
CA CCSS Math Connections: N-Q.1–3		
CA CCSS for ELA/Literacy Connections: RST.11–12.1; WHST.11–12.7, 9		
CA ELD Standards Connections: ELD.PI.11–12.1, 5, 6a–b, 9, 10, 11a		

High School Four-Course Model
Chemistry Instructional Segment 2: Structure of Matter

Guiding Questions:

- What is inside atoms and how does this affect how they interact?
- How does the structure of matter at the atomic scale explain patterns in the bulk properties of matter?

Performance Expectations

Students who demonstrate understanding can do the following:

HS-PS1-1. Use the periodic table as a model to predict the relative properties of elements based on the patterns of electrons in the outermost energy level of atoms. [Clarification Statement: Examples of properties that could be predicted from patterns could include reactivity of metals, types of bonds formed, numbers of bonds formed, and reactions with oxygen.] [Assessment Boundary: Assessment is limited to main group elements. Assessment does not include quantitative understanding of ionization energy beyond relative trends.]

HS-PS1-2 Construct and revise an explanation for the outcome of a simple chemical reaction based on the outermost electron states of atoms, trends in the periodic table, and knowledge of the patterns of chemical properties. [Clarification Statement: Examples of chemical reactions could include the reaction of sodium and chlorine, of carbon and oxygen, or of carbon and hydrogen.] [Assessment Boundary: Assessment is limited to chemical reactions involving main group elements and combustion reactions.]

The bundle of performance expectations above focuses on the following elements from the NRC document *A Framework for K–12 Science Education*:

Highlighted Science and Engineering Practices	Highlighted Disciplinary Core Ideas	Highlighted Crosscutting Concepts
[SEP-2] Developing and Using Models [SEP-6] Constructing Explanations (for science) and Designing Solutions (for engineering)	PS1.A: Structure and Properties of Matter PS1.B: Chemical Reactions	[CCC-1] Patterns [CCC-2] Cause and Effect: Mechanism and Explanation [CCC-6] Structure and Function

CA CCSS Math Connections: N-Q.1–3

CA CCSS for ELA/Literacy Connections: RST.9–10.7; WHST.11–12.2, 5

CA ELD Standards Connections: ELD.PI.11–12.1, 5, 6a–b, 9, 10, 11a

High School Four-Course Model

Chemistry Instructional Segment 3: Understanding Chemical Reactions

Guiding Questions:

- What holds atoms together to make molecules?
- Why do some combination of elements react and others do not?
- How do organisms harness energy from the chemical bonds in their food?

Performance Expectations

Students who demonstrate understanding can do the following:

HS-PS1-4 Develop a model to illustrate that the release or absorption of energy from a chemical reaction system depends upon the changes in total bond energy. [Clarification Statement: Emphasis is on the idea that a chemical reaction is a system that affects the energy change. Examples of models could include molecular-level drawings and diagrams of reactions, graphs showing the relative energies of reactants and products, and representations showing energy is conserved.] [Assessment Boundary: Assessment does not include calculating the total bond energy changes during a chemical reaction from the bond energies of reactants and products.]

HS-PS3-5 Develop and use a model of two objects interacting through electric or magnetic fields to illustrate the forces between objects and the changes in energy of the objects due to the interaction. [Clarification Statement: Examples of models could include drawings, diagrams, and texts, such as drawings of what happens when two charges of opposite polarity are near each other.] [Assessment Boundary: Assessment is limited to systems containing two objects.]

HS-PS2-4 Use mathematical representations of Newton’s Law of Gravitation and Coulomb’s Law to describe and predict the gravitational and electrostatic forces between objects. [Clarification Statement: Emphasis is on both quantitative and conceptual descriptions of gravitational and electric fields.] [Assessment Boundary: Assessment is limited to systems with two objects.]

The bundle of performance expectations above focuses on the following elements from the NRC document *A Framework for K–12 Science Education*:

Highlighted Science and Engineering Practices	Highlighted Disciplinary Core Ideas	Highlighted Crosscutting Concepts
[SEP-2] Developing and Using Models [SEP-5] Using Mathematics and Computational Thinking	PS1.A: Structure and Properties of Matter PS2.B: Types of Interactions PS3.C: Relationship Between Energy and Forces	[CCC-1] Patterns [CCC-2] Cause and Effect: Mechanism and Explanation [CCC-5] Energy and Matter: Flows, Cycles, and Conservation
CA CCSS Math Connections: A-SSE.1a–b, 3a–c; N-Q.1–3; MP.2, MP.4		
CA CCSS for ELA/Literacy Connections: SL.11–12.5; RST.11–12.1; WHST.11–12.1.a–e		
CA ELD Standards Connections: ELD.PI.11–12.1, 5, 6a–b, 9, 10, 11a		

Chemistry Instructional Segment 4: Modifying Chemical Reactions
<i>Guiding Questions:</i> • How can we alter chemical equilibrium and reaction rates? • How can we predict the relative quantities of products in a chemical reaction?
Performance Expectations
<i>Students who demonstrate understanding can do the following:</i> HS-PS1-5. Apply scientific principles and evidence to provide an explanation about the effects of changing the temperature or concentration of the reacting particles on the rate at which a reaction occurs. [Clarification Statement: Emphasis is on student reasoning that focuses on the number and energy of collisions between molecules.] [Assessment Boundary: Assessment is limited to simple reactions in which there are only two reactants; evidence from temperature, concentration, and rate data; and qualitative relationships between rate and temperature.] HS-PS1-6 Refine the design of a chemical system by specifying a change in conditions that would produce increased amounts of products at equilibrium.* [Clarification Statement: Emphasis is on the application of Le Châtelier’s Principle and on refining designs of chemical reaction systems, including descriptions of the connection between changes made at the macroscopic level and what happens at the molecular level. Examples of designs could include different

ways to increase product formation including adding reactants or removing products.] [Assessment Boundary: Assessment is limited to specifying the change in only one variable at a time. Assessment does not include calculating equilibrium constants and concentrations.]

HS-ETS1-2 Design a solution to a complex real-world problem by breaking it down into smaller, more manageable problems that can be solved through engineering.

HS-PS1-7 Use mathematical representations to support the claim that atoms, and therefore mass, are conserved during a chemical reaction. [Clarification Statement: Emphasis is on using mathematical ideas to communicate the proportional relationships between masses of atoms in the reactants and the products, and the translation of these relationships to the macroscopic scale using the mole as the conversion from the atomic to the macroscopic scale. Emphasis is on assessing students' use of mathematical thinking and not on memorization and rote application of problem-solving techniques.] [Assessment Boundary: Assessment does not include complex chemical reactions.]

*The performance expectations marked with an asterisk integrate traditional science content with engineering through a practice or disciplinary core idea.

The bundle of performance expectations above focuses on the following elements from the NRC document *A Framework for K–12 Science Education*:

Highlighted Science and Engineering Practices	Highlighted Disciplinary Core Ideas	Highlighted Crosscutting Concepts
[SEP-2] Developing and Using Models [SEP-5] Using Mathematics and Computational Thinking [SEP-6] Constructing Explanations (for science) and Designing Solutions (for engineering) [SEP-7] Engaging in Argument from Evidence	PS1.B: Chemical Reactions ETS1.C: Optimizing the Design Solution	[CCC-1] Patterns [CCC-2] Cause and Effect: Mechanism and Explanation [CCC-5] Energy and Matter: Flows, Cycles, and Conservation [CCC-7] Stability and Change

CA CCSS Math Connections: N-Q.1–3; MP.2, MP.4
CA CCSS for ELA/Literacy Connections: RST.11–12.1, 2; WHST.11–12.7
CA ELD Standards Connections: ELD.PI.11–12.1, 5, 6a–b, 9, 10, 11a

High School Four-Course Model Chemistry Instructional Segment 5: Energy Conservation, Transfer, and Applications
<i>Guiding Questions:</i> • How is energy transferred and conserved? • Why do some reactions release energy and others absorb it? • How can energy from chemical reactions be harnessed to perform useful tasks?
Performance Expectations
<i>Students who demonstrate understanding can do the following:</i> HS-PS3-1. Create a computational model to calculate the change in the energy of one component in a system when the change in energy of the other component(s) and energy flows in and out of the system are known. [Clarification Statement: Emphasis is on explaining the meaning of mathematical expressions used in the model.] [Assessment Boundary: Assessment is limited to basic algebraic expressions or computations; to systems of two or three components; and to thermal energy, kinetic energy, and/or the energies in gravitational, magnetic, or electric fields.] HS-ETS1-1. Analyze a major global challenge to specify qualitative and quantitative criteria and constraints for solutions that account for societal needs and wants. HS-ETS1-2. Design a solution to a complex real-world problem by breaking it down into smaller, more manageable problems that can be solved through engineering. HS-ETS1-3. Evaluate a solution to a complex real-world problem based on prioritized criteria and trade-offs that account for a range of constraints, including cost, safety, reliability, and aesthetics, as well as possible social, cultural, and environmental impacts. HS-ETS1-4. Use a computer simulation to model the impact of proposed solutions to a complex real-world problem with numerous criteria and constraints on interactions within and between systems relevant to the problem. HS-PS3-3 Design, build, and refine a device that works within given constraints to convert one form of energy into another form of energy.* [Clarification Statement: Emphasis is on both qualitative and quantitative evaluations of devices. Examples of devices could include Rube Goldberg devices, wind turbines, solar cells, solar ovens, and generators. Examples of constraints could include use

of renewable energy forms and efficiency.] [Assessment Boundary: Assessment for quantitative evaluations is limited to total output for a given input. Assessment is limited to devices constructed with materials provided to students.]

HS-PS3-4. Plan and conduct an investigation to provide evidence that the transfer of thermal energy when two components of different temperature are combined within a closed system results in a more uniform energy distribution among the components in the system (second law of thermodynamics). [Clarification Statement: Emphasis is on analyzing data from student investigations and using mathematical thinking to describe the energy changes both quantitatively and conceptually. Examples of investigations could include mixing liquids at different initial temperatures or adding objects at different temperatures to water.] [Assessment Boundary: Assessment is limited to investigations based on materials and tools provided to students.]

*The performance expectations marked with an asterisk integrate traditional science content with engineering through a practice or disciplinary core idea.

The bundle of performance expectations above focuses on the following elements from the NRC document *A Framework for K–12 Science Education*:

Highlighted Science and Engineering Practices	Highlighted Disciplinary Core Ideas	Highlighted Crosscutting Concepts
[SEP-2] Developing and Using Models	PS3.A: Definitions of Energy	[CCC-3] Scale, Proportion, and Quantity
[SEP-3] Planning and Carrying Out Investigations	PS3.D: Energy in Chemical Processes and Everyday Life	[CCC-4] Systems and system models
[SEP-4] Analyzing and Interpreting Data	ETS1.A: Defining and Delimiting Engineering Problems	[CCC-5] Energy and Matter: Flows, Cycles, and Conservation
[SEP-5] Using Mathematics and Computational Thinking	ETS1.B: Developing Possible Solutions	
[SEP-6] Constructing Explanations (for science) and Designing Solutions (for engineering)		

<i>Highlighted California Environmental Principles & Concepts:</i> Principle V Decisions affecting resources and natural systems are complex and involve many factors.
CA CCSS Math Connections: N-Q.1–3; MP.2, MP.4
CA CCSS for ELA/Literacy Connections: RST.11–12.1, 7, 8, 9; WHST.9–12.7, 8, 9; SL.11–12.5
CA ELD Standards Connections: ELD.PI.11–12.1, 5, 6a–b, 9, 10, 11a

High School Four-Course Model	
Physics Instructional Segment 1: Forces and Motion	
<i>Guiding Questions:</i>	
• How can Newton’s Laws be used to explain how and why things move? • How can the Newton’s Laws be used to solve engineering problems?	
Performance Expectations	
<i>Students who demonstrate understanding can do the following:</i>	
HS-PS2-1	Analyze data to support the claim that Newton’s second law of motion describes the mathematical relationship among the net force on a macroscopic object, its mass, and its acceleration. [Clarification Statement: Examples of data could include tables or graphs of position or velocity as a function of time for objects subject to a net unbalanced force, such as a falling object, an object rolling down a ramp, or a moving object being pulled by a constant force.] [Assessment Boundary: Assessment is limited to one-dimensional motion and to macroscopic objects moving at non-relativistic speeds.]
HS-PS2-2	Use mathematical representations to support the claim that the total momentum of a system of objects is conserved when there is no net force on the system. [Clarification Statement: Emphasis is on the quantitative conservation of momentum in interactions and the qualitative meaning of this principle.] [Assessment Boundary: Assessment is limited to systems of two macroscopic bodies moving in one dimension.]

- HS-PS2-3** Apply scientific and engineering ideas to design, evaluate, and refine a device that minimizes the force on a macroscopic object during a collision.* [Clarification Statement: Examples of evaluation and refinement could include determining the success of the device at protecting an object from damage and modifying the design to improve it. Examples of a device could include a football helmet or a parachute.] [Assessment Boundary: Assessment is limited to qualitative evaluations and/or algebraic manipulations.]
- HS-ETS1-1** Analyze a major global challenge to specify qualitative and quantitative criteria and constraints for solutions that account for societal needs and wants.
- HS-ETS1-4** Use a computer simulation to model the impact of proposed solutions to a complex real-world problem with numerous criteria and constraints on interactions within and between systems relevant to the problem.

* This performance expectation integrates traditional science content with engineering through a practice or disciplinary core idea.

The bundle of performance expectations above focuses on the following elements from the NRC document *A Framework for K–12 Science Education*:

Highlighted Science and Engineering Practices	Highlighted Disciplinary Core Ideas	Highlighted Crosscutting Concepts
[SEP-1] Asking Questions and Defining Problems [SEP-4] Analyzing and Interpreting Data [SEP-5] Using Mathematics and Computational Thinking	PS2.A: Forces and Motion ETS1.A: Defining and Delimiting Engineering Problems ETS1.B: Developing Possible Solutions	[CCC-2] Cause and Effect: Mechanism and Explanation [CCC-4] Systems and System Models Influence of Science, Engineering, and Technology on Society and the Natural World

[SEP-6] Constructing Explanations (for science) and Designing Solutions (for engineering)	ETS1.C: Optimizing the Design Solution	
Highlighted California Environmental Principles & Concepts: Principle V Decisions affecting resources and natural systems are based on a wide range of considerations and decision-making processes.		
CA CCSS Math Connections: A-SSE.1.a–b, 3a–c; A-CED.1, 2, 4; F-IF.7a–e; S-ID.1; N-Q.1–3; MP.2, MP.4		
CA CCSS for ELA/Literacy Connections: RST.11–12.1, 7, 8, 9; WHST.9–12.7, 9		
CA ELD Standards Connections: ELD.PI.11–12.1, 5, 6a–b, 9, 10, 11a		

High School Four-Course Model Physics Instructional Segment 2: Types of Interactions
<i>Guiding Questions:</i> • At the most fundamental level, how do objects interact with one another? • How do interactions affect the structure, nature, and properties of matter?
Performance Expectations
<i>Students who demonstrate understanding can do the following:</i> HS-PS2-4 Use mathematical representations of Newton’s Law of Gravitation and Coulomb’s Law to describe and predict the gravitational and electrostatic forces between objects. [Clarification Statement: Emphasis is on both quantitative and conceptual descriptions of gravitational and electric fields.] [Assessment Boundary: Assessment is limited to systems with two objects.] HS-PS2-5 Plan and conduct an investigation to provide evidence that an electric current can produce a magnetic field and that a changing magnetic field can produce an electric current.

[Assessment Boundary: Assessment is limited to designing and conducting investigations with provided materials and tools.]

HS-PS2-6 Communicate scientific and technical information about why the molecular-level structure is important in the functioning of designed materials.* [Clarification Statement: Emphasis is on the attractive and repulsive forces that determine the functioning of the material. Examples could include why electrically conductive materials are often made of metal, flexible but durable materials are made up of long chained molecules, and pharmaceuticals are designed to interact with specific receptors.] [Assessment Boundary: Assessment is limited to provided molecular structures of specific designed materials.]

** This performance expectation integrates traditional science content with engineering through a practice or disciplinary core idea.*

The bundle of performance expectations above focuses on the following elements from the NRC document *A Framework for K–12 Science Education*:

Highlighted Science and Engineering Practices	Highlighted Disciplinary Core Ideas	Highlighted Crosscutting Concepts
[SEP-3] Planning and Carrying Out Investigations [SEP-5] Using Mathematics and Computational Thinking [SEP-8] Obtaining, Evaluating, and Communicating Information	PS2.B: Types of Interaction PS3.A: Definitions of Energy ETS1.A: Defining and Delimiting Engineering Problems	[CCC-1] Patterns [CCC-2] Cause and Effect: Mechanism and Explanation [CCC-6] Structure and Function Influence of Science, Engineering, and Technology on Society and the Natural World Interdependence of Science, Engineering, and Technology

CA CCSS Math Connections: ASSE.1a–b, 3a–c; N-Q.1–3; MP.2, MP.4

CA CCSS for ELA/Literacy Connections: RST.11–12.1; WHST.9–12.2a–e, 7; WHST.11–12.8,9

CA ELD Standards Connections: ELD.PI.11–12.1, 5, 6a–b, 9, 10, 11a

High School Four-Course Model

Physics Instructional Segment 3: Energy

Guiding Questions:

- What is energy and how does it relate to and interact with matter?
- How do nuclear reactions illustrate conservation of energy and mass?
- What is the relationship between energy and force?
- How can you model force fields?

Performance Expectations

Students who demonstrate understanding can do the following:

HS-PS1-8 Develop models to illustrate the changes in the composition of the nucleus of the atom and the energy released during the processes of fission, fusion, and radioactive decay.

[Clarification Statement: Emphasis is on simple qualitative models, such as pictures or diagrams, and on the scale of energy released in nuclear processes relative to other kinds of transformations.]

[Assessment Boundary: Assessment does not include quantitative calculation of energy released. Assessment is limited to alpha, beta, and gamma radioactive decays.]

HS-PS3-2 Develop and use models to illustrate that energy at the macroscopic scale can be accounted for as a combination of energy associated with the motions of particles (objects) and energy associated with the relative position of particles (objects). [Clarification Statement: Examples of phenomena at the macroscopic scale could include the conversion of kinetic energy to thermal energy, the energy stored due to position of an object above the Earth, and the energy stored between two electrically-charged plates. Examples of models could include diagrams, drawings, descriptions, and computer simulations.]

HS-PS3-5 Develop and use a model of two objects interacting through electric or magnetic fields to illustrate the forces between objects and the changes in energy of the objects due to the interaction. [Clarification Statement: Examples of models could include drawings, diagrams, and texts, such as drawings of what happens when two charges of opposite polarity are near each other.] [Assessment Boundary: Assessment is limited to systems containing two objects.]

HS-PS3-3 Design, build, and refine a device that works within given constraints to convert one form of energy into another form of energy.* [Clarification Statement: Emphasis is on both qualitative and quantitative evaluations of devices. Examples of devices could include Rube Goldberg devices, wind turbines, solar cells, solar ovens, and generators. Examples of constraints could include use of renewable energy forms and efficiency.] [Assessment Boundary: Assessment for quantitative evaluations is limited to total output for a given input. Assessment is limited to devices constructed with materials provided to students.]

HS-ETS1-1 Analyze a major global challenge to specify qualitative and quantitative criteria and constraints for solutions that account for societal needs and wants.

HS-ETS1-2. Design a solution to a complex real-world problem by breaking it down into smaller, more manageable problems that can be solved through engineering.

HS-ETS1-3. Evaluate a solution to a complex real-world problem based on prioritized criteria and trade-offs that account for a range of constraints, including cost, safety, reliability, and aesthetics, as well as possible social, cultural, and environmental impacts.

** This performance expectation integrates traditional science content with engineering through a practice or disciplinary core idea.*

The bundle of performance expectations above focuses on the following elements from the NRC document *A Framework for K–12 Science Education*:

Highlighted Science and Engineering Practices	Highlighted Disciplinary Core Ideas	Highlighted Crosscutting Concepts
[SEP-1] Asking Questions and Defining Problems [SEP-2] Developing and Using Models [SEP-6] Constructing Explanations (for science) and Designing Solutions (for engineering)	PS1.C: Nuclear Processes PS3.A: Definitions of Energy PS3.C: Relationship Between Energy and Forces PS3.D: Energy in Chemical Processes ETS1.A: Defining and Delimiting an Engineering Problem ETS1.B: Developing Possible Solutions ETS1.C: Optimizing the Design Solution	[CCC-2] Cause and Effect: Mechanism and Explanation [CCC-5] Energy and Matter: Flows, Cycles, and Conservation Influence of Science, Engineering, and Technology on Society and the Natural World
Highlighted California Environmental Principles & Concepts: Principle V Decisions affecting resources and natural systems are based on a wide range of considerations and decision-making processes.		
CA CCSS Math Connections: N-Q.1–3; MP.2, MP.4		
CA CCSS for ELA/Literacy Connections: SL.11–12.5; RST.11–12.1, 7, 8, 9; WHST.9–12.7, 9; WHST.11–12.8,		
CA ELD Standards Connections: ELD.PI.11–12.1, 5, 6a–b, 9, 10, 11a		

High School Four-Course Model

Physics Instructional Segment 4: Waves and Electromagnetic Radiation

Guiding Questions:

- How can we describe waves in a way that helps us predict their behavior?
- How have humans harnessed electromagnetic radiation to fuel innovations, both literally and figuratively?
- Is electromagnetic radiation dangerous?

Performance Expectations

Students who demonstrate understanding can do the following:

HS-PS4-1 Use mathematical representations to support a claim

regarding relationships among the frequency, wavelength, and speed of waves traveling in various media. [Clarification Statement: Examples of data could include electromagnetic radiation traveling in a vacuum and glass, sound waves traveling through air and water, and seismic waves traveling through the Earth.] [Assessment Boundary: Assessment is limited to algebraic relationships and describing those relationships qualitatively.]

HS-PS4-3 Evaluate the claims, evidence, and reasoning behind the idea that electromagnetic radiation can be described either by a wave model or a particle model, and that for some situations one model is more useful than the other. [Clarification Statement: Emphasis is on how the experimental evidence supports the claim and how a theory is generally modified in light of new evidence. Examples of a phenomenon could include resonance, interference, diffraction, and photoelectric effect.] [Assessment Boundary: Assessment does not include using quantum theory.]

HS-PS4-4 Evaluate the validity and reliability of claims in published materials of the effects that different frequencies of

electromagnetic radiation have when absorbed by matter.

[Clarification Statement: Emphasis is on the idea that photons associated with different frequencies of light have different energies, and the damage to living tissue from electromagnetic radiation depends on the energy of the radiation. Examples of published materials could include trade books, magazines, Web resources, videos, and other passages that may reflect bias.] [Assessment Boundary: Assessment is limited to qualitative descriptions.]

HS-PS4-5 Communicate technical information about how some technological devices use the principles of wave behavior and wave interactions with matter to transmit and capture information and energy.* [Clarification Statement: Examples could include solar cells capturing light and converting it to electricity; medical imaging; and communications technology.] [Assessment Boundary: Assessments are limited to qualitative information. Assessments do not include band theory.]

HS-PS4-2 Evaluate questions about the advantages of using a digital transmission and storage of information. [Clarification Statement: Examples of advantages could include that digital information is stable because it can be stored reliably in computer memory, transferred easily, and copied and shared rapidly. Disadvantages could include issues of easy deletion, security, and theft.]

*The performance expectations marked with an asterisk integrate traditional science content with engineering through a practice or disciplinary core idea.

The bundle of performance expectations above focuses on the following elements from the NRC document *A Framework for K–12 Science Education*:

Highlighted Science and Engineering Practices	Highlighted Disciplinary Core Ideas	Highlighted Crosscutting Concepts
[SEP-1] Asking Questions and Defining Problems [SEP-5] Using Mathematics and Computational Thinking [SEP-7] Engaging in Argument from Evidence [SEP-8] Obtaining, Evaluating, and Communicating Information	PS3.D: Energy in Chemical Processes PS4.A: Wave Properties PS4.B: Electromagnetic Radiation PS4.C: Information Technologies and Instrumentation	CCC-2. Cause and Effect: Mechanism and Explanation CCC-4. Systems and System Models CCC-7. Stability and Change Influence of Engineering, Technology, and Science on Society and the Natural World
CA CCSS Math Connections: A-SSE.1a–b, 3a–c; A.CED.4; MP.2, MP.4		
CA CCSS for ELA/Literacy Connections: RST.11–12.1, 7, 8; RST.9–10.8; WHST.9–12.2a–e; WHST.11–12.8		
CA ELD Standards Connections: ELD.PI.11–12.1, 5, 6a–b, 9, 10, 11a		

High School Four-Course Model

Earth and Space Sciences Instructional Segment 1: Oil and Gas

Guiding Questions:

- Where do oil and gas come from?
- How are gas and oil deposits related to carbon cycling and Earth systems?
- What is the impact of driving cars and using other fossil fuels on the Earth systems?

Performance Expectations

Students who demonstrate understanding can do the following:

HS-ESS2-1. Develop a model to illustrate how Earth's internal and surface processes operate at different spatial and temporal scales to form continental and ocean-floor features. [Clarification Statement:

Emphasis is on how the appearance of land features (such as mountains, valleys, and plateaus) and sea-floor features (such as trenches, ridges, and seamounts) are a result of both constructive forces (such as volcanism, tectonic uplift, and orogeny) and destructive mechanisms (such as weathering, mass wasting, and coastal erosion).] [Assessment Boundary: Assessment does not include memorization of the details of the formation of specific geographic features of Earth's surface.] (Introduced here, but revisited from in IS3 and again in IS4)

HS-ESS2-6. Develop a quantitative model to describe the cycling of carbon among the hydrosphere, atmosphere, geosphere, and biosphere. [Clarification Statement: **The carbon cycle is a property of the Earth system that arises from interactions among the hydrosphere, atmosphere, geosphere, and biosphere.** Emphasis is on modeling biogeochemical cycles that include the cycling of carbon through the ocean, atmosphere, soil, and biosphere (including humans), providing the foundation for living organisms.]

HS-ESS2-7. Construct an argument based on evidence about the simultaneous coevolution of Earth's systems and life on Earth.

[Clarification Statement: Emphasis is on the dynamic causes, effects, and feedbacks between the biosphere and Earth's other systems, whereby geoscience factors control the evolution of life, which in turn continuously alters Earth's surface. Examples include: how photosynthetic life altered the atmosphere through the production of oxygen, which in turn increased weathering rates and allowed for the evolution of animal life; how microbial life on land increased the formation of soil, which in turn allowed for the evolution of land plants; or how the evolution of corals created reefs that altered patterns of erosion and deposition along coastlines and provided habitats for the evolution of new life forms.] [Assessment Boundary: Assessment does

not include a comprehensive understanding of the mechanisms of how the biosphere interacts with all of Earth's other systems.]		
The bundle of performance expectations above focuses on the following elements from the NRC document <i>A Framework for K–12 Science Education</i> :		
Highlighted Science and Engineering Practices	Highlighted Disciplinary Core Ideas	Highlighted Crosscutting Concepts
[SEP-2] Developing and Using Models [SEP-7] Engaging in Argument from Evidence	ESS2.A: Earth Materials and Systems ESS2.B: Plate Tectonics and Large-Scale System Interactions ESS2.D: Weather and Climate ESS2.E: Biogeology ESS3.A: Natural Resources	[CCC-5] Energy and Matter: Flows, Cycles, and Conservation [CCC-7] Stability and Change
<i>Highlighted California Environmental Principles & Concepts:</i> Principle III Natural systems proceed through cycles that humans depend upon, benefit from, and can alter. Principle IV: The exchange of matter between natural systems and human societies affects the long-term functioning of both.		
CA CCSS Math Connections: N-Q.1–3; MP.2, MP.4		
CA CCSS for ELA/Literacy Connections: WHST.9–12.1a–e		
CA ELD Standards Connections: ELD.PI.11–12.1, 5, 6a–b, 9, 10, 11a		

High School Four-Course Model
Earth and Space Sciences Instructional Segment 2: Climate
<i>Guiding Questions:</i> • What regulates weather and climate? • What effects are humans having on the climate?
Performance Expectations
<i>Students who demonstrate understanding can do the following:</i> HS-ESS2-2. Analyze geoscience data to make the claim that one change to Earth's surface can create feedbacks that cause changes to other Earth systems. [Clarification Statement: Examples should

include climate feedbacks, such as how an increase in greenhouse gases causes a rise in global temperatures that melts glacial ice, which reduces the amount of sunlight reflected from Earth's surface, increasing surface temperatures and further reducing the amount of ice. Examples could also be taken from other system interactions, such as how the loss of ground vegetation causes an increase in water runoff and soil erosion; how dammed rivers increase groundwater recharge, decrease sediment transport, and increase coastal erosion; or how the loss of wetlands causes a decrease in local humidity that further reduces the wetland extent.]

HS-ESS2-4. Use a model to describe how variations in the flow of energy into and out of Earth's systems result in changes in climate.

[Clarification Statement: Examples of the causes of climate change differ by timescale, over 1-10 years: large volcanic eruption, ocean circulation; 10-100s of years: changes in human activity, ocean circulation, solar output; 10-100s of thousands of years: changes to Earth's orbit and the orientation of its axis; and 10-100s of millions of years: long-term changes in atmospheric composition.]

[Assessment Boundary: Assessment of the results of changes in climate is limited to changes in surface temperatures, precipitation patterns, glacial ice volumes, sea levels, and biosphere distribution.]

HS-ESS2-6. Develop a quantitative model to describe the cycling of carbon among the hydrosphere, atmosphere, geosphere, and biosphere. [Clarification Statement: **The carbon cycle is a property of the Earth system that arises from interactions among the hydrosphere, atmosphere, geosphere, and biosphere.** Emphasis is on modeling biogeochemical cycles that include the cycling of carbon through the ocean, atmosphere, soil, and biosphere (including humans), providing the foundation for living organisms.]

HS-ESS3-2. Evaluate competing design solutions for developing, managing, and utilizing energy and mineral resources based on cost-benefit ratios.* [Clarification Statement: Emphasis is on the conservation, recycling, and reuse of resources (such as minerals and metals) where possible, and on minimizing impacts where it is not. Examples include developing best practices for agricultural soil use, mining (for coal, tar sands, and oil shales), and pumping (for petroleum and natural gas). Science knowledge indicates what can happen in natural systems—not what should happen.]

HS-ESS3-5. Analyze geoscience data and the results from global climate models to make an evidence-based forecast of the current rate of global or regional climate change and associated future impacts to Earth systems.

HS-ESS3-6. Use a computational representation to illustrate the relationships among Earth systems and how those relationships

are being modified due to human activity.* [Clarification Statement: Examples of Earth systems to be considered are the hydrosphere, atmosphere, cryosphere, geosphere, and/or biosphere. An example of the far-reaching impacts from a human activity is how an increase in atmospheric carbon dioxide results in an increase in photosynthetic biomass on land and an increase in ocean acidification, with resulting impacts on sea organism health and marine populations.]

[Assessment Boundary: Assessment does not include running computational representations but is limited to using the published results of scientific computational models.]

HS-ETS1-1. Analyze a major global challenge to specify qualitative and quantitative criteria and constraints for solutions that account for societal needs and wants.

* This performance expectation integrates traditional science content with engineering through a practice or disciplinary core idea.

The bundle of performance expectations above focuses on the following elements from the NRC document *A Framework for K–12 Science Education*:

Highlighted Science and Engineering Practices	Highlighted Disciplinary Core Ideas	Highlighted Crosscutting Concepts
[SEP-1] Asking Questions and Defining Problems	ESS1.B: Earth and the Solar System	[CCC-2] Cause and Effect
[SEP-2] Developing and Using Models	ESS2.A: Earth Materials and Systems	[CCC-4] Systems and System Models
[SEP-4] Analyzing and Interpreting Data	ESS2.D: Weather and Climate	[CCC-5] Energy and Matter: Flows, Cycles, and Conservation
[SEP-5] Using Mathematics and Computational Thinking	ESS3.A: Natural Resources	[CCC-7] Stability and Change
[SEP-7] Engaging in Argument from Evidence	ESS3.D: Global Climate Change	Influence of Science, Engineering, and Technology on Society and the Natural World
	ETS1.A: Defining and Delimiting Engineering Problems	Connections to Nature of Science
	ETS1.B: Developing Possible Solutions	Science Addresses Questions About the Natural and Material World

Highlighted California Environmental Principles & Concepts:
Principle I: The continuation and health of individual human lives and of human communities and societies depend on the health of the natural systems that provide essential goods and ecosystem services.
Principle II: The long-term functioning and health of terrestrial, freshwater, coastal and marine ecosystems are influenced by their relationships with human societies.
Principle III: Natural systems proceed through cycles that humans depend upon, benefit from, and can alter.
Principle IV: The exchange of matter between natural systems and human societies affects the long-term functioning of both.
Principle V: Decisions affecting resources and natural systems are based on a wide range of considerations and decision-making processes.
CA CCSS Math Connections: N-Q.1–3; MP.2, MP.4
CA CCSS for ELA/Literacy Connections: SL.11–12.5; RST.11–12.1,2, 7, 8
CA ELD Standards Connections: ELD.PI.11–12.1, 5, 6a–b, 9, 10, 11a

High School Four-Course Model

Earth and Space Sciences Instructional Segment 3: Mountains, Valleys, and Coasts

Guiding Questions:

- How did California's landscape get to look the way it does today?
- What forces shape the Earth's surface?
- How do those processes affect humans?

Performance Expectations

Students who demonstrate understanding can do the following:

HS-ESS2-1. Develop a model to illustrate how Earth's internal and surface processes operate at different spatial and temporal scales to form continental and ocean-floor features. [Clarification Statement: Emphasis is on how the appearance of land features (such as mountains, valleys, and plateaus) and sea-floor features (such as trenches, ridges, and seamounts) are a result of both constructive forces (such as volcanism, tectonic uplift, and orogeny) and destructive mechanisms (such as weathering, mass wasting, and coastal erosion).] [Assessment Boundary: Assessment does not include memorization of the details of the formation of specific geographic features of Earth's surface.] (Revisited from IS1 and again in IS4)

HS-ESS2-5. Plan and conduct an investigation of the properties of water and its effects on Earth materials and surface processes.

[Clarification Statement: Emphasis is on mechanical and chemical investigations with water and a variety of solid materials to provide the evidence for connections between the hydrologic cycle and system interactions commonly known as the rock cycle. Examples of

mechanical investigations include stream transportation and deposition using a stream table, erosion using variations in soil moisture content, or frost wedging by the expansion of water as it freezes. Examples of chemical investigations include chemical weathering and recrystallization (by testing the solubility of different materials) or melt generation (by examining how water lowers the melting temperature of most solids).]

HS-ESS3-1. Construct an explanation based on evidence for how the availability of natural resources, occurrence of natural hazards, and changes in climate have influenced human activity.

[Clarification Statement: Examples of key natural resources include access to fresh water (such as rivers, lakes, and groundwater), regions of fertile soils such as river deltas, and high concentrations of minerals and fossil fuels. Examples of natural hazards can be from interior processes (such as volcanic eruptions and earthquakes), surface processes (such as tsunamis, mass wasting and soil erosion), and severe weather (such as hurricanes, floods, and droughts). Examples of the results of changes in climate that can affect populations or drive mass migrations include changes to sea level, regional patterns of temperature and precipitation, and the types of crops and livestock that can be raised.]

HS-ETS1-3. Evaluate a solution to a complex real-world problem based on prioritized criteria and trade-offs that account for a range of constraints, including cost, safety, reliability, and aesthetics as well as possible social, cultural, and environmental impacts.

HS-ESS3-4. Evaluate or refine a technological solution that reduces impacts of human activities on natural systems.* [Clarification Statement: Examples of data on the impacts of human activities could include the quantities and types of pollutants released, changes to biomass and species diversity, or areal changes in land surface use (such as for urban development, agriculture and livestock, or surface mining). Examples for limiting future impacts could range from local efforts (such as reducing, reusing, and recycling resources) to large-scale geoengineering design solutions (such as altering global temperatures by making large changes to the atmosphere or ocean).]

HS-ESS3-5. Analyze geoscience data and the results from global climate models to make an evidence-based forecast of the current rate of global or regional climate change and associated future impacts to Earth systems. [Clarification Statement: Examples of evidence, for both data and climate model outputs, are for climate changes (such as precipitation and temperature) and their associated impacts (such as on sea level, glacial ice volumes, or atmosphere and ocean composition).] [Assessment Boundary: Assessment is limited to one example of a climate change and its associated impacts.]

* This performance expectation integrates traditional science content with engineering through a practice or disciplinary core idea.

The bundle of performance expectations above focuses on the following elements from the NRC document *A Framework for K–12 Science Education*:

Highlighted Science and Engineering Practices	Highlighted Disciplinary Core Ideas	Highlighted Crosscutting Concepts
[SEP-2] Developing and Using Models [SEP-3] Planning and Carrying Out Investigations [SEP-4] Analyzing and Interpreting Data [SEP-6] Constructing Explanations (for science) and Designing Solutions (for engineering)	ESS2.A: Earth Materials and Systems ESS2.B: Plate Tectonics and Large-Scale System Interactions ESS2.C: The Role of Water in Earth’s Surface Processes ESS3.A: Natural Resources ESS3.B: Natural Hazards ESS3.C: Human Impacts on Earth Systems ESS3.D: Global Climate Change ETS1.B: Developing Possible Solutions	[CCC-2] Cause and Effect [CCC-7] Stability and Change Connections to Engineering, Technology, and Applications of Science Influence of Science, Engineering, and Technology on Society and the Natural World

Highlighted California Environmental Principles & Concepts:

Principle I: The continuation and health of individual human lives and of human communities and societies depend on the health of the natural systems that provide essential goods and ecosystem services.

Principle II: The long-term functioning and health of terrestrial, freshwater, coastal, and marine ecosystems are influenced by their relationships with human societies.

Principle III: Natural systems proceed through cycles that humans depend upon, benefit from, and can alter.

Principle IV: The exchange of matter between natural systems and human societies affects the long-term functioning of both.

Principle V: Decisions affecting resources and natural systems are based on a wide range of considerations and decision-making processes.
CA CCSS Math Connections: N-Q.1–3; MP.2, MP.4
CA CCSS for ELA/Literacy Connections: SL.11–12.5; RST.11–12.1, 2, 7, 8, 9; WHST.9–12.2a–e,7
CA ELD Standards Connections: ELD.PI.11–12.1, 5, 6a–b, 9, 10, 11a

High School Four-Course Model
Earth and Space Sciences Instructional Segment 4: Water and Farming
<i>Guiding Questions:</i> • Why do droughts have such a strong impact on California and other parts of the world? • How will changes in climate affect our water resources?
Performance Expectations
<i>Students who demonstrate understanding can do the following:</i> HS-ESS3-1. Construct an explanation based on evidence for how the availability of natural resources, occurrence of natural hazards, and changes in climate have influenced human activity. [Clarification Statement: Examples of key natural resources include access to fresh water (such as rivers, lakes, and groundwater), regions of fertile soils such as river deltas, and high concentrations of minerals and fossil fuels. Examples of natural hazards can be from interior processes (such as volcanic eruptions and earthquakes), surface processes (such as tsunamis, mass wasting and soil erosion), and severe weather (such as hurricanes, floods, and droughts). Examples of the results of changes in climate that can affect populations or drive mass migrations include changes to sea level, regional patterns of temperature and precipitation, and the types of crops and livestock that can be raised.] HS-ESS3-3. Create a computational simulation to illustrate the relationships among management of natural resources, the sustainability of human populations, and biodiversity. [Clarification Statement: Examples of factors that affect the management of natural resources include costs of resource extraction and waste management, per-capita consumption, and the development of new technologies. Examples of factors that affect human sustainability include agricultural efficiency, levels of conservation, and urban planning.] [Assessment Boundary: Assessment for computational simulations is limited to using provided multi-parameter programs or constructing simplified spreadsheet calculations.] HS-ESS3-5. Analyze geoscience data and the results from global climate models to make an evidence-based forecast of the current rate of global or regional climate change and associated future impacts

to Earth system. [Clarification Statement: Examples of evidence, for both data and climate model outputs, are for climate changes (such as precipitation and temperature) and their associated impacts (such as on sea level, glacial ice volumes, or atmosphere and ocean composition).] [Assessment Boundary: Assessment is limited to one example of a climate change and its associated impacts.] HS-ETS1-1. Analyze a major global challenge to specify qualitative and quantitative criteria and constraints for solutions that account for societal needs and wants. HS-ETS1-4. Use a computer simulation to model the impact of proposed solutions to a complex real-world problem with numerous criteria and constraints on interactions within and between systems relevant to the problem.		
The bundle of performance expectations above focuses on the following elements from the NRC document <i>A Framework for K–12 Science Education</i>:		
Highlighted Science and Engineering Practices	Highlighted Disciplinary Core Ideas	Highlighted Crosscutting Concepts
[SEP-1] Asking Questions and Defining Problems	ESS2.C: The Roles of Water in Earth’s Surface Processes	[CCC-2] Cause and Effect: Mechanism and Explanation
[SEP-3] Planning and Carrying Out Investigations	ESS3.A: Natural Resources	[CCC-7] Stability and Change
[SEP-4] Analyzing and Interpreting Data	ESS3.B: Natural Hazards	Connections to Engineering, Technology, and Applications of Science
[SEP-5] Using Mathematics and Computational Thinking	ESS3.C: Human Impacts on Earth Systems	Influence of Science, Engineering, and Technology on Society and the Natural World
SEP-6. Constructing Explanations (for science) and Designing Solutions (for engineering)	ESS3.D: Global Climate Change	
	ETS1.A: Defining and Delimiting Engineering Problems	
	ETS1.C: Developing Possible Solutions	

Highlighted California Environmental Principles & Concepts: Principle I: The continuation and health of individual human lives and of human communities and societies depend on the health of the natural systems that provide essential goods and ecosystem services. Principle II: The long-term functioning and health of terrestrial, freshwater, coastal, and marine ecosystems are influenced by their relationships with human societies. Principle III: Natural systems proceed through cycles that humans depend upon, benefit from and can alter. Principle IV: The exchange of matter between natural systems and human societies affects the long-term functioning of both. Principle V: Decisions affecting resources and natural systems are based on a wide range of considerations and decision-making processes.
CA CCSS Math Connections: N-Q.1–3; MP.2, MP.4
CA CCSS for ELA/Literacy Connections: RST.11–12.1, 2, 7, 8, 9; WHST.9–12.2a–e, 7
CA ELD Standards Connections: ELD.PI.11-12.1, 5, 6a-b, 9, 10, 11a

High School Four-Course Model
Earth and Space Sciences Instructional Segment 5: Causes and Effects of Earthquakes
<i>Guiding Questions:</i> • What causes earthquakes?
Performance Expectations
<i>Students who demonstrate understanding can do the following:</i> HS-ESS1-5. Evaluate evidence of the past and current movements of continental and oceanic crust and the theory of plate tectonics to explain the ages of crustal rocks. [Clarification Statement: Emphasis is on the ability of plate tectonics to explain the ages of crustal rocks. Examples include evidence of the ages oceanic crust increasing with distance from mid-ocean ridges (a result of plate spreading) and the ages of North American continental crust increasing with distance away from a central ancient core (a result of past plate interactions).] HS-ESS2-1. Develop a model to illustrate how Earth’s internal and surface processes operate at different spatial and temporal scales to form continental and ocean-floor features. [Clarification Statement: Emphasis is on how the appearance of land features (such as mountains, valleys, and plateaus) and sea-floor features (such as trenches, ridges, and seamounts) are a result of both constructive forces (such as volcanism, tectonic uplift, and orogeny) and destructive mechanisms (such as weathering, mass wasting, and coastal erosion).] [Assessment Boundary: Assessment does

not include memorization of the details of the formation of specific geographic features of Earth's surface.]

HS-ESS2-3. Develop a model based on evidence of Earth's interior to describe the cycling of matter by thermal convection. [Clarification Statement: Emphasis is on both a one-dimensional model of Earth, with radial layers determined by density, and a three-dimensional model, which is controlled by mantle convection and the resulting plate tectonics. Examples of evidence include maps of Earth's three-dimensional structure obtained from seismic waves, records of the rate of change of Earth's magnetic field (as constraints on convection in the outer core), and identification of the composition of Earth's layers from high-pressure laboratory experiments.]

HS-ESS3-1. Construct an explanation based on evidence for how the availability of natural resources, occurrence of natural hazards, and changes in climate have influenced human activity. [Clarification Statement: Examples of key natural resources include access to fresh water (such as rivers, lakes, and groundwater), regions of fertile soils such as river deltas, and high concentrations of minerals and fossil fuels. Examples of natural hazards can be from interior processes (such as volcanic eruptions and earthquakes), surface processes (such as tsunamis, mass wasting and soil erosion), and severe weather (such as hurricanes, floods, and droughts). Examples of the results of changes in climate that can affect populations or drive mass migrations include changes to sea level, regional patterns of temperature and precipitation, and the types of crops and livestock that can be raised.]

The bundle of performance expectations above focuses on the following elements from the NRC document *A Framework for K–12 Science Education*:

Highlighted Science and Engineering Practices	Highlighted Disciplinary Core Ideas	Highlighted Crosscutting Concepts
[SEP-2] Developing and Using Models SEP-6. Constructing Explanations (for science) and Designing Solutions (for engineering) [SEP-7] Engaging in Argument from Evidence	ESS1.C: The History of Planet Earth ESS2.A: Earth Materials and Systems ESS2.B: Plate Tectonics and Large-Scale System Interactions ESS3.A: Natural Resources ESS3.B: Natural Hazards PS1.C: Nuclear Processes PS4.A: Wave Properties	[CCC-1 Patterns] [CCC-2] Cause and Effect: Mechanism and Explanation [CCC-5] Energy and Matter [CCC-7] Stability and Change

CA CCSS Math Connections: MP.2, MP.4; N-Q 1–3

CA CCSS for ELA/Literacy Connections: SL.11–12.5; RST.11–12.1; WHST.9–12.2.a–e

CA ELD Standards Connections: ELD.PI.11–12.1, 5, 6a–b, 9, 10, 11a

High School Four-Course Model

Earth and Space Sciences Instructional Segment 6: Urban Geoscience

Guiding Questions:

- How do Earth's natural systems influence our cities?
- How do cities affect Earth's natural systems?

Performance Expectations

Students who demonstrate understanding can do the following:

HS-ESS3-1. Construct an explanation based on evidence for how the availability of natural resources, occurrence of natural hazards, and changes in climate have influenced human activity.

[Clarification Statement: Examples of key natural resources include access to fresh water (such as rivers, lakes, and groundwater), regions of fertile soils such as river deltas, and high concentrations of minerals and fossil fuels. Examples of natural hazards can be from interior processes (such as volcanic eruptions and earthquakes), surface processes (such as tsunamis, mass wasting and soil erosion), and severe weather (such as hurricanes, floods, and droughts). Examples of the results of changes in climate that can affect populations or drive mass migrations include changes to sea level, regional patterns of temperature and precipitation, and the types of crops and livestock that can be raised.]

HS-ESS3-4. Evaluate or refine a technological solution that reduces impacts of human activities on natural systems.* [Clarification Statement: Examples of data on the impacts of human activities could include the quantities and types of pollutants released, changes to biomass and species diversity, or areal changes in land surface use (such as for urban development, agriculture and livestock, or surface mining). Examples for limiting future impacts could range from local efforts (such as reducing, reusing, and recycling resources) to large-scale geoengineering design solutions (such as altering global temperatures by making large changes to the atmosphere or ocean).]

HS-ETS1-2. Design a solution to a complex real-world problem by breaking it down into smaller, more manageable problems that can be solved through engineering.

HS-ETS1-4. Use a computer simulation to model the impact of proposed solutions to a complex real-world problem with numerous criteria and constraints on interactions within and between systems relevant to the problem.

* This performance expectation integrates traditional science content with engineering through a practice or disciplinary core idea.

The bundle of performance expectations above focuses on the following elements from the NRC document *A Framework for K–12 Science Education*:

Highlighted Science and Engineering Practices	Highlighted Disciplinary Core Ideas	Highlighted Crosscutting Concepts
[SEP-5] Using Mathematics and Computational Thinking [SEP-6] Constructing Explanations (for science) and Designing Solutions (for engineering)	ESS3.A: Natural Resources ESS3.B: Natural Hazards ESS3.C: Human Impacts on Earth Systems ETS1.B Developing Possible Solutions ETS1.C Optimizing Design Solutions	[CCC-2] Cause and Effect [CCC-4] Systems and system models [CCC-7] Stability and Change
Highlighted California Environmental Principles & Concepts: Principle I: The continuation and health of individual human lives and of human communities and societies depend on the health of the natural systems that provide essential goods and ecosystem services. Principle II: The long-term functioning and health of terrestrial, freshwater, coastal and marine ecosystems are influenced by their relationships with human societies. Principle III: Natural systems proceed through cycles that humans depend upon, benefit from and can alter. Principle IV: The exchange of matter between natural systems and human societies affects the long-term functioning of both. Principle V: Decisions affecting resources and natural systems are based on a wide range of considerations and decision-making processes.		
CA CCSS Math Connections: N-Q.1-3; MP.2, MP.4		
CA CCSS for ELA/Literacy Connections: RST.11-12.1, 2, 7, 8, 9		
CA ELD Standards Connections: ELD.PI.11-12.1, 5, 6a-b, 9, 10, 11a		

High School Four-Course Model

Earth and Space Sciences Instructional Segment 7: Star Stuff

Guiding Questions:

- How do we know what stars are made of?
- What fuels our Sun? Will it ever run out of that fuel?
- Do other stars work the same way as our Sun?

Performance Expectations

Students who demonstrate understanding can do the following:

HS-ESS1-1. Develop a model based on evidence to illustrate the life span of the Sun and the role of nuclear fusion in the Sun's core to release energy in the form of radiation. [Clarification Statement: Emphasis is on the energy transfer mechanisms that allow energy from nuclear fusion in the Sun's core to reach Earth. Examples of evidence for the model include observations of the masses and lifetimes of other stars, as well as the ways that the Sun's radiation varies due to sudden solar flares ("space weather"), the 11- year sunspot cycle, and non-cyclic variations over centuries.] [Assessment Boundary: Assessment does not include details of the atomic and sub-atomic processes involved with the Sun's nuclear fusion.]

HS-ESS1-3. Communicate scientific ideas about the way stars, over their life cycle, produce elements. [Clarification Statement: Emphasis is on the way nucleosynthesis, and therefore the different elements created, varies as a function of the mass of a star and the stage of its lifetime.] [Assessment Boundary: Details of the many different nucleosynthesis pathways for stars of differing masses are not assessed.]

HS-ESS1-6. Apply scientific reasoning and evidence from ancient Earth materials, meteorites, and other planetary surfaces to construct an account of Earth's formation and early history. [Clarification Statement: Emphasis is on using available evidence within the solar system to reconstruct the early history of Earth, which formed along with the rest of the solar system 4.6 billion years ago. Examples of evidence include the absolute ages of ancient materials (obtained by radiometric dating of meteorites, Moon rocks, and Earth's oldest minerals), the sizes and compositions of solar system objects, and the impact cratering record of planetary surfaces.] (Repeated in IS8)

* This performance expectation integrates traditional science content with engineering through a practice or disciplinary core idea.

The bundle of performance expectations above focuses on the following elements from the NRC document *A Framework for K–12 Science Education*:

Highlighted Science and Engineering Practices	Highlighted Disciplinary Core Ideas	Highlighted Crosscutting Concepts
[SEP-2] Developing and Using Models [SEP-6] Constructing Explanations (for science) and Designing Solutions (for engineering) [SEP-8] Obtaining, Evaluating, and Communicating Information	ESS1.A: The Universe and Its Stars ESS1.C: The History of Planet Earth PS1.C: Nuclear Processes PS3.D: Energy in Chemical Processes and Everyday Life	[CCC-1] Patterns [CCC-2] Cause and Effect: Mechanism and Explanation [CCC-3] Scale, Proportion, and Quantity [CCC-5] Energy and Matter: Flows, Cycles, and Conservation [CCC-7] Stability and Change
Highlighted California Environmental Principles & Concepts: Principle I: The continuation and health of individual human lives and of human communities and societies depend on the health of the natural systems that provide essential goods and ecosystem services. Principle II: The long-term functioning and health of terrestrial, freshwater, coastal and marine ecosystems are influenced by their relationships with human societies. Principle III: Natural systems proceed through cycles that humans depend upon, benefit from, and can alter. Principle IV: The exchange of matter between natural systems and human societies affects the long-term functioning of both. Principle V: Decisions affecting resources and natural systems are based on a wide range of considerations and decision-making processes.		
CA CCSS Math Connections: A-SSE.1a–b, A-CED.2, 4; F-IF.5; S-ID.6.a–c; N-Q.1–3; MP.2, MP.4		
CA CCSS for ELA/Literacy Connections: SL.11–12.4; RST.11–12.1, 2, 7, 8, 9; WHST.9–12.1a–e, 2a–e, 7		
CA ELD Standards Connections: ELD.PI.11–12.1, 5, 6a–b, 9, 10, 11a		

High School Four-Course Model
Earth and Space Sciences Instructional Segment 8: Motion in the Universe
<i>Guiding Questions:</i> • What are the predictable patterns of movement in our solar system and beyond? • What can those motions tell us about the origin of the universe and our planet?
Performance Expectations
<i>Students who demonstrate understanding can do the following:</i> HS-ESS1-2. Construct an explanation of the Big Bang theory based on astronomical evidence of light spectra, motion of distant galaxies, and composition of matter in the universe. [Clarification Statement: Emphasis is on the astronomical evidence of the red shift of light from galaxies as an indication that the universe is currently expanding, the cosmic microwave background as the remnant radiation from the Big Bang, and the observed composition of ordinary matter of the universe, primarily found in stars and interstellar gases (from the spectra of electromagnetic radiation from stars), which matches that predicted by the Big Bang theory (3/4 hydrogen and 1/4 helium).] HS-ESS1-4. Use mathematical or computational representations to predict the motion of orbiting objects in the solar system. [Clarification Statement: Emphasis is on Newtonian gravitational laws governing orbital motions, which apply to human-made satellites as well as planets and moons.] [Assessment Boundary: Mathematical representations for the gravitational attraction of bodies and Kepler's Laws of orbital motions should not deal with more than two bodies, nor involve calculus.] HS-ESS1-6. Apply scientific reasoning and evidence from ancient Earth materials, meteorites, and other planetary surfaces to construct an account of Earth's formation and early history. [Clarification Statement: Emphasis is on using available evidence within the solar system to reconstruct the early history of Earth, which formed along with the rest of the solar system 4.6 billion years ago. Examples of evidence include the absolute ages of ancient materials (obtained by radiometric dating of meteorites, Moon rocks, and Earth's oldest minerals), the sizes and compositions of solar system objects, and the impact cratering record of planetary surfaces.] (Revisited from IS 7)
The bundle of performance expectations above focuses on the following elements from the NRC document <i>A Framework for K–12 Science Education</i>:

Highlighted Science and Engineering Practices	Highlighted Disciplinary Core Ideas	Highlighted Crosscutting Concepts
[SEP-5] Using Mathematics and Computational Thinking [SEP-6] Constructing Explanations (for science) and Designing Solutions (for engineering)	ESS1.A: The Universe and Its Stars ESS1.B: Earth and the Solar System ESS1.C: The History of Planet Earth PS4.B: Electromagnetic Radiation	[CCC-3] Scale, Proportion, and Quantity [CCC-5] Energy and Matter: Flows, Cycles, and Conservation [CCC-6] Structure and Function [CCC-7] Stability and Change Connections to Engineering, Technology, and Applications of Science Interdependence of Science, Engineering, and Technology Connections to Nature of Science Scientific Knowledge Assumes an Order and Consistency in Natural Systems
Highlighted California Environmental Principles & Concepts: Principle I: The continuation and health of individual human lives and of human communities and societies depend on the health of the natural systems that provide essential goods and ecosystem services. Principle II: The long-term functioning and health of terrestrial, freshwater, coastal, and marine ecosystems are influenced by their relationships with human societies. Principle III: Natural systems proceed through cycles that humans depend upon, benefit from and can alter.		

Principle IV: The exchange of matter between natural systems and human societies affects the long-term functioning of both.
Principle V: Decisions affecting resources and natural systems are based on a wide range of considerations and decision-making processes.
CA CCSS Math Connections: A-SSE.1a–b, A-CED.2, 4; F-IF.5; S-ID.6.a–c; N-Q.1–3; MP.2, MP.4
CA CCSS for ELA/Literacy Connections: RST.11–12.1, 2, 7, 9; WHST.9–12.2a–e, 7
CA ELD Standards Connections: ELD.PI.11–12.1, 5, 6a–b, 9, 10, 11a

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